BIO 280 · PATHOPHYSIOLOGY · FINAL EXAM · JOYCE UNIVERSITY

FINAL EXAM STUDY GUIDE

The final covers **every module**, weighted most heavily toward **Modules 13 and 14**. Those two get a full deep dive here; Modules 1 through 12 get a focused rapid review of the mechanisms and discriminators you still have to know.

Dr. Akash Garg | All 14 modules | 30 topics | 102 practice questions

Module 13
Gastrointestinal — heavily weighted

Module 14
Liver, biliary & reproductive — heavily weighted

Modules 1–12
Rapid review of everything else

HOW TO USE THIS GUIDE

1. **Start with Modules 13 and 14.** They carry the most weight and they are the newest material, so they are also the least rehearsed.
2. **Then sweep the rapid reviews.** If a discriminator in those sections surprises you, that is your signal to go back to that module's own study guide.
3. **Drill, don't reread.** Open **Exam Prep HQ** and run the simulator in cumulative mode, then work the weakest exam it names.
4. **Answer the mechanism, not the disease.** When a question asks which change explains something, the right option is a process, not a diagnosis.

DEEP DIVE

Module 13 — Gastrointestinal

1 GI Motility and Secretion: Which Cell Makes What, and What Its Loss Causes

> **Key idea —** Every gastrointestinal symptom in this module traces back to one of three failures: a **cell stopped making its secretion**, a **nerve stopped telling muscle when to contract or relax**, or a **barrier stopped protecting the epithelium from what the epithelium itself produces**. Learn the cell, its product, and its feedback loop, and the disease writes itself.

The gastric gland is best understood as a small factory with five workers. **Parietal cells** in the body and fundus make hydrochloric acid through the H+/K+ ATPase proton pump, and they also make **intrinsic factor** — one cell, two products, which is why a single autoimmune attack produces both achlorhydria and B12 deficiency. **Chief cells** release pepsinogen, which acid itself cleaves into active pepsin. **Enterochromaffin-like (ECL) cells** release histamine, the final common amplifier of acid secretion. **G cells** in the antrum release gastrin into the blood. **D cells** release somatostatin, the universal off-switch. Acid secretion has three inputs — acetylcholine from the vagus, gastrin from G cells, and histamine from ECL cells — and because histamine amplifies the other two, H2 blockers blunt all three pathways indirectly while proton pump inhibitors shut the pump itself and therefore act more completely.

The feedback loop is the single most testable piece of physiology here. Acid arriving in the antrum stimulates D cells to release somatostatin, which switches off the G cells. Acid is therefore **the brake on its own production**. Anything that removes acid removes the brake, and gastrin rises. This is why the same laboratory value — a high serum gastrin — means completely different diseases depending on the gastric pH sitting next to it.

Cell (location)	Product	What its loss (or excess) causes
Parietal cell (gastric body and fundus)	HCl and intrinsic factor	Achlorhydria plus B12 deficiency; pernicious anaemia and subacute combined degeneration; reactive hypergastrinaemia and ECL hyperplasia
Chief cell (gastric body)	Pepsinogen	Reduced protein digestion, but pancreatic proteases compensate, so it rarely presents alone
ECL cell (gastric body)	Histamine	The amplifier of acid secretion; chronically driven by high gastrin it hyperplases and can form gastric carcinoids
G cell (antrum)	Gastrin	Loss removes the main hormonal drive to acid; ectopic excess (gastrinoma) gives Zollinger-Ellison syndrome
D cell (antrum and body)	Somatostatin	Loss removes the brake on gastrin, so acid runs unopposed — the antral H. pylori pattern
Surface mucous and neck cells	Mucus and bicarbonate gel	Loss of the alkaline barrier leaves epithelium bathed in acid and pepsin — the NSAID lesion
S cell (duodenum)	Secretin	Pancreatic bicarbonate is not called for, so the duodenum stays acidic and lipase is inactivated
I cell (duodenum)	Cholecystokinin	Gallbladder does not contract and enzymes are not released, so fat is maldigested
Pancreatic acinar cell	Lipase, amylase, proteases (as zymogens)	Steatorrhoea and deficiency of the fat-soluble vitamins A, D, E and K once lipase falls below about 10 per cent of normal
Pancreatic ductal cell (CFTR)	Bicarbonate-rich alkaline fluid	Thick inspissated secretions and an acid duodenum that inactivates whatever lipase remains — the cystic fibrosis pattern
Enterocyte brush border	Lactase and other disaccharidases	Undigested lactose holds water in the lumen and is fermented by colonic flora: osmotic diarrhoea, bloating and flatus

Interstitial cells of Cajal	Electrical slow waves (the pacemaker)	Loss of the basic rhythm produces gastroparesis, early satiety, and dysmotility
Myenteric (Auerbach) ganglion cells	Nitric oxide and VIP (inhibitory), acetylcholine (excitatory)	Loss of the inhibitory neurons means sphincters cannot relax: achalasia above, Hirschsprung aganglionic segment below

Autoimmune destruction of parietal cells → no intrinsic factor → no B12 uptake in the terminal ileum → megaloblastic anaemia + demyelination of dorsal columns and corticospinal tracts

> **Pearl —** Never interpret a serum gastrin without the gastric pH beside it. **High gastrin with a very low pH** means an autonomous source that is ignoring feedback — a gastrinoma. **High gastrin with a high pH** means the G cells are shouting appropriately at a stomach that can no longer make acid — atrophic gastritis, or simply a patient taking a proton pump inhibitor. Same number, opposite diseases.

Below the stomach, chyme acidity and fat content drive the next layer of control. **S cells** sense acid and release secretin, which calls for pancreatic and biliary bicarbonate to raise duodenal pH into the range where lipase works. **I cells** sense fatty acids and amino acids and release cholecystokinin, which contracts the gallbladder, stimulates acinar enzyme release, and slows gastric emptying so the duodenum is not overwhelmed. This is why a patient with Zollinger-Ellison syndrome has not only ulcers but **diarrhoea and steatorrhoea**: the acid load overwhelms bicarbonate buffering and denatures pancreatic lipase in the lumen.

MEMORY HOOK **CCK does three C things: C**ontracts the gallbladder, **C**alls out the pancreatic enzymes, and **C**lamps down gastric emptying.

Motility depends on a pacemaker and a wiring diagram. The **interstitial cells of Cajal** generate slow waves that set the maximum contraction frequency. The **myenteric plexus** controls motility; the **submucosal (Meissner) plexus** controls secretion and local blood flow. Between meals the **migrating motor complex** sweeps residue and bacteria distally roughly every 90 to 120 minutes — the housekeeper wave — and when it fails, bacteria are not swept out and small intestinal bacterial overgrowth follows. Diabetic autonomic neuropathy damages vagal fibres and the ICC network, producing gastroparesis with early satiety, postprandial vomiting of recognisable food eaten hours earlier, and erratic glycaemic control because absorption no longer matches insulin timing.

- **Achalasia:** loss of inhibitory NO and VIP neurons in the lower oesophageal sphincter, so the sphincter cannot relax and the body loses peristalsis — dysphagia to solids **and** liquids from the outset, regurgitation of undigested food, bird-beak tapering on barium.
- **Hirschsprung disease:** failure of neural crest migration leaves a distal aganglionic segment that is tonically contracted — failure to pass meconium in 48 hours, with dilated normal bowel proximal to the narrow abnormal segment.
- **Gastroparesis:** vagal and ICC injury from diabetes, surgery or medication, with delayed emptying and bezoar risk.
- **Ileus:** the enteric nervous system is intact but suppressed — by surgery, opioids, hypokalaemia, sepsis or peritonitis.
- **Bacterial overgrowth:** loss of the migrating motor complex or of gastric acid, giving bloating, B12 deficiency (bacteria consume it) and a paradoxically **elevated** folate (bacteria make it).

Loss of inhibitory NO/VIP neurons → lower oesophageal sphincter cannot relax → food column stalls and the body dilates → regurgitation, weight loss, aspiration risk

RED FLAG

A macrocytic anaemia must have **B12 measured before folate is given**. Folate will correct the blood picture in B12 deficiency while the spinal cord degeneration silently progresses. New paraesthesia, loss of vibration and proprioception, or an unsteady gait in a patient with atrophic gastritis or after ileal resection is a neurological emergency, not an anaemia to be treated at the next visit.

RED FLAG

Dysphagia of rapid onset in an adult over 60, especially with weight loss, is **pseudoachalasia from malignancy at the gastro-oesophageal junction until endoscopy proves otherwise**. True achalasia declares itself slowly over years.

PRACTICE

Q1. A 62-year-old woman with vitiligo and autoimmune hypothyroidism has a macrocytic anaemia and numbness in both feet with loss of vibration sense. Gastric biopsy shows destruction of the oxyntic glands of the body and fundus. Which single cellular loss explains BOTH the anaemia and the neurological findings?

A. Chief cell loss, so pepsinogen is no longer available to digest dietary protein

B. Parietal cell loss, so intrinsic factor is not produced and vitamin B12 cannot be absorbed in the terminal ileum

C. G cell loss in the antrum, so gastrin no longer stimulates acid secretion

D. D cell loss, so somatostatin no longer restrains gastrin release

Show answer

B is correct. The parietal cell makes two products — acid and intrinsic factor — so a single autoimmune attack causes achlorhydria and B12 malabsorption together. B12 requires intrinsic factor for receptor-mediated uptake in the terminal ileum; without it the patient develops megaloblastic anaemia plus subacute combined degeneration of the dorsal columns and corticospinal tracts. Chief cell loss is real in atrophic gastritis but pancreatic proteases compensate, so it is clinically silent. Neither G cell nor D cell loss causes B12 deficiency, and note the trap: in atrophic gastritis gastrin actually rises, because the loss of acid removes the somatostatin-mediated brake.

PRACTICE

Q2. Two patients each have a serum gastrin above 1000 pg/mL. Patient A has a gastric pH of 1.2, multiple ulcers in the distal duodenum and jejunum, and watery diarrhoea with steatorrhoea. Patient B has a gastric pH of 6.8 and a macrocytic anaemia. Which interpretation is correct?

A. Both must have a gastrinoma, because gastrin this high has only one cause

B. Patient A has autoimmune atrophic gastritis and Patient B has a gastrinoma

C. Patient A has an autonomous gastrin source (gastrinoma) ignoring feedback, while Patient B has atrophic gastritis in which loss of acid removes the normal brake on G cells

D. Both have H. pylori, since urease raises pH and therefore raises gastrin in every infected patient

Show answer

C is correct. Acid is the physiological brake on gastrin, acting through antral D cells and somatostatin. High gastrin persisting in the face of a very low pH means the source is ignoring feedback and is therefore ectopic and autonomous — a gastrinoma, which classically causes multiple ulcers in unusual distal locations plus diarrhoea and steatorrhoea because the acid load inactivates pancreatic lipase. High gastrin with achlorhydria is an appropriate, expected response to an absent brake, and is seen in atrophic gastritis and in any patient on a proton pump inhibitor. The number alone is uninterpretable without the pH.

PRACTICE

Q3. Manometry shows absent peristalsis in the distal oesophagus and a lower oesophageal sphincter that fails to relax with swallowing; barium shows a dilated body tapering to a smooth 'bird's beak'. The primary lesion is:

A. Excess acetylcholine released from surviving excitatory neurons

B. Fibrosis and scarring of the sphincter from years of acid reflux

C. Loss of inhibitory myenteric neurons that release nitric oxide and vasoactive intestinal peptide

D. Loss of interstitial cells of Cajal, so slow waves can no longer be generated

Show answer

C is correct. Sphincter relaxation is not a passive event — it is an active inhibition delivered by myenteric neurons using nitric oxide and VIP. When those neurons are destroyed (idiopathically, or by Trypanosoma cruzi in Chagas disease), the excitatory tone is unopposed and the sphincter stays shut, while the denervated body loses coordinated peristalsis. A peptic stricture from reflux does narrow the distal oesophagus, but it follows a long heartburn history, causes dysphagia to solids only, and leaves peristalsis intact — that is the tempting distractor. Loss of interstitial cells of Cajal disturbs gastric pacemaking and causes gastroparesis, not failure of sphincter relaxation.

PRACTICE

Q4. A man with long-standing alcohol-related chronic pancreatitis has bulky, greasy, foul-smelling stools, easy bruising, and difficulty seeing at night. His serum amylase is normal. The best explanation is:

A. Loss of pancreatic ductal cells alone, so bicarbonate is not secreted

B. Loss of pancreatic acinar cells, so lipase output falls below the digestive threshold and the fat-soluble vitamins A, D, E and K are not absorbed

C. Loss of enterocyte lactase, producing an osmotic diarrhoea

D. Loss of duodenal I cells, so cholecystokinin no longer empties the gallbladder

Show answer

B is correct. Steatorrhoea appears only when lipase secretion falls below roughly 10 per cent of normal, which requires extensive acinar destruction. Because that acinar mass is destroyed and burnt out, the serum amylase is often normal — a normal amylase never excludes chronic pancreatitis. Fat malabsorption carries the fat-soluble vitamins with it, giving night blindness (A) and easy bruising from vitamin K-dependent factor deficiency. Ductal bicarbonate loss genuinely contributes, because an acid duodenum inactivates residual lipase, but by itself it does not explain the vitamin deficiencies. Lactase deficiency gives watery, gassy osmotic diarrhoea, not greasy stool with vitamin deficits.

2 GERD and Barrett Oesophagus: A Barrier Problem That Becomes a Cancer Problem

> **Key idea —** Gastro-oesophageal reflux disease is **not an acid-excess disease**. Most patients make a normal amount of acid; the failure is mechanical containment. Severity tracks **contact time** — how long acid, pepsin and bile sit against squamous epithelium that was never designed to meet them.

The antireflux barrier has several components acting together: the intrinsic tone of the lower oesophageal sphincter, the crural diaphragm squeezing it from outside, the acute angle of His acting as a flap valve, and the length of intra-abdominal oesophagus that transmits abdominal pressure to the sphincter rather than working against it. A **hiatal hernia** is so important because it dismantles several of these at once: the sphincter slides into the chest, separates from the crural pinch, blunts the angle of His, and creates a small acid reservoir above the diaphragm that empties back into the oesophagus with every swallow.

The dominant mechanism of individual reflux episodes is the **transient lower oesophageal sphincter relaxation** — a vagally mediated relaxation that is not triggered by swallowing and is provoked by gastric distension. That single fact explains most of the management advice: large meals, late meals, obesity, and delayed gastric emptying all increase fundic distension and therefore increase reflux episodes. It also explains why raising the head of the bed, avoiding recumbency after eating, and weight loss work mechanically rather than pharmacologically.

- **Increased intra-abdominal pressure:** obesity, pregnancy, ascites, tight garments, straining.
- **Reduced sphincter tone:** nicotine, alcohol, chocolate, peppermint, high-fat meals, caffeine, calcium channel blockers, nitrates, anticholinergics, and progesterone in pregnancy.
- **Delayed clearance:** reduced salivary bicarbonate (sleep, xerostomia, scleroderma), impaired peristalsis.
- **Delayed gastric emptying:** gastroparesis, gastric outlet narrowing, large fatty meals.
- **Anatomical:** hiatal hernia, prior surgery at the gastro-oesophageal junction.

Histologically, refluxate widens the intercellular spaces of the squamous epithelium first — this is why patients can have symptoms with a normal-looking endoscopy (non-erosive reflux disease). Persistent injury produces basal zone hyperplasia, elongated vascular papillae, and intraepithelial eosinophils and neutrophils. As injury deepens the ladder of complications climbs: erosive oesophagitis, then ulceration and bleeding, then fibrotic **peptic stricture** (progressive dysphagia to solids in a patient with a long heartburn history), then metaplasia.

Chronic acid and bile exposure → squamous injury → intestinal metaplasia with goblet cells (Barrett) → low-grade dysplasia → high-grade dysplasia → oesophageal adenocarcinoma

Barrett oesophagus is metaplasia: the stratified squamous epithelium of the distal oesophagus is replaced by intestinal-type columnar epithelium containing goblet cells. Endoscopically it appears as salmon-pink tongues extending proximally above the gastro-oesophageal junction. Metaplasia is an **adaptive** change — columnar epithelium survives acid far better than squamous — but it is bought at a price, because the new epithelium is a proliferative, genetically unstable population from which dysplasia and then adenocarcinoma can arise. Progression risk is low per year (roughly half a per cent) but cumulative over decades, which is why surveillance endoscopy with biopsy, not symptom control, is what determines outcome.

> **Pearl —** Many patients report that their heartburn **improved** as Barrett developed. This is not healing — columnar metaplastic epithelium simply lacks the acid-sensitive nerve endings of squamous mucosa. The most dangerous transformation in the oesophagus is often the point at which the patient stops complaining. Treat a change in symptom character in a long-standing refluxer as a reason to look, not a reason to relax.

Feature	Adenocarcinoma	Squamous cell carcinoma
Typical location	Distal third, at or just above the gastro-oesophageal junction	Upper and middle thirds
Precursor lesion	Barrett intestinal metaplasia then dysplasia	Squamous dysplasia; no Barrett involvement
Dominant risk factors	Chronic GERD, obesity, hiatal hernia, male sex, smoking	Smoking and alcohol together, hot liquids, caustic (lye) stricture, achalasia, nitrosamines, poor nutrition
Trend in Western populations	Rising, tracking the obesity and reflux burden	Falling where smoking rates have fallen
Presenting picture	Long heartburn history, then progressive dysphagia to solids and weight loss	Dysphagia and weight loss in a smoker and drinker, often with hoarseness

MEMORY HOOK **Barrett Becomes Bowel.** The oesophagus grows goblet cells it was never supposed to have — intestinal epithelium in the wrong postcode. Goblet cells are the histological price of admission for the diagnosis.

RED FLAG

Alarm features in a refluxer demand endoscopy rather than another trial of acid suppression: **dysphagia, odynophagia, unintentional weight loss, iron deficiency anaemia, overt GI bleeding or melaena, persistent vomiting, a palpable mass, or new-onset symptoms in an older adult**. Empirical treatment of alarm symptoms delays a cancer diagnosis into an incurable stage.

Reflux also causes disease outside the oesophagus, because refluxate can reach the pharynx and airway. Chronic cough, hoarseness and posterior laryngitis, adult-onset or worsening asthma, globus sensation, and **dental enamel erosion on the lingual surfaces** are all recognised extraoesophageal manifestations. In a debilitated or neurologically impaired patient, nocturnal reflux is a genuine aspiration risk and can present as recurrent pneumonia rather than as heartburn at all.

PRACTICE

Q1. Endoscopy in a 55-year-old man with 20 years of heartburn shows salmon-coloured tongues of mucosa extending 4 cm above the gastro-oesophageal junction. Biopsy shows columnar epithelium containing goblet cells. This change is best described as:

A. Dysplasia, an established premalignant disordered growth

B. Hyperplasia of the native squamous epithelium in response to injury

C. Metaplasia — replacement of stratified squamous by intestinal-type columnar epithelium, adaptive in the short term but premalignant over time

D. Anaplasia, indicating that invasive carcinoma is already present

Show answer

C is correct. Metaplasia is the replacement of one differentiated cell type by another that better tolerates the local stress; here squamous epithelium is replaced by intestinal-type columnar epithelium with goblet cells, which survives acid far better. It is adaptive and can regress if the stimulus is removed, but the new proliferative population is genetically unstable and is the substrate for adenocarcinoma. Dysplasia is the next step — disordered, atypical growth that has not yet invaded — and would be reported as such by the pathologist. Anaplasia means complete loss of differentiation and belongs to established malignancy. Hyperplasia is an increase in the number of the same cells, which is not what is described.

PRACTICE

Q2. A 58-year-old man with two decades of nightly heartburn reports that his symptoms have actually eased over the past two years, but he now regurgitates food and has lost 6 kg without trying. Why is the improvement in heartburn NOT reassuring?

A. Columnar metaplastic epithelium is far less acid-sensitive than squamous epithelium, so symptoms fade at exactly the point the highest-risk lesion is developing

B. Symptom improvement proves the lower oesophageal sphincter has recovered, so the weight loss must be unrelated

C. Acid suppression eliminates the risk of adenocarcinoma once symptoms resolve

D. Weight loss lowers intra-abdominal pressure and therefore reduces reflux, which explains the whole picture

Show answer

A is correct. Barrett epithelium lacks the acid-sensitive nerve endings of squamous mucosa, so heartburn commonly diminishes as metaplasia advances — a false reassurance. The combination of a change in symptoms with new regurgitation and unintentional weight loss is a cluster of alarm features requiring endoscopy now. Option 4 is genuinely tempting because deliberate weight loss does reduce reflux mechanically, but this weight loss is unintentional and is therefore a red flag, not a therapeutic effect. Acid suppression relieves symptoms and heals oesophagitis, but it does not abolish the cancer risk once metaplasia exists, which is why surveillance rather than symptom control determines outcome.

PRACTICE

Q3. Which pairing of location, risk factor and tumour type is correct?

A. Distal third with long-standing GERD and obesity gives adenocarcinoma; upper and middle thirds with smoking, alcohol, achalasia or a caustic stricture gives squamous cell carcinoma

B. Distal third with smoking and alcohol gives adenocarcinoma; upper third with GERD gives squamous cell carcinoma

C. Both tumours arise in the distal third; only the risk factors differ

D. Barrett oesophagus is the precursor lesion for squamous cell carcinoma

Show answer

A is correct. Adenocarcinoma follows the reflux pathway and therefore arises where the reflux is — the distal third, out of Barrett metaplasia, in patients with obesity, hiatal hernia and long-standing GERD. Squamous cell carcinoma arises from squamous dysplasia in the upper and middle thirds and is driven by mucosal irritants and stasis: smoking and alcohol acting together, very hot liquids, lye strictures, and achalasia, in which retained fermenting food chronically irritates the mucosa. Barrett is never the precursor of squamous carcinoma — by definition Barrett has already lost its squamous lining.

PRACTICE

Q4. Which mechanism accounts for most individual reflux episodes in typical GERD?

A. A permanently hypotensive lower oesophageal sphincter with a resting pressure near zero

B. Transient lower oesophageal sphincter relaxations that are not triggered by swallowing and are provoked by gastric distension

C. Failure of the oesophageal squamous epithelium to secrete a protective mucus layer

D. Acid output that is substantially above the normal range

Show answer

B is correct. Most patients with GERD have a normal resting sphincter pressure. The dominant event is the transient relaxation — a vagally mediated reflex triggered by fundic distension, unrelated to swallowing — which is precisely why large meals, obesity, delayed gastric emptying and a hiatal hernia matter so much. A permanently hypotensive sphincter does occur, notably in scleroderma, but it is the exception. Acid hypersecretion is not the usual problem: the acid is normal in quantity and simply in the wrong place, which is why the disease is defined by contact time, not by acid output.

3 Peptic Ulcer Disease: Gastric versus Duodenal

> **Key idea —** An ulcer is a **local failure of the balance between aggressive factors and mucosal defence**, deep enough to breach the muscularis mucosae — that depth is what separates a true ulcer from an erosion, and it is what allows an ulcer to reach an artery or the peritoneal cavity. Two causes dominate, and they attack from opposite directions: **H. pylori raises aggression**, while **NSAIDs remove defence**.

Mucosal defence is layered. A mucus-bicarbonate gel holds a pH gradient so that the luminal surface can sit at pH 2 while the epithelial cell surface stays near neutral. Surface phospholipids make the layer hydrophobic. Damaged cells are replaced within minutes by **restitution**, in which neighbouring cells flatten and migrate across the defect. Underneath all of this, mucosal blood flow removes back-diffused hydrogen ions and delivers the bicarbonate and oxygen the repair needs. **Prostaglandin E2 maintains almost all of these layers simultaneously** — mucus output, bicarbonate output, blood flow and restitution — which is why a single drug that blocks prostaglandin synthesis can dismantle the whole system.

Helicobacter pylori is a spiral, urease-producing organism that survives in the mucus layer by generating ammonia to buffer its immediate surroundings. Where it settles determines the disease, and this is the highest-yield concept in the topic. **Antral-predominant infection** inflames the D cells, reducing somatostatin, so gastrin rises, parietal cell mass expands and acid output climbs; the excess acid delivered into the duodenum induces gastric metaplasia in the bulb, which the organism then colonises — the **duodenal ulcer** pathway. **Body-predominant or pangastritis** destroys the oxyntic glands instead, producing atrophy, hypochlorhydria and intestinal metaplasia — the **gastric ulcer, gastric adenocarcinoma and MALT lymphoma** pathway. One organism, opposite acid consequences, decided by geography.

Antral H. pylori → D cell inhibition → somatostatin falls → gastrin rises → acid output rises → acid load into duodenal bulb → gastric metaplasia → colonisation and ulceration

NSAIDs injure by inhibiting COX-1, cutting prostaglandin E2 and stripping mucus, bicarbonate and mucosal blood flow. The effect is systemic, not merely topical, which is the point students most often miss: enteric-coated, rectal and intravenous NSAIDs still cause ulcers because the prostaglandin deficiency is delivered by the bloodstream. Risk multiplies with age over 65, high dose or two NSAIDs together,

concurrent corticosteroids, anticoagulants or SSRIs, a prior ulcer, and coexisting H. pylori. Because NSAID ulcers blunt pain perception through the same prostaglandin pathway, an appreciable number of NSAID users present for the first time with **bleeding or perforation rather than pain**.

Feature	Gastric ulcer	Duodenal ulcer
Peak age	Older, typically over 55	Younger, typically 30 to 55
Timing of pain	Soon after eating, roughly 15 to 30 minutes	2 to 5 hours after a meal and classically at 1 to 2 a.m.
Effect of food	Worsens the pain, so the patient avoids eating	Buffers the acid and relieves the pain, so the patient eats more
Weight trajectory	Weight loss because eating hurts	Weight stable or gaining
Acid output	Normal or low; the defect is defence	Normal or high; the defect is excess aggression
H. pylori association	Around 70 per cent	Over 90 per cent
Dominant mechanism	Broken mucosal barrier (NSAIDs, atrophic gastritis, bile reflux)	Acid and pepsin overwhelming a normal barrier
Malignant potential	Real — every gastric ulcer must be biopsied and healing confirmed	Essentially none; malignancy here is a rarity
Classic complication	Bleeding from the left gastric artery on the lesser curve	Posterior wall erosion into the gastroduodenal artery, or anterior wall perforation

Complications follow anatomy exactly. **Bleeding** is the most common; a **posterior** duodenal bulb ulcer lies directly against the gastroduodenal artery and produces brisk arterial haemorrhage, while a **gastric** ulcer on the lesser curve erodes branches of the left gastric artery. **Perforation** is the province of the **anterior** duodenal wall, which has nothing behind it but peritoneum: sudden severe pain, a rigid board-like abdomen, absent bowel sounds and free air under the diaphragm. **Penetration** is a walled-off

perforation into an adjacent organ, classically into the pancreas, converting the pain into a constant boring ache radiating to the back. **Gastric outlet obstruction** results from oedema or scarring near the pylorus and produces vomiting of undigested food and a succussion splash.

MEMORY HOOK **Anterior perforates (free Air); posterior bleeds (the artery is Behind).** Two walls of the same duodenal bulb, two entirely different emergencies.

MEMORY HOOK **Cushing = Cranium** (raised intracranial pressure drives vagal acid hypersecretion). **Curling = Charred** (burns cause hypovolaemia and mucosal ischaemia). Both are stress-related mucosal disease; the routes are opposite — too much acid versus too little blood.

Pearl — Every **gastric** ulcer is biopsied and its healing confirmed, because an ulcerated gastric carcinoma is indistinguishable from a benign ulcer on symptoms alone. **Duodenal** ulcers are not routinely biopsied, because malignancy there is vanishingly rare. If instead you find **multiple ulcers, ulcers distal to the duodenal bulb, ulcers that recur after eradication and acid suppression, or ulcers with diarrhoea and steatorrhoea**, stop thinking about common causes and think Zollinger-Ellison syndrome.

RED FLAG

Sudden, severe, diffuse abdominal pain in a known or suspected ulcer patient, with a rigid abdomen, guarding, absent bowel sounds and a patient lying perfectly still — this is **perforation with chemical peritonitis**. Free air under the diaphragm on an upright film confirms it. This is a surgical emergency, and giving analgesia and observing is the wrong instinct.

PRACTICE

Q1. A 41-year-old man describes burning epigastric pain that wakes him at 2 a.m. and is relieved within minutes by milk or an antacid. He has gained 2 kg because eating helps. The most likely lesion and its explanation is:

A. A gastric ulcer, because food stimulates acid secretion and worsens the pain

B. A duodenal ulcer, because acid delivered into an empty duodenum causes pain that food buffers

C. Oesophageal adenocarcinoma, because nocturnal symptoms indicate malignancy

D. Acute pancreatitis, because the pain is epigastric and relieved by antacid

Show answer

B is correct. Duodenal ulcer pain appears 2 to 5 hours after eating and characteristically at night, when acid is secreted into a duodenum that has no food left to buffer it; food or antacid relieves it within minutes, so patients eat more and maintain or gain weight. The gastric ulcer pattern is the mirror image — pain shortly after eating, food avoidance and weight loss — which makes it the tempting distractor, but it does not fit a man gaining weight because eating helps. Oesophageal malignancy presents with progressive dysphagia and weight loss, and pancreatitis produces constant boring pain radiating to the back that is not relieved by antacid.

PRACTICE

Q2. An 80-year-old woman taking daily naproxen for osteoarthritis develops a gastric ulcer. H. pylori testing is negative. The mechanism is:

A. Direct stimulation of parietal cell acid hypersecretion by the drug

B. Inhibition of COX-1 with loss of prostaglandin E2, reducing mucus and bicarbonate secretion and mucosal blood flow

C. Autoimmune destruction of parietal cells triggered by the drug

D. Bacterial urease neutralising the protective mucus layer

Show answer

B is correct. NSAIDs do not raise acid output; they dismantle defence. Prostaglandin E2 sustains mucus and bicarbonate secretion, mucosal blood flow and epithelial restitution all at once, so blocking COX-1 removes several protective layers simultaneously and normal acid becomes sufficient to ulcerate. Because the deficiency is delivered systemically, enteric-coated, rectal and intravenous NSAIDs are not protective — a widespread misconception. Age over 65, corticosteroids, anticoagulants, SSRIs and a prior ulcer all multiply the risk. Urease is the H. pylori mechanism, which has been excluded here.

PRACTICE

Q3. A patient with a known duodenal ulcer suddenly vomits a large volume of bright red blood and becomes hypotensive. Which anatomical explanation best fits?

A. An anterior duodenal ulcer has perforated into the free peritoneal cavity

B. A posterior duodenal ulcer has eroded into the gastroduodenal artery

C. A pyloric channel ulcer has caused gastric outlet obstruction

D. A Mallory-Weiss tear has occurred at the gastro-oesophageal junction

Show answer

B is correct. The posterior wall of the duodenal bulb lies directly against the gastroduodenal artery, so a posterior ulcer that penetrates gives brisk arterial haemorrhage and can exsanguinate a patient. Anterior ulcers have only peritoneum in front of them, so they perforate rather than bleed and present with sudden severe pain, a rigid abdomen and free air — not haematemesis. Gastric outlet obstruction causes vomiting of undigested food, not blood. A Mallory-Weiss tear is a genuine cause of haematemesis but follows retching or vomiting, is usually self-limited, and does not explain a known ulcer eroding into a vessel.

PRACTICE

Q4. How does antral-predominant H. pylori infection increase the risk of DUODENAL ulcer specifically?

A. Urease raises gastric pH, which activates pepsinogen more efficiently

B. Inflammation reduces antral D cell somatostatin, so gastrin and acid output rise, and the acid load induces gastric metaplasia in the duodenal bulb that the organism then colonises

C. It destroys parietal cells, producing achlorhydria and bacterial overgrowth in the duodenum

D. It blocks intrinsic factor secretion, and the resulting B12 deficiency impairs mucosal repair

Show answer

B is correct. Antral infection knocks out the somatostatin brake, so gastrin rises, parietal cell mass expands and acid output increases; the duodenal bulb responds to that acid load by undergoing gastric metaplasia, which gives H. pylori a foothold where it does not normally live, and ulceration follows. The trap is option D, which describes the opposite phenotype: body-predominant or pangastritis causes atrophy and hypochlorhydria and is the pattern associated with gastric ulcer, gastric adenocarcinoma and MALT lymphoma. The same organism produces opposite acid consequences depending on where in the stomach it settles. Pepsinogen is activated by low pH, not high.

4 Upper versus Lower GI Bleeding: Localising the Source

> **Key idea —** The dividing line is the **ligament of Treitz**, and the way you localise a bleed is by asking **what the blood has been through**, not how much of it there is. Colour, odour and the urea-to-creatinine ratio are all read-outs of transit time and digestion, and a very fast bleed can defeat all three.

Blood entering the stomach is degraded by acid and then by colonic bacteria; haemoglobin is converted to haematin and related pigments, producing stool that is **black, sticky, tarry and characteristically foul-smelling**. This requires both a volume threshold — usually 50 to 100 mL — and time, roughly 14 hours of transit. Blood that reaches the rectum without that exposure stays red or maroon. So melaena tells you the blood sat in acid and moved slowly; haematochezia tells you it did not — either because the source is distal, or because the bleed was so fast that it swept through before digestion could occur.

The **urea-to-creatinine ratio** is an elegant piece of physiology. Blood in the small intestine is a large protein meal; it is digested and absorbed, and its nitrogen is converted to urea, raising the BUN. Simultaneously, hypovolaemia reduces renal perfusion and drives proximal tubular urea reabsorption. Both effects push the same way, so a BUN:creatinine ratio above roughly 20 to 30 to 1 with a normal creatinine strongly favours an upper source. A lower GI bleed does not raise urea this way, because the blood never passes through the absorptive small bowel.

Clue	Upper GI bleed (proximal to the ligament of Treitz)	Lower GI bleed (distal to the ligament of Treitz)
Typical presentation	Haematemesis, or coffee-ground vomitus when blood has sat in acid	Haematochezia — bright red or maroon blood per rectum
Stool character	Black, tarry, sticky, malodorous melaena	Red or maroon blood, clots, blood streaking or mixed through stool
BUN:creatinine ratio	Raised, often above 20 to 30 to 1 with a normal creatinine	Usually normal
Bowel sounds	Often hyperactive as blood acts as a cathartic in the gut	Normal or unremarkable
Common causes	Peptic ulcer (most common), oesophageal varices, erosive gastritis, Mallory-Weiss tear, malignancy, Dieulafoy lesion	Diverticulosis (most common), angiodysplasia, haemorrhoids, ischaemic colitis, IBD, neoplasm
Diagnostic test of choice	Upper endoscopy, after resuscitation	Colonoscopy once stable, with CT angiography if bleeding is brisk

Pearl — Haemodynamics change before haemoglobin does. Whole blood is lost with its cells and plasma in the same proportion, so immediately after an acute bleed the haemoglobin concentration is unchanged. It falls only as interstitial fluid shifts in over the following hours, or as soon as crystalloid is given. Tachycardia, a narrowed pulse pressure, orthostatic change and delayed capillary refill are the early findings; a reassuring haemoglobin one hour into a major bleed is a laboratory artefact of timing.

RED FLAG

Haematochezia in a hypotensive, tachycardic patient must be treated as a massive UPPER bleed until endoscopy proves otherwise. Losing more than about a litre rapidly pushes red blood through the gut before it can be degraded, and haemodynamic instability with red blood per rectum predicts an upper source. Anchoring on 'red blood means colon' in an unstable patient is the classic and dangerous error. Resuscitate first, scope second.

Upper causes each have a mechanical signature. **Peptic ulcer** is the most common overall, bleeding when the ulcer base erodes a vessel. **Oesophageal varices** bleed

painlessly, massively, and in the context of chronic liver disease — this is the bleed that kills fastest. A **Mallory-Weiss tear** is a longitudinal mucosal laceration at the gastro-oesophageal junction produced by forceful retching, so the classic history is several episodes of non-bloody vomiting followed by blood; it is usually self-limited. A **Dieulafoy lesion** is an abnormally large submucosal artery eroding through otherwise normal mucosa, giving dramatic bleeding with no ulcer to find. An **aortoenteric fistula** should be considered in anyone with a prior aortic graft who has a small herald bleed before catastrophic haemorrhage.

> Portal hypertension → portosystemic collaterals open → oesophageal and gastric varices → thin-walled vessels rupture → painless massive haematemesis

Lower causes also separate by pattern. **Diverticulosis** is the most common cause of major lower bleeding: an artery (the vasa recta) is draped over the dome of the diverticulum and ruptures, producing painless, brisk, often large-volume bleeding that usually stops on its own. **Angiodysplasia** is a degenerative ectasia of mucosal vessels, favours the right colon and the elderly, and characteristically causes slow, recurrent, sometimes occult bleeding; it has a recognised association with aortic stenosis. **Ischaemic colitis** hits watershed zones such as the splenic flexure and gives **pain first, bleeding second**, with modest volumes. **Haemorrhoids and fissures** give small-volume bright red blood on the paper or coating the stool, with pain in the case of a fissure. Slow, occult bleeding presenting only as **iron deficiency anaemia in an adult over 50 is colorectal carcinoma until colonoscopy proves otherwise**. In a child, painless rectal bleeding should raise Meckel diverticulum with ectopic gastric mucosa.

> MEMORY HOOK **Painless, brisk and large-volume, stopping on its own = diverticulosis (diverticula are mostly sigmoid, but the ones that bleed are often right-sided). Painless and slow from the right colon = angiodysplasia. Pain first, then blood = ischaemic colitis.** Pain and blood together in a young patient = inflammatory bowel disease.

- **Upper endoscopy:** both diagnostic and therapeutic, and the first-line test in any suspected upper bleed once the patient is resuscitated.
- **Colonoscopy:** the test for lower bleeding, but it needs a prepared colon and reasonable stability.
- **CT angiography:** localises bleeding at roughly 0.3 to 0.5 mL per minute and is fast, so it suits the actively bleeding unstable patient.
- **Tagged red cell scintigraphy:** detects slower bleeding, around 0.1 mL per minute, but localises only approximately.

- **Capsule endoscopy:** for obscure small bowel bleeding between the two endoscopic reaches.

PRACTICE

Q1. A 66-year-old man passes black, tarry, foul-smelling stool. His BUN is 42 mg/dL and creatinine is 0.9 mg/dL. Which conclusion is best supported?

A. The source is the sigmoid colon and the blood is simply old

B. The source is proximal to the ligament of Treitz — haemoglobin has been degraded by acid and bacteria, and digested blood protein has been absorbed and converted to urea

C. The raised urea establishes intrinsic renal injury as the primary problem

D. Oral iron supplementation fully explains both the stool colour and the laboratory findings

Show answer

B is correct. Melaena requires blood to sit in acid and traverse the gut slowly, which places the source above the ligament of Treitz; roughly 50 to 100 mL and about 14 hours of transit are needed. The disproportionate rise in urea with a normal creatinine reflects two additive mechanisms — absorption of digested blood protein as a nitrogen load, and prerenal urea reabsorption from hypovolaemia. Intrinsic renal injury would raise the creatinine as well. Iron and bismuth are the classic trap: they darken stool, but the stool is not tarry or malodorous and they do not raise the BUN.

PRACTICE

Q2. A patient passes a large volume of bright red blood per rectum. Heart rate is 128, blood pressure 82/50. The most appropriate initial assumption is:

A. The source must be the lower GI tract because the blood is bright red

B. Haemorrhoids are the commonest cause of rectal bleeding, so conservative management is appropriate

C. A rapid upper GI bleed can transit before the pigment is degraded, so an upper source must be actively excluded

D. Colonoscopy should be performed immediately, before fluid resuscitation

Show answer

C is correct. Loss of more than about a litre of blood into the upper tract can transit in under four hours, arriving red rather than black; haemodynamic instability accompanying haematochezia actually predicts an upper source. Anchoring on colour in an unstable patient is the classic error and delays the correct intervention. Haemorrhoids are indeed the most common cause of rectal bleeding overall, but they do not cause shock, so using their frequency here is a base-rate trap. Resuscitation with large-bore access, fluids and blood always precedes endoscopy, and an unprepared colon in an unstable patient makes urgent colonoscopy both unsafe and uninformative.

PRACTICE

Q3. An 80-year-old woman passes a large volume of painless maroon blood per rectum. The bleeding stops spontaneously within an hour and she has had no abdominal pain, fever or change in bowel habit. The most likely source is:

A. Acute diverticulitis

B. Diverticulosis, with rupture of a vasa recta stretched over the dome of a diverticulum

C. Ischaemic colitis of the splenic flexure

D. An anal fissure

Show answer

B is correct. Diverticulosis is the most common cause of major lower GI bleeding and is characteristically painless, brisk and self-limiting, because an artery draped over the diverticular dome ruptures and then seals. The trap is diverticulitis: inflammation of a diverticulum causes pain, fever and leukocytosis but typically does NOT bleed significantly — the same organ, two opposite presentations, which is why the -osis versus -itis distinction is worth memorising. Ischaemic colitis produces pain before bleeding and the volumes are modest. An anal fissure gives scant bright blood with severe pain on defaecation, not a large painless volume.

PRACTICE

Q4. One hour after a witnessed large-volume haematemesis, a patient's haemoglobin is 13.8 g/dL. What does this value mean?

A. Whole blood is lost in the same proportion of cells to plasma, so the haemoglobin falls only after interstitial fluid shifts in or crystalloid is given — an early normal value cannot exclude major loss

B. The bleed was trivial and the patient can be discharged after observation

C. Adequate iron stores mean that ongoing bleeding is unlikely

D. The bone marrow has already compensated for the loss

Show answer

A is correct. An acute haemorrhage removes red cells and plasma together, so the concentration of haemoglobin does not change at the moment of loss. Equilibration by plasma refill takes hours and may not be complete for a day or more, and giving crystalloid accelerates the apparent drop by dilution. Clinical signs therefore lead the laboratory: tachycardia, a narrowed pulse pressure, orthostatic change and delayed capillary refill appear first. Bone marrow response takes days to weeks and produces reticulocytosis, not an immediately normal haemoglobin, and iron stores have nothing to do with acute haemodynamics.

5 Inflammatory Bowel Disease: Crohn Disease versus Ulcerative Colitis

> **Key idea —** Both diseases are the same fundamental error — a dysregulated mucosal immune response to normal gut flora in a genetically susceptible host — expressed through two different anatomies. **Crohn goes DEEP and PATCHY; ulcerative colitis stays SHALLOW and CONTINUOUS.** Almost every clinical difference between them is a consequence of depth and distribution, so derive the complications rather than memorising them.

The susceptibility genes differ between the two. **NOD2/CARD15** variants are specific to **Crohn disease** and predict ileal, stricturing disease; ulcerative colitis is associated with a different set of loci including HLA class II and barrier genes. NOD2/CARD15 impairs the intracellular sensing and handling of bacterial products, so the epithelial barrier and innate response fail to keep commensal flora at a respectful distance. The adaptive immune system then mounts a chronic, self-sustaining response against luminal antigens. Both diseases show a bimodal age distribution, peaking in the teens to thirties with a smaller later peak, and both run a relapsing and remitting course. Smoking behaves paradoxically: it **worsens Crohn disease** and increases post-operative recurrence, but is **associated with a lower risk of ulcerative colitis**, with UC sometimes presenting shortly after a patient stops smoking.

Feature	Crohn disease	Ulcerative colitis
Distribution	Anywhere from mouth to anus; terminal ileum is the favourite site; skip lesions with normal bowel between	Colon only, always starting at the rectum and extending proximally in a continuous sheet
Rectal involvement	Often spared	Always involved — this is a defining feature
Depth of inflammation	Transmural, through the entire wall thickness	Mucosa and submucosa only
Characteristic histology	Non-caseating granulomas, fissuring ulcers	Crypt abscesses, crypt distortion, goblet cell depletion
Gross appearance	Cobblestone mucosa, creeping mesenteric fat, thickened rigid wall, string sign on imaging	Pseudopolyps, friable granular mucosa, loss of haustra giving a lead-pipe colon
Typical stool	Non-bloody or intermittently bloody diarrhoea, often with right lower quadrant pain and a palpable mass	Bloody diarrhoea with mucus, urgency and tenesmus, and left-sided cramping
Signature complications	Fistulae, abscesses, fibrotic strictures with obstruction, perianal disease, malabsorption	Toxic megacolon, severe haemorrhage, dysplasia and colorectal carcinoma
Effect of smoking	Worsens disease and recurrence	Associated with reduced risk
Nutritional consequences	Prominent — B12 deficiency, bile salt loss, fat and vitamin malabsorption, growth failure in children	Less prominent; anaemia is mainly from blood loss
Does colectomy cure it?	No — disease can recur at the anastomosis and elsewhere	Yes — the disease is confined to the colon

Transmural inflammation → deep fissuring ulcers through the wall → sinus tracts → fistulae to skin, bladder, vagina or adjacent bowel, plus abscesses and fibrotic strictures

Crohn's nutritional consequences are dictated by its favourite site. The **terminal ileum** is the only place that reabsorbs bile salts and the only place with intrinsic factor-B12

receptors, so ileal disease or resection produces B12 deficiency and a depleted bile salt pool. Bile salt loss impairs micelle formation, so fat is malabsorbed, generating steatorrhoea and deficiency of vitamins A, D, E and K. Unabsorbed bile salts that reach the colon also irritate the mucosa and drive a secretory diarrhoea. A shrunken bile salt pool leaves bile lithogenic, so **gallstones** are more common. And unabsorbed fat binds luminal calcium, which would otherwise have bound oxalate, so free oxalate is absorbed by the colon and **calcium oxalate kidney stones** follow.

> Ileal disease or resection → bile salts lost → fat unabsorbed → luminal calcium binds fat instead of oxalate → free oxalate absorbed by colon → calcium oxalate renal stones

Ulcerative colitis behaves like a disease of the rectum that spreads. Because the rectum is inflamed and non-compliant, patients have urgency, frequency and **tenesmus** — the sensation of needing to pass stool with an already empty rectum. Bleeding is prominent because the inflamed mucosa is friable and superficially ulcerated. Extent predicts severity: proctitis may be a nuisance, whereas pancolitis is a systemic illness with fever, weight loss and anaemia. Long-standing extensive colitis carries a substantial **colorectal cancer risk** beginning roughly 8 to 10 years after diagnosis, which is why surveillance colonoscopy with biopsies for dysplasia — not symptom control — determines long-term outcome. Crohn colitis carries a similar risk over comparable amounts of involved colon.

- **Skin:** erythema nodosum (tender red shin nodules, tends to track disease activity) and pyoderma gangrenosum (a rapidly enlarging ulcer with a violaceous undermined border, often independent of activity).
- **Eye:** uveitis, episcleritis and iritis — eye pain, redness or visual change in an IBD patient is urgent.
- **Joints:** peripheral arthritis that follows bowel activity, and axial disease (ankylosing spondylitis, sacroiliitis) that does not.
- **Hepatobiliary:** primary sclerosing cholangitis, strongly linked to ulcerative colitis, with a rising alkaline phosphatase and a beaded biliary tree — it also raises cholangiocarcinoma and colorectal cancer risk further.
- **Haematological:** anaemia of chronic disease plus iron deficiency, and a genuine hypercoagulable state during flares, so venous thromboembolism must be considered in an unwell IBD inpatient.

MEMORY HOOK **Crohn's C's:** Cobblestones, Creeping fat, Chronic transmural inflammation, non-Caseating granulomas, sCattered skip lesions, Cutaneous fistulae. **UC's U's and C's:** Ulcers that are Continuous, Confined to the Colon, starting at the rectum, with Crypt abscesses and Cancer risk.

RED FLAG

Toxic megacolon. A patient with colitis who develops fever, tachycardia, hypotension, abdominal distension with a tympanitic, tender abdomen and a transverse colon dilated beyond 6 cm is on the edge of perforation, which carries very high mortality. Two interventions make it worse and must be avoided: **antimotility agents and opioids**, which paralyse an already atonic colon, and **colonoscopy or barium enema**, which insufflate a paper-thin wall. Paradoxical improvement in diarrhoea in an increasingly unwell colitis patient is an ominous sign, not progress.

Pearl — Colectomy **cures** ulcerative colitis, because the entire disease lives in the colon; it does **not** cure Crohn disease, which can recur at the anastomosis and anywhere else along the tract. This single fact explains why surgery is offered relatively readily in refractory UC and is used sparingly and conservatively in Crohn disease, where every resection costs bowel length and repeated resections risk short bowel syndrome.

PRACTICE

Q1. A 27-year-old with Crohn disease and a prior terminal ileal resection develops recurrent kidney stones. The mechanism is:

A. Uric acid stone formation from increased purine turnover during flares

B. Unabsorbed fat binds luminal calcium, leaving oxalate free to be absorbed by the colon and precipitate as calcium oxalate — enteric hyperoxaluria

C. Hypercalcaemia caused by long-term corticosteroid therapy

D. Struvite stone formation from urease-producing urinary organisms

Show answer

B is correct. Loss of the terminal ileum depletes the bile salt pool, so fat is not absorbed. In a normal gut, luminal calcium binds oxalate and the complex is excreted; when fatty acids occupy that calcium instead, oxalate remains soluble and is absorbed across the colon, raising urinary oxalate and forming calcium oxalate stones. Chronic diarrhoea compounds the problem by concentrating the urine and lowering urinary citrate. Corticosteroids cause bone loss and hyperglycaemia, not hypercalcaemia. Struvite stones require urea-splitting urinary infection, which is unrelated to ileal resection.

PRACTICE

Q2. A patient develops pneumaturia and recurrent urinary infections, and imaging confirms a fistula between the ileum and the bladder. Which pattern of inflammation makes this possible, and which diagnosis does it favour?

A. Mucosal and submucosal inflammation only; ulcerative colitis

B. Transmural inflammation extending through the full thickness of the bowel wall; Crohn disease

C. Serosal inflammation without mucosal involvement; ulcerative colitis

D. Epithelial dysplasia progressing to invasive carcinoma; colorectal cancer

Show answer

B is correct. A fistula requires inflammation to burrow through the entire wall and into an adjacent structure, which only transmural disease can do. Crohn's deep fissuring ulcers extend into sinus tracts that become fistulae — enterocutaneous, enterovesical, enterovaginal or enteroenteric — and abscesses, and the same transmural fibrosis produces strictures and the string sign. Ulcerative colitis cannot fistulise to other organs because it does not breach the submucosa; the single exception is toxic megacolon, where the whole wall becomes involved and the colon perforates freely rather than forming a tract. Malignant fistulae occur but would not be the first explanation in a young patient with known Crohn-pattern disease.

PRACTICE

Q3. A patient with known ulcerative colitis has 12 bloody stools per day, a temperature of 38.9°C, heart rate 122, and a distended, tympanitic, tender abdomen. Plain film shows the transverse colon dilated to 9 cm. Which action is MOST dangerous?

A. Intravenous fluid and electrolyte replacement with correction of hypokalaemia

B. Giving loperamide or an opioid for symptom control and arranging colonoscopy to assess disease extent

C. Serial abdominal examination with repeat imaging

D. Intravenous corticosteroids with early surgical consultation

Show answer

B is correct. This is toxic megacolon — colonic dilation beyond 6 cm with systemic toxicity. Antimotility agents and opioids further paralyse an atonic colon and are well recognised precipitants of both the condition and its progression, while colonoscopy insufflates gas into a thin, inflamed, dilated wall and can perforate it. The correct approach is aggressive resuscitation, correction of hypokalaemia (which itself worsens colonic atony), intravenous corticosteroids, broad-spectrum cover, bowel rest, serial examination and imaging, and early surgical involvement because perforation carries very high mortality. Note the trap in symptom control: a reduction in stool frequency in this setting can mean the colon has stopped moving, not that the patient is improving.

PRACTICE

Q4. Colonic biopsies show crypt abscesses and continuous inflammation limited to the mucosa and submucosa, beginning at the rectum and extending proximally without interruption. Which additional feature is most consistent with this diagnosis?

A. Non-caseating granulomas with skip lesions and terminal ileal involvement

B. Creeping mesenteric fat, cobblestoning and perianal fistulae

C. Pseudopolyps with loss of haustra producing a lead-pipe colon, and disease that is cured by total colectomy

D. A fibrotic small bowel stricture producing a string sign on imaging

Show answer

C is correct. Continuous, rectum-onward, mucosa-limited inflammation with crypt abscesses is ulcerative colitis. Regenerating islands of mucosa between ulcers form pseudopolyps, and chronic inflammation obliterates the haustral markings, giving the lead-pipe appearance; because the disease never leaves the colon, total colectomy is curative. All the other options are Crohn features and depend on transmural, discontinuous inflammation — granulomas and skip lesions, creeping fat and perianal fistulae, and fibrotic strictures with the string sign — none of which can arise from a process confined to the mucosa and submucosa.

6 The Acute Surgical Abdomen: Appendicitis, Diverticulitis, and Bowel Obstruction

> **Key idea —** Appendicitis, diverticulitis and strangulating obstruction are the same story told at three sites: **a lumen is obstructed → pressure rises → venous drainage fails before arterial inflow does → the wall becomes ischaemic → bacteria translocate → the wall perforates**. The tempo differs; the sequence never does. Once you know that sequence, the red flags are simply the point at which the wall stops being viable.

Appendicitis

The appendix has a narrow lumen and a single blind end, so any obstruction — a **faecolith** in adults, **lymphoid hyperplasia** after a viral illness in children and adolescents, occasionally a tumour in older adults — traps a secreting mucosa. The mucosa keeps producing mucus into a closed space, intraluminal pressure climbs above venous pressure, drainage fails, the wall becomes oedematous and then frankly ischaemic, resident bacteria invade the devitalised wall, and gangrene proceeds to perforation, typically within 48 to 72 hours of symptom onset.

Luminal obstruction → continued mucus secretion into a closed space → intraluminal pressure exceeds venous pressure → wall oedema and ischaemia → bacterial invasion → gangrene → perforation

> **Pearl —** The **migration of the pain** is the most teachable sign in the abdomen. Early distension is sensed by visceral afferents that enter the cord at T10, and visceral pain is poorly localised, so the patient reports vague, dull, crampy **periumbilical** pain. Once inflammation becomes transmural and irritates the adjacent **parietal peritoneum**, somatic fibres take over and the pain becomes sharp, constant and precisely localised to McBurney's point in the right lower quadrant. Anorexia is nearly universal, and the vomiting comes **after** the pain — the reverse order (vomiting first, then pain) points toward gastroenteritis instead.

- **Rovsing sign:** palpation of the left lower quadrant produces right lower quadrant pain — the pain follows the peritoneum, not the hand.
- **Psoas sign:** pain on passive hip extension, suggesting a retrocaecal appendix lying against the psoas muscle.

- **Obturator sign:** pain on internal rotation of the flexed hip, suggesting a pelvic appendix.
- **Rebound and guarding:** parietal peritoneal irritation; involuntary guarding is the more reliable of the two.
- **Laboratory:** mild leukocytosis with a left shift; a normal white count does not exclude early appendicitis, and a very high count suggests perforation.

Diverticular disease

Colonic diverticula are **pulsion pseudodiverticula**: mucosa and submucosa herniate through the muscular wall at the points where the vasa recta penetrate it, because those points are the weakest spots. They are 'pseudo' because they contain no muscularis. High intraluminal pressure drives them, which is why a low-fibre diet with small, hard stool and vigorous segmenting contractions in the narrow, high-pressure **sigmoid colon** is the classic setting. **Diverticulosis** is the presence of the pouches; **diverticulitis** is what happens when one becomes obstructed by inspissated stool, the trapped mucosa becomes inflamed and ischaemic, and it micro-perforates.

> **Pearl —** One organ, two opposite presentations. **Diverticulosis bleeds without pain**, because the artery is draped over the dome of the pouch. **Diverticulitis hurts without bleeding**, because inflammation obliterates the vessel rather than opening it. If a student can hold on to that contrast, most diverticular questions solve themselves.

Feature	Appendicitis	Diverticulitis
Typical age	10 to 30 years	Over 50 years, rising with age
Pain location	Begins periumbilical, then migrates and localises to the right lower quadrant	Left lower quadrant from the start (the sigmoid), earning the name 'left-sided appendicitis'
Initiating event	Faecolith or lymphoid hyperplasia obstructing the appendiceal lumen	Inspissated stool obstructing the neck of a sigmoid diverticulum
Bowel habit	Anorexia, nausea, vomiting after the pain begins	Change in bowel habit, constipation or loose stool, sometimes urinary symptoms from an adjacent inflamed loop
Best initial imaging	Ultrasound in children and pregnancy; CT in adults	CT of the abdomen and pelvis with contrast
Endoscopy in the acute phase	Not applicable	Contraindicated — insufflation of an inflamed, micro-perforated segment risks free perforation; colonoscopy is deferred 6 to 8 weeks
Signature complications	Perforation with localised abscess or generalised peritonitis, appendiceal mass	Abscess, colovesical or colovaginal fistula, stricture with obstruction, free perforation with faeculent peritonitis

> **RED FLAG**
> In appendicitis, **sudden relief of pain followed within hours by diffuse abdominal pain, rigidity and worsening systemic signs** means the appendix has perforated and the closed pressurised space has decompressed into the peritoneum. The 'better' interval is a trap. In diverticulitis, **pneumaturia or faecaluria** signals a colovesical fistula, and diffuse peritonitis with free air signals free perforation — both change the management immediately.

Bowel obstruction: mechanical versus functional

Mechanical obstruction means a physical blockage. In the small bowel, the order of frequency is adhesions from previous surgery, then incarcerated hernias, then malignancy; in the large bowel it is colorectal carcinoma, then volvulus, then

diverticular stricture. **Functional** obstruction, or ileus, means the lumen is open but peristalsis has stopped — after abdominal surgery, with opioids and anticholinergics, in hypokalaemia and other electrolyte disturbance, in sepsis and peritonitis, after spinal injury or retroperitoneal haemorrhage. Acute colonic pseudo-obstruction (Ogilvie syndrome) is the colonic version in a debilitated inpatient, in which massive dilation occurs with no mechanical lesion at all.

The physiology of a mechanical obstruction is a fluid problem long before it is a surgical one. Proximal to the block the bowel distends with swallowed air and with its own secretions, and the obstructed gut **keeps secreting while losing its capacity to reabsorb** — several litres a day can be sequestered. Rising wall tension then drives plasma into the bowel wall itself and out onto the peritoneal surface. All of this fluid is **inside the body but outside the circulation**: the classic **third space**. The result is a patient who is intravascularly depleted — tachycardic, oliguric, with a prerenal picture — while total body water may be nearly normal, and who has lost no blood and passed no diarrhoea to explain it. Persistent vomiting of gastric contents adds loss of hydrogen ion, chloride and potassium, producing a **hypochloraemic, hypokalaemic metabolic alkalosis** maintained by the volume depletion itself — the direct link back to Module 1.

Obstruction → proximal distension → ongoing secretion with failed reabsorption plus wall oedema → litres sequestered in the third space → intravascular volume falls → tachycardia, oliguria, prerenal azotaemia and shock

Feature	Mechanical obstruction	Functional obstruction (ileus)
Underlying problem	A physical blockage; peristalsis is intact and fighting against it	The lumen is open; neuromuscular activity has ceased
Pain	Colicky, coming in waves that match peristaltic effort, often every few minutes	Little or no colic; discomfort is a constant dull distension
Bowel sounds	Early: hyperactive, high-pitched, tinkling and rushing. Late: silent as the bowel fatigues	Silent or hypoactive from the very beginning
Distension	Depends on level — minimal if proximal, marked if distal or colonic	Generalised and often striking
Vomiting	Early and copious in proximal obstruction; late and faeculent in distal obstruction	Effortless, often small volume
Flatus and stool	Obstipation once distal contents have cleared	Reduced, but small amounts may still pass
Imaging	Dilated loops with air-fluid levels and a clear transition point, with collapsed bowel beyond	Gas distributed diffusely throughout small and large bowel with no transition point
Common causes	Adhesions, hernia, tumour, volvulus, intussusception, stricture, impaction	Post-operative state, opioids, hypokalaemia, sepsis, peritonitis, spinal or retroperitoneal injury
Direction of management	Decompression and fluid resuscitation, then surgery if it does not resolve or if strangulation is suspected	Correct the cause: replace potassium, stop the opioid, treat the sepsis; surgery does not help

MEMORY HOOK **The ABC of small bowel obstruction, in order of frequency: Adhesions, Bulges (hernias), Cancer.** For the large bowel, reverse the emphasis — **Cancer first**, then volvulus and diverticular stricture.

Strangulation is the transition from a plumbing problem to a dying bowel. As distension raises intramural pressure, the thin-walled **veins collapse first**, so venous outflow is obstructed while arterial inflow continues; the wall becomes engorged and

oedematous, which raises the pressure further, and eventually arterial supply fails. The mucosal barrier breaks down, bacteria and endotoxin translocate into the portal blood and peritoneum, and the patient becomes septic. A **closed loop** obstruction — a volvulus, or a hernia incarcerated at both ends of a loop — carries the highest risk, because there is no proximal escape for the pressure and the mesentery is often twisted at the same time, cutting the blood supply directly.

> **RED FLAG**
> Suspect **strangulation** when the character of the illness changes: colic gives way to **constant, unrelenting pain**, with fever, tachycardia out of proportion, localised tenderness or peritoneal signs, a rising white cell count, and — critically — a **rising lactate with a metabolic acidosis**. Note the acid-base pivot: an obstructed patient who was alkalotic from vomiting and becomes acidotic has almost certainly developed ischaemic bowel. This is an operation, not an observation.

- **Proximal small bowel obstruction:** early, copious, bilious vomiting; little distension; dehydration and alkalosis develop fast.
- **Distal small bowel obstruction:** marked distension, later vomiting that becomes feculent from bacterial overgrowth, more colic.
- **Large bowel obstruction:** gross distension and obstipation dominate; vomiting is late or absent.
- **Closed loop with a competent ileocaecal valve:** the colon cannot decompress backwards, so the thin-walled **caecum** distends most and is the usual site of perforation.

PRACTICE

Q1. A 19-year-old has 12 hours of dull, poorly localised periumbilical pain with anorexia and one episode of vomiting that began after the pain. The pain is now sharp, constant and localised to the right lower quadrant with rebound tenderness. What explains the migration?

A. The appendix has physically shifted position within the abdomen

B. Early distension stimulates visceral afferents referred to the T10 dermatome; once transmural inflammation reaches the adjacent parietal peritoneum, somatic fibres localise the pain precisely

C. The migration itself indicates that the appendix has already perforated

D. The pain is referred from a ureteric stone passing down the right ureter

Show answer

B is correct. Visceral afferents from the midgut enter the cord around T10 and carry poorly localised, crampy sensation, so early appendiceal distension is felt vaguely around the umbilicus. When inflammation becomes transmural and irritates the richly innervated parietal peritoneum overlying the appendix, somatic fibres take over and the pain becomes sharp and precisely located. The order of symptoms matters too — pain first, then vomiting, whereas gastroenteritis usually vomits first. Perforation is suggested by transient relief followed by diffuse pain and rigidity, not by crisp localisation, and ureteric colic produces severe loin-to-groin pain with haematuria and a patient who cannot lie still.

PRACTICE

Q2. A 68-year-old man has three days of left lower quadrant pain, temperature 38.4°C and a white cell count of 15,000. He now reports passing air bubbles in his urine and has a new urinary infection. Which statement is correct?

A. This is diverticulosis bleeding and urgent colonoscopy is required to localise the source

B. Acute diverticulitis has produced a contained perforation with a colovesical fistula; CT is the diagnostic test and colonoscopy is deferred until the acute inflammation resolves

C. Pneumaturia indicates an isolated gas-forming urinary tract infection and the abdominal findings are incidental

D. This is an atypical presentation of appendicitis with the appendix lying on the left

Show answer

B is correct. Pneumaturia and faecaluria mean a communication has formed between bowel and bladder; colovesical fistula is the most common diverticular fistula and it results from a walled-off perforation of an inflamed sigmoid diverticulum eroding into the bladder dome. CT with contrast is the investigation of choice; colonoscopy during acute inflammation risks converting a contained perforation into a free one, so it is delayed six to eight weeks and then performed to exclude an underlying carcinoma. Diverticulosis bleeds painlessly and does not cause fever or leukocytosis — the -osis versus -itis trap. A gas-forming urinary infection is possible but does not explain three days of localised left lower quadrant pain with fever.

PRACTICE

Q3. A patient with a three-day small bowel obstruction and repeated vomiting has a heart rate of 118, blood pressure 96/60 and urine output of 15 mL/h. Laboratory studies show hypochloraemia, hypokalaemia and a metabolic alkalosis. There has been no blood loss and no diarrhoea. Where has the fluid gone?

A. It has been lost as insensible loss alone, so total body water is critically depleted

B. It has been sequestered into the bowel lumen, the oedematous bowel wall and the peritoneal cavity — a third space that lies inside the body but outside the circulation

C. The alkalosis proves that the primary disturbance is respiratory

D. Intrinsic renal failure is causing both the oliguria and the electrolyte abnormalities

Show answer

B is correct. Obstructed bowel continues to secrete but cannot reabsorb, and rising wall tension drives plasma into the wall and onto the peritoneal surface; several litres can be sequestered where they no longer support the circulation. Total body water may be near normal while the intravascular compartment is depleted, producing tachycardia, hypotension and prerenal oliguria — which is why third-spacing is such a favourite examination concept. The electrolyte pattern comes from vomiting gastric contents rich in hydrogen ion, chloride and potassium, and the alkalosis is then maintained by the volume depletion itself, making it chloride-responsive and correctable with saline and potassium. The oliguria is prerenal, and would reverse with volume, so intrinsic renal failure is the tempting but incorrect read.

PRACTICE

Q4. Which finding most strongly suggests that a mechanical bowel obstruction has STRANGULATED rather than remaining simple?

A. High-pitched tinkling bowel sounds with colicky pain arriving in waves

B. Constant, unrelenting pain that has replaced the colic, with fever, tachycardia, localised peritoneal tenderness and a rising lactate with metabolic acidosis

C. Absent bowel sounds and generalised distension without colic in a patient two days after abdominal surgery

D. Vomiting of gastric contents shortly after the pain began

Show answer

B is correct. Colic with high-pitched tinkling sounds is the signature of a simple mechanical obstruction — peristalsis is intact and fighting a blockage. When the wall becomes ischaemic, peristalsis stops, the pain becomes constant, systemic inflammatory signs appear, and anaerobic metabolism in the ischaemic bowel produces lactate and a metabolic acidosis. The acid-base shift is the most useful single pivot: an obstructed patient who was alkalotic from vomiting and turns acidotic has dying bowel until proved otherwise. option D describes post-operative ileus — functional, silent from the outset and without colic — and option 4 simply indicates a proximal level of obstruction, not ischaemia.

7 Bilirubin Handling and the Three Jaundices

> **Key idea —** Jaundice is a plumbing problem in a three-stage pipeline: **production** of pigment, **conjugation** inside the hepatocyte, and **excretion** into bile. Where the block sits determines whether the bilirubin backing up into blood is fat-soluble (unconjugated) or water-soluble (conjugated), and that single physical property predicts the urine color, the stool color, and the enzyme pattern. If you can reason from solubility, you never have to memorize the tables.

The normal bilirubin pipeline

Red cells live about 120 days. When they are removed, splenic and hepatic macrophages strip the heme ring and convert it to biliverdin and then to **unconjugated (indirect) bilirubin**. This molecule is lipid-soluble and travels through plasma bound tightly to albumin, which means two things at once: it can cross lipid membranes including the blood-brain barrier, and it cannot be filtered by the glomerulus, so it never appears in urine. The hepatocyte takes it up and the enzyme UDP-glucuronosyltransferase attaches glucuronic acid, producing **conjugated (direct) bilirubin**, which is now water-soluble. Conjugated pigment is actively pumped into the bile canaliculi, travels down the biliary tree, is stored and concentrated in the gallbladder, and is released into the duodenum after a meal. Colonic bacteria then reduce it to urobilinogen; most becomes stercobilin, which is what makes stool brown. A small fraction of urobilinogen is reabsorbed, recirculates through the liver, and a trace is excreted by the kidney as urobilin, which is what makes urine yellow.

Heme → unconjugated bilirubin (albumin-bound) → hepatocyte conjugation → bile → gut bacteria → stercobilin (brown stool) + urobilinogen (yellow urine)

> **Pearl —** Conjugated means water-soluble, and water-soluble means the kidney can dump it. So **dark tea- or cola-colored urine only occurs when conjugated bilirubin is in the blood**. A patient who is visibly yellow but whose urine is a normal straw color has an unconjugated problem. Any answer choice that pairs unconjugated hyperbilirubinemia with bilirubin in the urine is internally impossible.

Prehepatic jaundice: too much pigment arriving

Here the liver is healthy but is being flooded. Sickle cell crisis, G6PD deficiency, hereditary spherocytosis, autoimmune hemolysis, a hemolytic transfusion reaction, or resorption of a large hematoma all deliver heme faster than the conjugating enzyme can keep up. Unconjugated bilirubin rises. Because it is albumin-bound, urine bilirubin is **negative**. Because more pigment does eventually reach the gut, more urobilinogen is made and reabsorbed, so **urine urobilinogen goes up** and stool is normal or even darker than usual. The supporting labs are the hemolysis panel: high LDH, low haptoglobin, high reticulocyte count, and anemia. In a neonate, immature UGT activity plus a physiologically higher red cell turnover makes this the common form of jaundice, and the danger is that lipid-soluble unconjugated bilirubin crosses the immature blood-brain barrier and deposits in the basal ganglia, causing kernicterus.

Intrahepatic jaundice: the hepatocyte itself is sick

Viral hepatitis, alcoholic hepatitis, acetaminophen toxicity, drug reactions, and cirrhosis damage the cell that does both the uptake and the export. The result is a **mixed** picture: uptake and conjugation are impaired so unconjugated bilirubin rises, but the injured cell also leaks already-conjugated pigment back into the sinusoids, so conjugated bilirubin usually rises more. The signature is enzymatic: **ALT and AST climb into the hundreds or thousands** because those enzymes live in the hepatocyte cytoplasm and spill out when cells die, while ALP stays relatively modest. Urine bilirubin is positive and the urine darkens. Stool is normal to pale depending on how much bile still gets out.

Posthepatic jaundice: the pipe is blocked

A stone in the common bile duct, a carcinoma of the pancreatic head, a biliary stricture, or cholangiocarcinoma stops conjugated bile from reaching the gut, so it refluxes back into the blood. Conjugated bilirubin is water-soluble, so it is filtered freely and the urine turns dark. Nothing reaches the colon, so no stercobilin is made and the stool becomes pale, putty-colored and greasy (acholic steatorrhea), and **urine urobilinogen disappears**. Without bile salts, fat and the fat-soluble vitamins A, D, E and K are malabsorbed, so the prothrombin time lengthens from vitamin K deficiency, which unlike hepatocellular failure will correct with vitamin K. Bile salts deposited in skin cause intense pruritus. The enzyme signature is cholestatic: **ALP and GGT rise markedly and out of proportion to ALT and AST**, and GGT is what confirms the ALP is coming from bile duct epithelium rather than bone.

Finding	Prehepatic (hemolytic)	Intrahepatic (hepatocellular)	Posthepatic (obstructive)
Bilirubin fraction that rises	Unconjugated (indirect)	Both, usually conjugated-predominant	Conjugated (direct)
Urine bilirubin	Negative — albumin-bound, cannot be filtered	Positive	Strongly positive; dark tea-colored urine
Urine urobilinogen	Increased	Normal or slightly increased	Decreased to absent
Stool color	Normal to dark	Normal to pale	Clay/putty pale (acholic), greasy
Enzyme signature	ALT, AST, ALP all normal; LDH high, haptoglobin low, reticulocytes high	ALT and AST markedly elevated, ALP only mildly up	ALP and GGT markedly elevated, ALT and AST only mildly up
Pruritus	Absent	Variable	Prominent, from bile salts in the skin
Classic causes	Sickle cell crisis, G6PD, transfusion reaction, large hematoma	Viral or alcoholic hepatitis, acetaminophen, cirrhosis	Common bile duct stone, pancreatic head carcinoma, stricture

MEMORY HOOK **PRE**hepatic comes **PRE** (before) the liver, so nothing has been conjugated yet and nothing can spill into urine — yellow patient, normal urine. **POST**hepatic is **PAST** the liver, so everything backing up is already conjugated — dark urine, pale stool. Intrahepatic is the messy middle and looks like a bit of both.

Pearl — Let the enzymes tell you the pattern before the bilirubin does. A ratio question is really asking which cell type is injured: **ALT and AST are hepatocyte enzymes** (injury pattern), while **ALP and GGT are bile-duct epithelial enzymes** (cholestatic pattern). An ALP that is five times normal with an ALT only slightly up says obstruction, no matter how yellow the patient is.

RED FLAG

Painless jaundice with weight loss and a palpable, non-tender gallbladder (Courvoisier sign) in an older adult is pancreatic head carcinoma until proven otherwise — a stone-obstructed gallbladder is scarred and does not distend, so a big soft gallbladder implies a slow-growing external compression. In a newborn, jaundice appearing in the first 24 hours of life, or a rapidly rising unconjugated level, is an emergency because of kernicterus risk.

PRACTICE

Q1. A 22-year-old with sickle cell disease is admitted in vaso-occlusive crisis and is visibly jaundiced. Her urine is a normal straw color and her stool is dark brown. Which laboratory set best fits her jaundice?

A. Elevated unconjugated bilirubin, negative urine bilirubin, increased urine urobilinogen

B. Elevated conjugated bilirubin, positive urine bilirubin, absent urine urobilinogen

C. Elevated conjugated bilirubin with alkaline phosphatase four times the upper limit

D. Elevated unconjugated bilirubin with strongly positive urine bilirubin

Show answer

A is correct. Hemolysis floods a healthy liver with unconjugated pigment. Unconjugated bilirubin is lipid-soluble and travels bound to albumin, so the glomerulus cannot filter it and urine bilirubin must be negative — which is exactly why her urine is normal in color. Because more pigment ultimately does reach the colon, more urobilinogen is produced, reabsorbed and excreted, so urine urobilinogen rises and stool darkens. Option 4 is the seductive trap and is physiologically impossible: unconjugated bilirubin can never appear in urine. Options 2 and 3 describe obstruction, which would give her pale stool, not dark.

PRACTICE

Q2. A 68-year-old man has three weeks of painless jaundice, a 15-pound weight loss, severe itching, dark urine and pale clay-colored stools. Alkaline phosphatase and GGT are five times normal; ALT is only mildly elevated. What is the most likely mechanism?

A. Accelerated red cell destruction overwhelming hepatic conjugation

B. Widespread hepatocyte necrosis from acute viral hepatitis

C. Obstruction of the common bile duct by a pancreatic head mass

D. Reduced UDP-glucuronosyltransferase activity from Gilbert syndrome

Show answer

C is correct. The stool tells the story: acholic pale stool means no conjugated bile is reaching the colon, so no stercobilin is formed, and that requires a mechanical block distal to the liver. Dark urine confirms that the bilirubin backing up is conjugated and therefore filterable. The enzyme pattern is cholestatic (ALP and GGT far out of proportion to ALT), pointing to bile-duct epithelium rather than hepatocyte necrosis. Acute hepatitis is the tempting distractor, but it would drive ALT into the hundreds or thousands with only a modest ALP rise. Gilbert syndrome causes mild unconjugated hyperbilirubinemia with entirely normal urine and stool.

PRACTICE

Q3. Why is a newborn with severe unconjugated hyperbilirubinemia at risk of kernicterus, while an adult with equally severe conjugated hyperbilirubinemia from a bile duct stone is not at risk of bilirubin brain injury?

A. The newborn's higher hematocrit delivers more pigment to the brain

B. Unconjugated bilirubin is lipid-soluble and crosses the immature blood-brain barrier, while conjugated bilirubin is water-soluble and does not

C. Conjugated bilirubin is more neurotoxic but is cleared too quickly by the kidney to accumulate

D. The blood-brain barrier admits only albumin-bound molecules, and only conjugated bilirubin binds albumin

Show answer

B is correct. Solubility drives the whole answer. Conjugation with glucuronic acid is what makes bilirubin water-soluble; before that step it is lipid-soluble and can dissolve through membranes, including an immature blood-brain barrier, where it deposits in the basal ganglia and causes the encephalopathy called kernicterus. The newborn is doubly vulnerable because UGT activity is low at birth and red cell turnover is high. Option 4 inverts the facts: it is the unconjugated form that is albumin-bound, and albumin binding actually protects the brain until the binding sites are saturated.

PRACTICE

Q4. A jaundiced patient's urinalysis shows undetectable urobilinogen. What does this finding tell you?

A. Red cell destruction is accelerated

B. Essentially no conjugated bilirubin is reaching the intestine

C. The kidney has lost the ability to excrete urobilinogen

D. The hepatocytes have lost the ability to take up bilirubin from the blood

Show answer

B is correct. Urobilinogen is made only when colonic bacteria act on conjugated bilirubin delivered in bile. If none reaches the gut, none is formed, none is reabsorbed, and none appears in urine — the fingerprint of complete biliary obstruction, which pairs with acholic stools. Hemolysis does the opposite and raises urobilinogen. Option 4 is the thoughtful distractor: in hepatocellular disease urobilinogen is typically normal or even increased, because the damaged liver cannot re-extract the urobilinogen returning through the enterohepatic circulation, so more of it spills into the urine.

8 Cirrhosis: Portal Hypertension Versus Hepatocellular Failure

> **Key idea —** Every single sign of cirrhosis traces back to one of exactly two consequences of fibrosis: **blood cannot get THROUGH the liver** (portal hypertension) or **the liver cannot DO its jobs** (hepatocellular or synthetic failure). If you sort each finding into one of those two buckets, you never have to memorize a list — you derive it. A handful of findings, notably ascites and encephalopathy, sit in both buckets, and knowing which ones those are is usually the exam question.

How the architecture fails

Chronic injury from alcohol, hepatitis B or C, or metabolic fatty liver disease activates the **hepatic stellate cells** that sit in the space of Disse. Normally quiet vitamin A storage cells, they transform into contractile myofibroblasts and lay down dense collagen. Two things happen simultaneously. First, the sinusoids are stiffened and the regenerating hepatocytes reorganize into **regenerative nodules** that compress the vasculature, so resistance to portal blood flow rises. Second, functional hepatocyte mass is progressively replaced by scar, so synthetic and detoxifying capacity falls. Compounding the first problem, splanchnic arterioles vasodilate under the influence of nitric oxide, driving more blood into an already obstructed portal system. Cirrhosis is therefore irreversible architectural distortion, not simply dead cells.

Chronic injury → stellate cell activation → collagen in the space of Disse → regenerative nodules → increased portal resistance + lost hepatocyte mass → two failure modes

Failure mode 1: blood cannot get through (portal hypertension)

The portal vein normally runs at low pressure. When the hepatic venous pressure gradient exceeds about 5 mmHg, portal hypertension exists; above roughly 10 mmHg it becomes clinically significant, and above about 12 mmHg varices are at real risk of bleeding. Blood that cannot pass through the liver seeks out any pre-existing connection between the portal and systemic circulations. Those **portosystemic collaterals** are thin-walled, unsupported veins never designed for that pressure, and each one produces a recognizable sign. Meanwhile the spleen, which drains into the portal system, becomes congested and enlarges, and an enlarged spleen sequesters and destroys blood cells.

Failure mode 2: the liver cannot do its jobs (hepatocellular failure)

The liver synthesizes albumin and nearly all clotting factors, conjugates and excretes bilirubin, metabolizes hormones and drugs, converts ammonia to urea, stores glycogen and performs gluconeogenesis, and houses the Kupffer cells that filter portal blood. Lose hepatocyte mass and you lose each of those functions in turn. Because **factor VII has the shortest half-life** of the clotting factors, the prothrombin time and INR are the earliest and most sensitive markers of failing synthesis; albumin, with a half-life of about 20 days, moves more slowly and reflects chronicity. Estrogen that is normally cleared by the liver accumulates, producing the classic feminization findings in men.

Sign	Traces to	Mechanism
Esophageal and gastric varices	Portal hypertension	Left gastric vein blood is forced backward into the azygos system; thin submucosal veins dilate and can rupture catastrophically
Caput medusae	Portal hypertension	Recanalized paraumbilical veins carry portal blood into abdominal wall veins radiating from the umbilicus
Internal hemorrhoids	Portal hypertension	Collateral flow from the superior rectal vein (portal) into the middle and inferior rectal veins (systemic)
Splenomegaly with thrombocytopenia	Portal hypertension	Congestive splenic enlargement sequesters and destroys platelets, white cells and red cells (hypersplenism)
Ascites	Both	Elevated portal hydrostatic pressure pushes fluid out, low albumin removes the oncotic pull holding it in, and splanchnic vasodilation activates RAAS and ADH, causing sodium and water retention
Spontaneous bacterial peritonitis	Portal hypertension	Gut bacteria translocate across a congested bowel wall into protein-poor ascitic fluid that has little opsonic activity
Hepatorenal syndrome	Portal hypertension	Extreme splanchnic vasodilation drops effective arterial blood volume; compensatory renal vasoconstriction causes functional, not structural, kidney failure
Peripheral edema	Hepatocellular failure	Loss of albumin synthesis drops plasma oncotic pressure throughout the body
Prolonged PT/INR, easy bruising, gum bleeding	Hepatocellular failure	Clotting factors, especially the short-half-life factor VII, are no longer synthesized; vitamin K will not correct this
Jaundice and scleral icterus	Hepatocellular failure	Failure of hepatocyte uptake, conjugation and canalicular excretion of bilirubin

Gynecomastia, spider angiomas, palmar erythema, testicular atrophy, loss of body hair	Hepatocellular failure	Estrogen is no longer metabolized and accumulates, producing cutaneous vasodilation and feminization
Hypoglycemia	Hepatocellular failure	Depleted glycogen stores and lost gluconeogenic capacity
Prolonged or exaggerated drug effects	Hepatocellular failure	Loss of cytochrome P450 metabolism and first-pass extraction
Hepatic encephalopathy	Both	Too few hepatocytes to run the urea cycle, plus portosystemic shunts that carry gut ammonia straight to the brain
Fetor hepaticus	Both	Volatile sulfur compounds from the gut bypass hepatic clearance through shunts and are exhaled

Pearl — Ascites is the classic "both buckets" answer, and exam items love it. It needs the hydrostatic push of portal hypertension AND the lost oncotic pull of hypoalbuminemia AND the sodium retention driven by a kidney that senses low effective arterial volume. Any answer choice that offers only one of the three is incomplete; the best choice names the combination.

Pearl — Do not read a normal or only mildly elevated AST and ALT as reassurance in a cirrhotic. Transaminases measure **hepatocyte leakage**, and a burnt-out, mostly fibrotic liver has few hepatocytes left to leak. Albumin, INR and bilirubin measure **function**, and those are what actually stage the disease. An AST-to-ALT ratio above 2:1 points toward alcohol as the etiology.

MEMORY HOOK For the estrogen-excess cluster in a male cirrhotic, picture the **four hands and chest findings**: spider angiomas, palmar erythema, gynecomastia and testicular atrophy. All four come from the same failure to metabolize estrogen, so they travel together — sort them into the hepatocellular bucket, never the portal one.

RED FLAG

Hematemesis in a known cirrhotic is a variceal bleed until proven otherwise: the patient is bleeding from a high-pressure vein while simultaneously unable to make clotting factors and short on platelets from hypersplenism. Fever, diffuse abdominal tenderness or unexplained encephalopathy in a patient with ascites means spontaneous bacterial peritonitis — get a diagnostic paracentesis promptly rather than waiting on imaging.

PRACTICE

Q1. Which finding in a patient with alcoholic cirrhosis is explained by loss of hepatic synthetic function rather than by portal hypertension?

A. Caput medusae

B. Platelet count of 62,000

C. An INR of 2.4

D. Bleeding internal hemorrhoids

Show answer

C is correct. The liver manufactures nearly all clotting factors, so a rising INR is a direct readout of lost synthetic capacity, and factor VII's short half-life makes it the earliest signal. The other three are all pressure phenomena: caput medusae and hemorrhoids are portosystemic collaterals that open because blood cannot cross the fibrotic liver, and the thrombocytopenia comes from congestive splenomegaly sequestering platelets, not from failed production. Thrombocytopenia is the most common trap here because students associate low platelets with "liver disease" generically rather than with hypersplenism specifically.

PRACTICE

Q2. A cirrhotic patient with an INR of 2.6 receives three days of vitamin K with no improvement in the INR. What best explains this?

A. The deficiency is a loss of synthetic machinery, not a lack of the cofactor needed to make it work

B. Vitamin K requires bile salts for absorption and none are being produced

C. The INR in cirrhosis reflects platelet dysfunction, which vitamin K does not affect

D. Ongoing consumption of clotting factors by disseminated intravascular coagulation

Show answer

A is correct. Vitamin K is a cofactor for the gamma-carboxylation of factors II, VII, IX and X; it corrects a coagulopathy only when the hepatocytes are healthy but the vitamin is missing. In hepatocellular failure the factories themselves are gone, so supplying the cofactor changes nothing. This is the classic discriminator between the two liver coagulopathies: in **obstructive** jaundice, bile salts are absent, fat-soluble vitamin K is malabsorbed, and giving parenteral vitamin K promptly corrects the INR — which makes option C the tempting near-miss, since it describes cholestasis rather than cirrhosis.

PRACTICE

Q3. Which statement best explains the formation of ascites in cirrhosis?

A. Elevated portal hydrostatic pressure alone forces plasma into the peritoneal cavity

B. Hypoalbuminemia alone reduces plasma oncotic pressure and allows fluid to escape

C. Portal hydrostatic pressure, hypoalbuminemia, and RAAS-driven sodium and water retention triggered by splanchnic vasodilation all contribute

D. Right-sided heart failure raises central venous pressure and transmits it to the hepatic veins

Show answer

C is correct. Ascites is the textbook example of a finding that requires both failure modes plus a renal response. Portal hypertension supplies the hydrostatic push, hypoalbuminemia removes the oncotic force that would hold fluid inside the vessel, and splanchnic arterial vasodilation drops effective arterial blood volume so the kidney senses underfilling and activates renin-angiotensin-aldosterone and ADH, avidly retaining sodium and water that feeds the ascites. Options 1 and 2 are each true but incomplete. Option 4 describes cardiac ascites, a different mechanism entirely.

PRACTICE

Q4. A patient with cirrhosis and long-standing ascites presents with a temperature of 38.4 C, diffuse abdominal tenderness and new confusion. What is the priority action?

A. Increase the diuretic dose to reduce the ascitic volume

B. Perform a diagnostic paracentesis and send fluid for cell count and culture

C. Restrict dietary protein to under 20 grams per day

D. Obtain a CT of the abdomen before any intervention

Show answer

B is correct. Fever, abdominal tenderness and unexplained encephalopathy in a patient with ascites is spontaneous bacterial peritonitis, which arises when gut bacteria translocate into protein-poor ascitic fluid with little opsonic capacity. Diagnosis requires fluid, and an ascitic neutrophil count at or above 250 cells per microliter establishes it, so paracentesis both diagnoses and directs therapy. Increasing diuretics in a septic, potentially underfilled patient risks precipitating hepatorenal syndrome. Severe protein restriction is outdated and worsens sarcopenia. CT delays the one test that changes management.

9 Hepatic Encephalopathy and the Ammonia Story

> **Key idea —** Hepatic encephalopathy is what happens when nitrogenous waste generated in the gut reaches the brain instead of the urea cycle. Cirrhosis breaks that route in two independent ways at once: there are **too few functioning hepatocytes** to convert ammonia to urea, and **portosystemic shunts** physically route portal blood around the liver. Understanding it as a bypass problem, not just a "high ammonia" problem, explains both the precipitants and the treatment.

Colonic bacteria deaminate dietary and secreted protein, producing ammonia that is absorbed into the portal vein. In a healthy person, essentially all of it is extracted on first pass through the liver and converted to urea for renal excretion. In cirrhosis, blood shunts through collaterals and the residual hepatocytes cannot keep up, so ammonia and other gut-derived neurotoxins such as mercaptans, short-chain fatty acids and false neurotransmitters reach the systemic circulation. Ammonia is small and lipid-permeable and crosses the blood-brain barrier readily.

Inside the brain, **astrocytes** are the only cells that can detoxify ammonia. Glutamine synthetase combines ammonia with glutamate to form glutamine. Glutamine is osmotically active, so it draws water into the astrocyte, which swells; pathologically these are called Alzheimer type II astrocytes. The consequences are low-grade cerebral edema, disturbed astrocyte support of neurons, depletion of glutamate as an excitatory transmitter, and enhanced GABA-ergic inhibitory tone. Clinically that produces a fluctuating, reversible impairment of consciousness and motor control rather than a fixed focal deficit — which is precisely how it is distinguished from stroke.

Gut protein → bacterial ammonia → portal blood shunted past the liver → ammonia crosses the blood-brain barrier → astrocyte glutamine accumulates → astrocyte swelling and cerebral edema → encephalopathy

Grading, and what the exam findings mean

Grade	Mental status	Motor and exam findings
I	Subtle: shortened attention span, irritability, euphoria or anxiety, reversed sleep-wake cycle	Mild tremor, impaired handwriting and construction of simple figures
II	Lethargy, disorientation to time, obvious personality change, inappropriate behavior	Asterixis clearly present, slurred speech, ataxia
III	Somnolent but rousable, gross disorientation to place, marked confusion	Asterixis usually present, hyperreflexia, rigidity, nystagmus
IV	Coma, unresponsive to verbal or painful stimuli	Asterixis absent because the patient cannot sustain posture; decerebrate posturing may appear

Asterixis is worth understanding rather than memorizing. Ask the patient to extend the arms and dorsiflex the wrists. The metabolic encephalopathy interrupts the tonic postural drive intermittently, so the hands drop and are jerked back — it is a negative myoclonus, a brief loss of muscle tone, not a true tremor. That is why it disappears in grade IV: a comatose patient cannot hold the posture that reveals it. **Fetor hepaticus**, a sweet musty breath odor, comes from volatile sulfur compounds that shunt past the liver and are exhaled.

Precipitants: something almost always tipped the patient over

- Gastrointestinal bleeding, the single most important precipitant — a liter of blood in the gut is an enormous protein meal that colonic bacteria convert to ammonia
- Infection of any kind, especially spontaneous bacterial peritonitis, which increases catabolism and nitrogen load
- Constipation, which lengthens the contact time between bacteria and luminal protein
- Dehydration and overdiuresis, which concentrate toxins and reduce renal ammonia clearance
- Hypokalemia, which drives renal ammoniagenesis, and metabolic alkalosis, which shifts the equilibrium from ionized NH4+ toward diffusible NH3 that can cross membranes and the blood-brain barrier

- Sedatives, opioids and benzodiazepines, which are both poorly metabolized by a failing liver and additive to enhanced GABA-ergic tone
- Azotemia and renal failure, which increase urea delivery to gut bacteria that hydrolyze it back to ammonia
- A TIPS procedure or the progressive enlargement of spontaneous portosystemic shunts

MEMORY HOOK Precipitants sort into **"more nitrogen in"** (GI bleed, high protein load, azotemia, constipation), **"less clearance out"** (dehydration, renal failure, worsening liver function), and **"brain more sensitive"** (sedatives, hypokalemia with alkalosis). Any new confusion in a cirrhotic should trigger a hunt through all three columns.

Why lactulose works

Lactulose is a nonabsorbable disaccharide that passes intact to the colon, where bacteria ferment it into lactic and acetic acid. Acidifying the colonic lumen shifts the ammonia equilibrium toward **NH4+, the charged ammonium ion, which cannot diffuse across the gut wall** and is therefore trapped and excreted. The osmotic load simultaneously speeds transit, giving bacteria less time to produce ammonia and physically removing nitrogen from the gut. The therapeutic target is two to three soft stools per day; too few means underdosing, and too many risks dehydration and hypokalemia, which paradoxically worsen encephalopathy. Rifaximin works by a different route: it is a nonabsorbed antibiotic that reduces the population of urease-producing colonic bacteria, and it is typically added to lactulose rather than replacing it.

> **Pearl —** The serum ammonia level correlates poorly with the depth of encephalopathy, because what matters is intracerebral glutamine, not plasma nitrogen. **Treat the patient, not the number.** A normal ammonia does not exclude encephalopathy, and a high ammonia in an alert patient does not require escalation. The diagnosis is clinical, made by exam and by excluding other causes of altered mental status.

> **Pearl —** The old teaching of severe protein restriction has been abandoned. Cirrhotic patients are already catabolic and sarcopenic, and muscle is a major extrahepatic site of ammonia disposal, so starving them of protein removes the very tissue that helps clear ammonia. Current practice maintains adequate protein, often favoring vegetable and dairy sources.

RED FLAG

New confusion in a cirrhotic is never simply "his usual encephalopathy" until you have looked for the trigger: check for occult GI bleeding with a rectal exam, tap the ascites for spontaneous bacterial peritonitis, review sedative and diuretic doses, and check potassium, sodium and renal function. Grade III to IV encephalopathy threatens the airway from loss of protective reflexes, and a rapidly rising INR alongside encephalopathy in a previously healthy liver signals acute liver failure and a transplant referral.

PRACTICE

Q1. A patient with cirrhosis is admitted with hematemesis from esophageal varices. Twelve hours later he is disoriented and has asterixis. What is the primary mechanism linking the bleed to his confusion?

A. Blood loss has caused cerebral hypoperfusion and hypoxic injury

B. Hemoglobin in the gut is a large protein load that colonic bacteria convert into ammonia, which bypasses the liver through portosystemic shunts

C. The transfused blood products contain citrate, which is neurotoxic in liver disease

D. Acute anemia has reduced oxygen delivery, depressing the reticular activating system

Show answer

B is correct. Blood in the gastrointestinal lumen is the highest-yield precipitant of hepatic encephalopathy precisely because it is protein. Colonic bacteria deaminate it, generating a nitrogen surge that a shunted, low-functioning liver cannot extract, so ammonia reaches the brain and drives astrocyte glutamine accumulation. Options 1 and 4 describe hypoperfusion, which would produce a globally depressed but not asterixis-predominant picture and would not explain the flapping tremor. Citrate toxicity causes hypocalcemia, not encephalopathy.

PRACTICE

Q2. How does lactulose reduce ammonia levels in hepatic encephalopathy?

A. It chelates ammonia in the bloodstream so the kidney can excrete it

B. Colonic bacteria ferment it to acids that trap ammonia as non-absorbable ammonium, and its osmotic effect speeds transit

C. It kills the urease-producing bacteria that generate ammonia in the colon

D. It stimulates hepatic urea cycle enzymes to increase ammonia conversion

Show answer

B is correct. Lactulose is not absorbed, so it reaches the colon intact and is fermented into lactic and acetic acid. Acidifying the lumen protonates NH3 into NH4+, and the charged ammonium ion cannot diffuse back across the gut wall, so nitrogen is trapped and evacuated; the osmotic catharsis shortens the time bacteria have to make more. option D is the mechanism of rifaximin, a nonabsorbed antibiotic often used alongside lactulose, and is the most tempting distractor because both drugs treat the same condition through the same organ. Nothing available stimulates a urea cycle in hepatocytes that no longer exist.

PRACTICE

Q3. Which combination of laboratory abnormalities would be expected to worsen hepatic encephalopathy?

A. Hypokalemia with metabolic alkalosis

B. Hyperkalemia with metabolic acidosis

C. Hypernatremia with respiratory acidosis

D. Hypercalcemia with metabolic acidosis

Show answer

A is correct. This pairing hits the ammonia system twice. Hypokalemia drives renal tubular cells to increase ammoniagenesis, adding nitrogen to the circulation, and it also promotes intracellular shifting of hydrogen ions that sustains alkalosis. Alkalosis then shifts the NH3/NH4+ equilibrium toward uncharged, lipid-permeable NH3, which crosses membranes and the blood-brain barrier far more readily than the ionized form. This is exactly why aggressive diuresis, which causes both potassium loss and contraction alkalosis, so often precipitates encephalopathy in a cirrhotic being treated for ascites. Acidosis has the opposite effect on the equilibrium, favoring the trapped ionized form.

10 Viral Hepatitis A, B and C

> **Key idea —** The hepatitis viruses are only weakly cytotoxic on their own. Nearly all of the hepatocyte destruction is **immune-mediated**: cytotoxic T lymphocytes recognize viral antigen displayed on infected hepatocytes and kill them. That single fact explains the central paradox of the topic — a vigorous immune response produces a sicker, more jaundiced patient who clears the virus, while a weak response produces a nearly asymptomatic patient who becomes a chronic carrier. It is why a neonate infected with hepatitis B has almost no acute illness and a roughly 90 percent chance of lifelong infection.

Hepatitis A

A non-enveloped RNA picornavirus spread by the **fecal-oral route**: contaminated water, undercooked shellfish, food handled by an infected person, daycare centers, travel to endemic regions. Incubation is about two to six weeks. Because the virus is shed in stool in the two weeks before symptoms begin, transmission is largely complete by the time the patient turns yellow. Hepatitis A causes an acute, self-limited illness and **never becomes chronic and never causes cirrhosis**, though it can rarely be fulminant in older adults or those with pre-existing liver disease. **IgM anti-HAV** confirms acute infection; IgG anti-HAV means past infection or vaccination and confers lifelong immunity. A killed-virus vaccine exists. Hepatitis E behaves similarly, is fecal-oral and self-limited, but carries a strikingly high mortality when it strikes a pregnant woman in the third trimester.

Hepatitis B

The only DNA virus among the hepatitides, transmitted through **blood, sexual contact and perinatal exposure** — needle sticks, shared injection equipment, unscreened transfusion, and vertical transmission at delivery. Incubation is long, roughly one to six months, so the exposure is often forgotten. About 5 to 10 percent of infected adults fail to clear it and become chronic, but that figure rises to roughly 90 percent in infants infected at birth. Chronic infection carries the risk of cirrhosis and hepatocellular carcinoma, and hepatitis B is distinctive in that **it can cause hepatocellular carcinoma without first causing cirrhosis**, because the viral DNA integrates into the host genome and acts as an oncogenic insertion. Extrahepatic immune-complex disease, such as polyarteritis nodosa and membranous nephropathy, can accompany it. The vaccine is recombinant surface antigen only, which is the key to reading serology.

Serologic pattern	HBsAg	Anti-HBs	Anti-HBc	Interpretation
Susceptible	Negative	Negative	Negative	Never infected and never vaccinated
Immune from vaccination	Negative	Positive	Negative	The vaccine contains surface antigen only, so core antibody is never produced — this is the discriminator
Immune from resolved infection	Negative	Positive	Positive (IgG)	Virus cleared; core antibody persists for life as a permanent footprint of true infection
Acute infection	Positive	Negative	Positive (IgM)	IgM anti-HBc is the marker of recent infection
Window period	Negative	Negative	Positive (IgM)	Surface antigen already cleared but surface antibody not yet detectable; IgM anti-HBc is the only positive marker
Chronic infection	Positive for more than 6 months	Negative	Positive (IgG)	Persistent surface antigen; add HBeAg and HBV DNA to judge active replication and infectivity

> **Pearl —** Two shortcuts carry most hepatitis B serology questions. First, **anti-HBc separates vaccination from infection**: only someone who met the whole virus makes core antibody, so anti-HBs alone means vaccinated. Second, **HBeAg is the infectivity flag** — e antigen indicates active viral replication and high transmissibility, and seroconversion to anti-HBe signals that replication has quieted.

Hepatitis C

A blood-borne RNA flavivirus; injection drug use is now the dominant route, with transfusion before 1992 and needle-stick injury as historical and occupational sources. Sexual and perinatal transmission occur but are far less efficient than with hepatitis B. Acute infection is usually silent, which is exactly the problem: roughly 75 to 85 percent of those infected become **chronic**, the highest chronicity rate of the group. The reason is virologic — the RNA polymerase lacks proofreading, so the virus mutates rapidly

and circulates as a swarm of **quasispecies** that continually outruns the neutralizing antibody response. That same variability is why **there is no hepatitis C vaccine**. Over decades chronic infection produces cirrhosis and hepatocellular carcinoma, and extrahepatic manifestations include mixed cryoglobulinemia, membranoproliferative glomerulonephritis and porphyria cutanea tarda. Direct-acting antiviral regimens now cure well over 95 percent of cases in eight to twelve weeks, so the emphasis has shifted decisively to screening and detection.

Feature	Hepatitis A	Hepatitis B	Hepatitis C
Transmission	Fecal-oral: water, shellfish, food handlers, daycare, travel	Blood, sexual contact, perinatal	Blood, chiefly injection drug use and pre-1992 transfusion
Genome	RNA	DNA (the only DNA hepatitis virus)	RNA
Incubation	2 to 6 weeks	1 to 6 months	2 weeks to 6 months
Progresses to chronic infection	Never	About 5 to 10 percent of adults; about 90 percent of infected neonates	About 75 to 85 percent
Vaccine available	Yes	Yes (recombinant surface antigen)	No, because of rapid mutation into quasispecies
Cirrhosis and hepatocellular carcinoma risk	None	Yes, and can cause carcinoma even without cirrhosis via genomic integration	Yes, typically after decades of chronic inflammation
Marker of acute infection	IgM anti-HAV	HBsAg with IgM anti-HBc	HCV RNA (antibody may take weeks to appear)
Treatment approach	Supportive; vaccine and immune globulin for prophylaxis	Antivirals for chronic active disease; vaccination prevents it	Direct-acting antivirals cure more than 95 percent

The clinical course of an acute attack

Acute viral hepatitis moves through three phases. The **prodromal** phase brings malaise, low-grade fever, myalgia, anorexia, nausea and, classically, a sudden aversion to cigarettes in smokers; right upper quadrant discomfort reflects a stretched hepatic capsule. The **icteric** phase follows as bilirubin rises: jaundice, dark urine, pale stool, pruritus, and often a paradoxical improvement in the systemic symptoms. The **convalescent** phase brings resolution of jaundice with lingering fatigue for weeks. Laboratory hallmarks are ALT and AST in the hundreds to thousands with ALT usually exceeding AST, a pattern that reverses in alcoholic hepatitis where AST exceeds ALT by more than two to one and the transaminases rarely exceed 400.

MEMORY HOOK **The vowels hit your bowels**: hepatitis **A** and **E** are fecal-oral and acute only. The consonants **B** and **C** travel in blood and body fluids and are the ones that go chronic and cause cancer. And **E is Especially dangerous in Expecting mothers**.

RED FLAG

Rising INR together with new confusion in a patient with acute hepatitis means **fulminant hepatic failure**, not simply a bad case. Transaminases may actually be falling at that point — not because the patient is improving but because there are few hepatocytes left to leak enzyme. This warrants urgent transfer to a transplant center. Separately, an HBsAg-positive mother's newborn must receive hepatitis B immune globulin plus the first vaccine dose within 12 hours of birth, because that intervention is what prevents a roughly 90 percent chronicity rate.

PRACTICE

Q1. A nursing student's pre-employment serology shows: HBsAg negative, anti-HBs positive, anti-HBc negative. How should this be interpreted?

A. Immune from previous hepatitis B infection

B. Immune from hepatitis B vaccination

C. Chronic hepatitis B infection with low infectivity

D. In the window period of acute hepatitis B infection

Show answer

B is correct. The hepatitis B vaccine contains recombinant surface antigen and nothing else, so a vaccinated person develops surface antibody but never encounters the nucleocapsid core and therefore never makes core antibody. Anti-HBc is the permanent footprint of having met the actual virus, and it persists for life after natural infection, so option B would require a positive anti-HBc. Chronic infection requires persistent HBsAg. The window period is defined by anti-HBc IgM being the only positive marker, with both surface antigen and surface antibody negative.

PRACTICE

Q2. Why does hepatitis C progress to chronic infection in roughly 80 percent of cases while hepatitis A never does?

A. Hepatitis C integrates its genome into host hepatocyte DNA, making eradication impossible

B. Hepatitis C is an RNA virus whose error-prone replication generates quasispecies that continually escape the neutralizing antibody response

C. Hepatitis C is directly cytotoxic while hepatitis A is cleared by cytotoxic T cells

D. Hepatitis C is acquired parenterally, and parenteral exposure delivers a viral load too large for the immune system to clear

Show answer

B is correct. Hepatitis C's polymerase lacks proofreading, so within a single patient the virus exists as a rapidly shifting swarm of variants. By the time neutralizing antibody is produced against one variant, the dominant population has already changed, so the humoral response is always chasing a moving target — the same feature that has defeated vaccine development. option B is the classic mix-up: genomic integration is a hepatitis B property and is why hepatitis B can cause hepatocellular carcinoma without cirrhosis. option D inverts the biology, since damage in viral hepatitis is immune-mediated rather than directly cytotoxic.

PRACTICE

Q3. An infant born to an HBsAg-positive mother has a roughly 90 percent chance of developing chronic hepatitis B, compared with about 5 to 10 percent for an infected adult. What accounts for this difference?

A. The neonate's immature immune response fails to mount effective cytotoxic T-cell clearance, producing minimal acute illness but persistent infection

B. The neonate receives a much higher viral inoculum across the placenta

C. Maternal antibodies transferred across the placenta protect the virus from the infant's immune system

D. The neonatal liver lacks the enzymes needed to metabolize and clear viral particles

Show answer

A is correct. Hepatocyte injury and viral clearance in hepatitis B are both accomplished by cytotoxic T lymphocytes. An immature immune system mounts a weak response, which produces the paradox at the heart of this topic: the infant looks clinically well because there is little immune-mediated hepatocyte destruction, yet fails to eliminate the virus and becomes a lifelong carrier at high risk for cirrhosis and hepatocellular carcinoma. This is precisely why immune globulin plus vaccine within 12 hours of birth is so effective — it supplies the response the infant cannot generate.

PRACTICE

Q4. A previously healthy 30-year-old with acute hepatitis had an ALT of 2,800 four days ago. Today the ALT is 900, but the INR has risen to 2.8 and he is disoriented. What is the most accurate interpretation?

A. He is entering the convalescent phase, as shown by the falling transaminases

B. He has developed fulminant hepatic failure; the falling ALT reflects loss of viable hepatocytes

C. The confusion is from dehydration secondary to vomiting during the prodrome

D. The elevated INR reflects vitamin K malabsorption and will correct with supplementation

Show answer

B is correct. Transaminases measure hepatocyte leakage, not liver function. A falling ALT alongside a rising INR and new encephalopathy means there is not much liver left to leak enzyme — the ominous combination that defines acute liver failure and mandates urgent transplant-center referral. INR and mental status are the functional measures, and both are moving the wrong way. Option 4 is the thoughtful distractor: vitamin K deficiency from cholestasis does raise the INR and does correct with supplementation, but in massive hepatocyte loss the synthetic machinery itself is gone and vitamin K will not help.

11 Gallbladder Disease and Pancreatitis

> **Key idea —** Two organs share one exit. The common bile duct and the main pancreatic duct converge at the **ampulla of Vater** before emptying into the duodenum, so a single gallstone can produce entirely different diseases depending on exactly where it lodges: the cystic duct gives you biliary colic or cholecystitis, the common bile duct gives you obstructive jaundice or cholangitis, and the ampulla gives you pancreatitis. Locate the stone and the syndrome follows.

Why stones form

About 80 percent of stones are **cholesterol stones**, which form when bile becomes supersaturated with cholesterol relative to the bile salts and phospholipids that keep it in solution, and when gallbladder hypomotility gives the crystals time to nucleate and grow. Estrogen increases hepatic cholesterol secretion into bile and progesterone slows gallbladder emptying, which is why female sex, pregnancy, and oral contraceptives all raise risk. Obesity raises biliary cholesterol, while rapid weight loss, prolonged fasting and total parenteral nutrition cause stones through stasis, because a gallbladder that is never stimulated to contract is never emptied. **Pigment stones** come in two flavors: black stones form from chronic hemolysis, sickle cell disease and cirrhosis, where excess unconjugated bilirubin precipitates as calcium bilirubinate; brown stones form in infected or parasitized ducts where bacterial enzymes deconjugate bilirubin in situ. Crohn disease and ileal resection cause stones by interrupting the terminal-ileal reabsorption of bile salts, depleting the pool that keeps cholesterol soluble.

MEMORY HOOK The classic **4 F's** — Female, Fat, Forty, Fertile — still works, but add the mechanism-based ones the exam actually tests: **Fasting** and rapid weight loss (stasis), **Family** history, and Native American ancestry. And remember the counterintuitive one: crash dieting causes gallstones rather than preventing them.

The spectrum of biliary obstruction

Biliary colic is the mildest end and is a misnomer — the pain is steady rather than colicky, because it comes from a distending viscus rather than peristaltic waves. A fatty meal releases cholecystokinin, the gallbladder contracts against a stone temporarily wedged in the cystic duct, pressure rises, and the pain builds over an hour and then subsides once the stone falls back. Nothing is inflamed, so there is no fever and the labs are normal. If the stone stays put, the trapped bile becomes an irritant,

the wall becomes inflamed and edematous, secondary bacterial infection follows, and the picture converts to acute cholecystitis. Ultrasound is the first imaging test throughout: it detects stones, wall thickening and pericholecystic fluid, and it is fast, cheap and radiation-free.

Condition	Where the problem is	Hallmark presentation	Laboratory and imaging
Biliary colic	Stone transiently plugs the cystic duct during contraction	Steady right upper quadrant or epigastric pain 30 minutes to a few hours after a fatty meal, radiating to the right scapula, then resolving completely	Normal white count, normal liver enzymes; ultrasound shows stones
Acute cholecystitis	Stone persistently obstructs the cystic duct	Constant right upper quadrant pain lasting more than 6 hours, fever, nausea, positive Murphy sign (inspiratory arrest on palpation)	Leukocytosis; bilirubin and ALP normal or mildly elevated; ultrasound shows a thickened wall and pericholecystic fluid
Choledocholithiasis	Stone in the common bile duct	Jaundice, dark urine, pale stool, right upper quadrant pain, no fever	Direct bilirubin, ALP and GGT markedly elevated; duct dilation on imaging
Ascending cholangitis	Infected, obstructed common bile duct	Charcot triad of fever with rigors, right upper quadrant pain and jaundice; Reynolds pentad adds hypotension and altered mental status	Leukocytosis, cholestatic enzymes, positive blood cultures; needs urgent ERCP for decompression
Gallstone pancreatitis	Stone impacted at the ampulla of Vater	Severe epigastric pain boring straight through to the back, relieved by leaning forward, with vomiting	Lipase more than three times the upper limit; a concurrently elevated ALT points to a gallstone rather than alcohol as the cause

RED FLAG

Reynolds pentad — fever, right upper quadrant pain, jaundice, hypotension and confusion — means pus is under pressure in an obstructed biliary tree and bacteria are seeding the bloodstream. This is suppurative cholangitis, a true emergency requiring fluids, broad-spectrum antibiotics and emergent biliary decompression by ERCP. Antibiotics alone do not fix an obstructed, infected duct.

Acute pancreatitis: the gland digests itself

The pancreas manufactures enzymes powerful enough to dissolve tissue, so it stores them as inactive zymogens and does not activate them until they reach the duodenal lumen. Acute pancreatitis begins when **trypsinogen is prematurely converted to trypsin inside the acinar cell**. Trypsin is the master switch: once active, it activates chymotrypsinogen, proelastase, prophospholipase A2 and the kallikrein-kinin cascade. Elastase digests vessel walls and causes hemorrhage, lipase and phospholipase digest fat, and the released inflammatory mediators produce a systemic inflammatory response. The two dominant causes, together accounting for about 80 percent, are gallstones obstructing the ampulla and alcohol, which increases the protein concentration of pancreatic secretions and has direct toxic effects on acinar cells.

Stone impacted at the ampulla → pancreatic duct obstruction → trypsinogen activated inside the acinar cell → enzyme cascade → autodigestion and retroperitoneal inflammation → third-spacing, shock and SIRS

MEMORY HOOK Causes of pancreatitis: **I GET SMASHED** — Idiopathic, Gallstones, Ethanol, Trauma, Steroids, Mumps, Autoimmune, Scorpion sting, Hypertriglyceridemia and Hypercalcemia, ERCP, Drugs. In practice, always check gallstones and alcohol first, then triglycerides (a level above 1,000 mg/dL) and calcium.

- Hypovolemic shock: the inflamed retroperitoneum and the systemic capillary leak sequester liters of fluid outside the vascular space, which is why aggressive intravenous fluid resuscitation is the single most important early treatment
- Hypocalcemia: lipase liberates free fatty acids from peripancreatic fat, and those fatty acids bind calcium to form insoluble soaps (saponification), producing Chvostek and Trousseau signs and, when severe, tetany
- Acute respiratory distress syndrome: circulating phospholipase A2 degrades pulmonary surfactant, causing alveolar collapse and hypoxemia that often appears on day two to three

- Grey Turner sign (flank ecchymosis) and Cullen sign (periumbilical ecchymosis): blood from hemorrhagic necrosis tracking through retroperitoneal planes, both indicating severe disease
- Pancreatic pseudocyst: a collection of enzyme-rich fluid walled off by granulation tissue rather than true epithelium, typically appearing weeks later and presenting as a persistent mass, pain or early satiety
- Infected necrosis: devitalized pancreatic tissue seeded by gut organisms, the leading cause of late mortality
- Hyperglycemia: islet destruction plus a stress-driven glucagon surge

> **Pearl — Lipase beats amylase**: it is more specific to the pancreas (amylase also comes from salivary glands and rises in bowel infarction, ectopic pregnancy and parotitis) and it stays elevated for seven to ten days, so it still catches a patient who presents late. Critically, **enzyme magnitude does not predict severity** — a lipase of 4,000 is not necessarily worse than 900. Severity is judged by organ failure, hemodynamics, oxygenation and imaging, not by the enzyme number.

Chronic pancreatitis

Repeated injury, most often from sustained alcohol use, replaces acinar tissue with fibrosis, calcification and ductal strictures. The clinical sequence follows the anatomy: the **exocrine** pancreas, which is 80 to 90 percent of the gland's mass, fails first, so patients develop steatorrhea (bulky, greasy, foul-smelling stools that float), weight loss despite adequate intake, and deficiencies of the fat-soluble vitamins A, D, E and K. The **endocrine** islets are a small fraction of the gland and fail later, producing a brittle diabetes that is difficult to control because glucagon-secreting cells are lost alongside the beta cells. Pain is chronic epigastric and back pain that may paradoxically diminish as the gland burns out. The counterintuitive laboratory point is that **amylase and lipase are often normal in chronic pancreatitis**, because so few acinar cells remain to release them; diagnosis rests on imaging showing calcifications and ductal changes, and on fecal elastase testing.

> **RED FLAG**
> In acute pancreatitis, hypotension, tachycardia, oliguria, a falling oxygen saturation, or flank and periumbilical bruising all signal severe necrotizing disease that belongs in an intensive care setting. A rising hematocrit early on is not reassuring — it means hemoconcentration from massive third-spacing and predicts necrosis.

PRACTICE

Q1. A 44-year-old woman has had constant right upper quadrant pain for 9 hours, a temperature of 38.3 C, and inspiratory arrest when the right upper quadrant is palpated. Her total bilirubin is 0.9 mg/dL and alkaline phosphatase is normal. What is the most likely diagnosis?

A. Biliary colic

B. Acute cholecystitis

C. Choledocholithiasis

D. Ascending cholangitis

Show answer

B is correct. Persistent obstruction of the cystic duct produces gallbladder distension, wall inflammation and a positive Murphy sign, along with fever and leukocytosis. The duration is the key discriminator from biliary colic, which resolves within a few hours once the stone falls back and produces neither fever nor leukocytosis. Choledocholithiasis is the most tempting wrong answer, but a stone in the common bile duct obstructs the drainage of the entire liver and would raise bilirubin and alkaline phosphatase, which are normal here. Cholangitis requires that obstructed duct plus infection, and would present with jaundice and rigors.

PRACTICE

Q2. On hospital day two, a patient with acute pancreatitis develops perioral tingling and a positive Chvostek sign. What is the mechanism?

A. Pancreatic inflammation suppresses parathyroid hormone secretion

B. Lipase releases free fatty acids that bind calcium into insoluble soaps in the peripancreatic tissue

C. Aggressive intravenous fluid resuscitation has caused renal calcium wasting

D. Low serum albumin has reduced the total calcium while ionized calcium remains normal

Show answer

B is correct. This is saponification. Pancreatic lipase digesting peripancreatic and mesenteric fat liberates free fatty acids, which chelate calcium to form insoluble calcium soaps that are deposited in the necrotic fat and removed from the circulating pool. The resulting hypocalcemia is genuine ionized hypocalcemia and can cause tetany, which is why option 4 is the elegant trap — a low albumin does lower total calcium harmlessly, but it would not produce a positive Chvostek sign, since that reflects true ionized hypocalcemia. Hypocalcemia here is also a marker of severity, appearing in the standard prognostic scoring systems.

PRACTICE

Q3. Why does severe acute pancreatitis frequently cause hypotension and tachycardia within the first 24 hours?

A. Vomiting alone produces enough volume loss to cause shock

B. Retroperitoneal inflammation and a systemic capillary leak sequester several liters of fluid into the third space

C. Hemorrhage into the pancreatic duct causes acute blood loss into the gastrointestinal tract

D. Autonomic damage from the celiac plexus causes neurogenic vasodilation

Show answer

B is correct. Activated enzymes and inflammatory mediators produce an intense retroperitoneal inflammatory response and a generalized capillary leak, so plasma moves out of the vasculature into the retroperitoneum, the bowel wall and the peritoneal cavity. The patient is profoundly intravascularly depleted while total body water is unchanged or increased, which is exactly why early aggressive crystalloid resuscitation is the highest-yield intervention and why a rising hematocrit predicts necrosis. Vomiting contributes but is nowhere near sufficient to explain the volume deficit.

PRACTICE

Q4. A 52-year-old man with a 20-year history of heavy alcohol use reports chronic epigastric pain, a 25-pound weight loss and bulky greasy stools that float. His serum lipase is normal. What best explains the normal lipase?

A. A normal lipase excludes pancreatic disease, so another cause of malabsorption should be sought

B. Chronic fibrosis has destroyed most acinar cells, so little enzyme remains to be released into the blood

C. Lipase is only elevated when the pancreatic duct is obstructed, and his duct is patent

D. Alcohol directly inhibits the assay used to measure lipase

Show answer

B is correct. Serum lipase reflects enzyme leaking from injured but still-present acinar cells. In chronic pancreatitis the acinar mass has been replaced by fibrosis and calcification, so there is little enzyme left to spill and the level is often normal or even low. His steatorrhea, weight loss and floating stools are the direct clinical expression of that same lost exocrine mass. option B is the trap the case is built to expose: a normal lipase never excludes chronic pancreatitis. Diagnosis rests on imaging showing calcifications and ductal irregularity, and on a low fecal elastase, with pancreatic enzyme replacement and fat-soluble vitamin supplementation as treatment. Expect diabetes to follow later, when the islets finally fail.

12 Reproductive Pathophysiology: Cycle and PCOS, Prostate, and Torsion

> **Key idea —** Reproductive pathology on this exam reduces to three recurring shapes: a **feedback loop that lost its rhythm** (polycystic ovary syndrome), a **gland growing in the wrong zone** (benign prostatic hyperplasia versus prostate cancer), and a **blood supply that twisted shut** (testicular torsion). Each one is a mechanism you can reason through rather than a list to memorize.

The menstrual cycle in two halves

The **follicular (proliferative)** phase runs from day 1 to ovulation. FSH from the anterior pituitary recruits a cohort of follicles whose granulosa cells produce estradiol. Rising estradiol does two things: it drives the endometrium to proliferate, thickening the functional layer and regrowing the spiral arteries, and it exerts negative feedback on FSH, which starves the smaller follicles so that only the dominant follicle, with the most FSH receptors, survives. Then the feedback flips. Once estradiol has been sustained at a high level for roughly 48 hours, it switches from negative to **positive** feedback on the pituitary, producing the LH surge. Ovulation follows the surge by about 36 hours — day 14 in an idealized 28-day cycle.

The **luteal (secretory)** phase begins when the ruptured follicle luteinizes into the corpus luteum, which secretes progesterone. Progesterone converts the proliferated endometrium into a secretory, glycogen-rich bed capable of supporting implantation, thickens cervical mucus into a barrier, and raises the basal body temperature by about half a degree. The corpus luteum has a **fixed lifespan of about 14 days** unless rescued by hCG from an implanting embryo. Without hCG it involutes, progesterone and estrogen fall, the spiral arteries constrict, the functional endometrium becomes ischemic and sloughs, and menstruation begins.

FSH → follicle growth → sustained high estradiol → feedback flips positive → LH surge → ovulation → corpus luteum → progesterone → no hCG → luteolysis → progesterone withdrawal → menses

> **Pearl —** Because the corpus luteum's lifespan is fixed at about 14 days, **the luteal phase is essentially constant and all cycle-length variation comes from the follicular phase**. A woman with a 35-day cycle ovulates around day 21, not day 17. The second implication matters more clinically: progesterone is literally "pro-gestation," and its job is to hold and stabilize the lining. Anything that abolishes ovulation abolishes the corpus luteum, which abolishes progesterone, which leaves the endometrium under **unopposed estrogen** — continuous proliferation with no organized withdrawal bleed. That single idea is the bridge to PCOS and to endometrial cancer risk.

Polycystic ovary syndrome

PCOS is diagnosed by the Rotterdam criteria, requiring two of three: **oligo-ovulation or anovulation**, **hyperandrogenism** (clinical or biochemical), and **polycystic-appearing ovaries** on ultrasound. The mechanism is a self-reinforcing loop. An increased GnRH pulse frequency preferentially favors LH over FSH secretion, so the LH-to-FSH ratio rises. LH drives the ovarian theca cells to overproduce androgens, while the relatively low FSH means granulosa cells never fully mature a dominant follicle. Follicles arrest at the small antral stage — those are the "cysts," which are actually immature follicles arranged around the periphery, not true cysts. No dominant follicle means no LH surge trigger, no ovulation, and no corpus luteum. Layered on top is **insulin resistance**, present in a majority of patients and independent of obesity: hyperinsulinemia directly stimulates theca androgen synthesis and simultaneously suppresses hepatic production of sex hormone-binding globulin, so a greater fraction of the circulating testosterone is free and biologically active. Adipose tissue aromatizes androgens to estrone, sustaining the estrogen exposure of the endometrium.

High GnRH pulse frequency → LH:FSH ratio rises → theca androgen excess + follicular arrest → anovulation → no corpus luteum → no progesterone → unopposed estrogen → endometrial hyperplasia

- Menstrual: oligomenorrhea, amenorrhea, or unpredictable heavy bleeding when a thick unstable lining finally breaks down
- Androgenic: hirsutism in a male pattern, acne, androgenic alopecia, and rarely clitoromegaly (frank virilization should prompt a search for an androgen-secreting tumor instead)
- Metabolic: central obesity, acanthosis nigricans as a visible marker of hyperinsulinemia, dyslipidemia, impaired glucose tolerance and type 2 diabetes, obstructive sleep apnea

- Reproductive: anovulatory infertility, higher rates of miscarriage, gestational diabetes and preeclampsia
- Long-term: endometrial hyperplasia and endometrial carcinoma from chronic unopposed estrogen
- Treatment logic follows the mechanism: weight loss and metformin address insulin resistance; a combined oral contraceptive restores a regular withdrawal bleed, protects the endometrium and raises SHBG to lower free testosterone; letrozole induces ovulation for those seeking pregnancy

> **RED FLAG**
> A woman with PCOS who has gone many months without a period and then bleeds heavily, or any patient with PCOS and abnormal bleeding, needs evaluation for **endometrial hyperplasia or carcinoma**. Years of proliferation with no progesterone-driven shedding is the exact setup for endometrial neoplasia, and this is one of the few gynecologic cancers with a clear, preventable, mechanistic pathway.

BPH versus prostate cancer: it is all about zones

The prostate has distinct zones, and the two major diseases arise in different ones — which is why they present so differently. The **transition zone** wraps directly around the prostatic urethra and is where benign hyperplasia occurs, so BPH compresses the urethra early and announces itself with urinary symptoms. The **peripheral zone** lies posteriorly against the rectum and gives rise to roughly 70 to 80 percent of adenocarcinomas, so cancer grows away from the urethra and stays silent early, but is palpable on digital rectal exam. BPH is driven by dihydrotestosterone, produced when 5-alpha-reductase converts testosterone within prostatic stromal cells, and it is a hyperplasia of both glandular and stromal elements. It is not premalignant, and having BPH neither causes nor protects against carcinoma; the two simply coexist commonly in aging men.

Feature	Benign prostatic hyperplasia	Prostate adenocarcinoma
Zone of origin	Transition zone, wrapped around the urethra	Peripheral zone, at the posterior surface against the rectum
Why symptoms appear when they do	Early: growing tissue compresses the prostatic urethra	Late: growth is directed away from the urethra, so early disease is silent
Digital rectal exam	Smooth, symmetric, rubbery, uniformly enlarged	Hard, irregular, asymmetric nodule or induration
Typical symptoms	Obstructive (hesitancy, weak stream, intermittency, incomplete emptying, terminal dribbling) plus irritative from detrusor hypertrophy (frequency, urgency, nocturia)	Often none; later obstruction, hematuria, or bone pain from metastases
Driving hormone	Dihydrotestosterone generated by 5-alpha-reductase	Androgen-dependent growth, which is why androgen deprivation is a treatment mainstay
PSA behavior	Normal or mildly elevated; 5-alpha-reductase inhibitors lower measured PSA by roughly half	Often elevated, with a rising velocity over serial measurements being more meaningful than any single value
Spread	Does not metastasize and is not a premalignant lesion	Osteoblastic metastases to the lumbar spine and pelvis, spreading via the Batson vertebral venous plexus
Supporting findings	Elevated post-void residual, recurrent UTIs, bladder stones and diverticula, hydronephrosis with post-renal acute kidney injury	Elevated alkaline phosphatase with bone metastases; Gleason score grades histologic aggressiveness

Pearl — Remember the geometry: **BPH grows inward and announces itself early; cancer grows outward and stays silent**. That is also why prostate metastases are **osteoblastic** (bone-forming and sclerotic on x-ray) when most other cancers produce lytic lesions — so alkaline phosphatase rises with prostate bone metastases while serum calcium often stays normal, unlike the hypercalcemia of lytic disease.

RED FLAG

New back pain in an older man with a hard prostate nodule is vertebral metastasis until proven otherwise. If it progresses to leg weakness, saddle anesthesia, or urinary or fecal incontinence, that is **spinal cord compression** and a neurologic emergency requiring immediate corticosteroids and imaging. Separately, acute urinary retention in BPH — a distended, tender suprapubic bladder with an inability to void — needs prompt catheterization to prevent post-renal kidney injury, and is often precipitated by anticholinergics, antihistamines or decongestants.

Testicular torsion

Most patients have a congenital **bell-clapper deformity**, in which the tunica vaginalis inserts abnormally high on the spermatic cord so that the testis hangs freely inside it like a clapper in a bell rather than being anchored posteriorly. The testis can then rotate on the cord. The sequence matters: **thin-walled veins collapse first**, so venous outflow is obstructed while arterial inflow continues, producing rapid congestion, massive edema and pain out of proportion to the exam; as pressure builds, arterial inflow is finally choked off and infarction begins. This is why the clock is unforgiving — salvage exceeds 90 percent within 6 hours, falls to roughly 50 percent by 12 hours, and drops below 10 percent after 24 hours. Peak incidence is bimodal, in neonates and in boys aged 12 to 18, where a rapidly growing testis and increased activity raise the risk.

Feature	Testicular torsion	Epididymitis
Onset	Sudden and severe, often waking the patient from sleep or following activity or minor trauma	Gradual, over hours to days
Fever and urinary symptoms	Absent; urinalysis is typically normal	Common: fever, dysuria, urethral discharge, pyuria on urinalysis
Testis position and lie	High-riding with a horizontal (transverse) lie	Normal position, with maximal tenderness localized to the epididymis posteriorly
Cremasteric reflex	Absent on the affected side, a highly suggestive finding	Preserved
Prehn sign (elevating the testis)	Does not relieve, and may worsen, the pain	Relieves the pain
Peak age	Neonates and adolescents aged 12 to 18	Sexually active young adults and older men with urinary tract infection or obstruction
Management	Immediate surgical exploration with detorsion and bilateral orchiopexy	Antibiotics targeting the likely organism, scrotal support and NSAIDs

MEMORY HOOK **Time is testicle.** Torsion **T**wists, is sudden, has a **T**ransverse high-riding lie, and needs the operating room; epididymitis is gradual, febrile, and gets antibiotics. If the cremasteric reflex is gone and Prehn is negative, stop reasoning and call surgery.

RED FLAG

Acute scrotal pain in an adolescent is torsion until surgically disproven. A Doppler ultrasound may be obtained only if it does not delay the operation, because a negative or equivocal study never overrides a convincing clinical picture — the definitive diagnostic test is surgical exploration. Both testes are fixed at surgery, since the bell-clapper deformity is almost always bilateral and the remaining testis is at risk too.

PRACTICE

Q1. A 24-year-old with PCOS has not menstruated in 5 months. Which mechanism best explains her amenorrhea?

A. Her ovaries have stopped producing estrogen, leaving the endometrium atrophic

B. She does not ovulate, so no corpus luteum forms and no progesterone is produced to permit a synchronized withdrawal bleed

C. Elevated FSH has depleted her follicular reserve prematurely

D. The peripheral ovarian cysts physically obstruct the release of the ovum and the flow of menses

Show answer

B is correct. Menstruation is a progesterone-withdrawal event. Without ovulation there is no corpus luteum, without a corpus luteum there is no progesterone, and without progesterone the endometrium simply keeps proliferating under continuous estrogen with no organized synchronized shedding. option B inverts the physiology and describes premature ovarian insufficiency, where estrogen is low and FSH is high; PCOS patients have plenty of estrogen, including estrone aromatized from androgens in adipose tissue, which is precisely why they are at risk for endometrial hyperplasia rather than atrophy. In PCOS the FSH is relatively low, not elevated, and the peripheral follicles are arrested immature follicles that obstruct nothing.

PRACTICE

Q2. Which mechanism best explains the hirsutism and elevated free testosterone seen in PCOS?

A. Adrenal cortical hyperplasia producing excess cortisol precursors

B. Hyperinsulinemia stimulating ovarian theca cell androgen production while suppressing hepatic sex hormone-binding globulin

C. Deficient 5-alpha-reductase preventing testosterone from being inactivated

D. Elevated prolactin driving ovarian androgen synthesis

Show answer

B is correct. Insulin resistance is central to PCOS in a majority of patients, and hyperinsulinemia acts on two fronts simultaneously: it acts as a co-gonadotropin with LH to push theca cells toward androgen synthesis, and it suppresses hepatic SHBG production so that a larger fraction of the circulating testosterone is unbound and biologically active. Both effects push in the same direction, which is why insulin-directed treatment such as weight loss and metformin improves the androgenic features, and why a combined oral contraceptive helps partly by raising SHBG. option D has the physiology backward — 5-alpha-reductase converts testosterone to the more potent DHT rather than inactivating it.

PRACTICE

Q3. A 72-year-old man has a hard, irregular nodule on the right lobe of the prostate on digital rectal exam and reports 6 weeks of low back pain. He denies urinary hesitancy or a weak stream. Alkaline phosphatase is elevated. Which statement best explains the absence of urinary symptoms?

A. The tumor arises in the peripheral zone and grows away from the urethra, so obstruction occurs only late

B. Prostate cancer secretes factors that relax the bladder neck, preserving urinary flow

C. Urinary symptoms are absent because the tumor has already metastasized and regressed locally

D. Prostate cancer arises in the transition zone but grows too slowly to obstruct

Show answer

A is correct. Zone anatomy drives the entire clinical picture. Roughly 70 to 80 percent of prostate adenocarcinomas arise in the peripheral zone, which sits posteriorly against the rectum — palpable on exam but far from the urethra, so the tumor can grow and even metastasize before it causes any obstructive symptom. That is the mirror image of BPH, which begins in the transition zone encircling the urethra and therefore causes symptoms early while remaining entirely benign. Option 4 is the tempting near-miss because it gets the anatomy of BPH and applies it to cancer. His back pain with an elevated alkaline phosphatase points to osteoblastic vertebral metastases arriving via the Batson venous plexus, and any progression to leg weakness or incontinence would signal cord compression.

PRACTICE

Q4. A 15-year-old boy wakes at 3 a.m. with sudden severe left scrotal pain and vomiting. He is afebrile, the left testis is high-riding with a horizontal lie, the cremasteric reflex is absent on the left, elevating the testis does not relieve the pain, and the urinalysis is normal. What is the appropriate action?

A. Begin antibiotics for epididymitis and arrange urology follow-up in 48 hours

B. Obtain a Doppler ultrasound and reassess after the results are available in the morning

C. Arrange immediate surgical exploration for detorsion with bilateral orchiopexy

D. Prescribe NSAIDs and scrotal elevation, as the pain will likely resolve spontaneously

Show answer

C is correct. Every feature here is torsion: abrupt nocturnal onset, vomiting, a high-riding transverse testis, an absent cremasteric reflex, a negative Prehn sign, and no fever or pyuria to suggest infection. Because venous outflow occludes before arterial inflow, congestion and edema mount quickly and infarction follows, so salvage falls from over 90 percent within 6 hours to roughly 50 percent by 12 hours. Imaging is acceptable only if it does not delay surgery, which makes option C the most dangerous distractor — a negative or equivocal Doppler never overrides a convincing clinical picture, and surgical exploration is the definitive test. Both testes are fixed because the underlying bell-clapper deformity is nearly always bilateral.

RAPID REVIEW

Modules 1–2 — Fluids, Acid–Base & Immunity

13 Module 1 — Fluid Compartments, Tonicity, Oedema & Electrolytes

> **Key idea —** Sodium decides **where the water sits**; potassium and calcium decide **whether membranes fire**. Almost every fluid-and-electrolyte question on the final is secretly asking one of those two things — a volume/osmolality problem, or an excitability problem.

The three compartments and the 60–40–20 rule

About 60% of adult body weight is water. Two-thirds of it is **intracellular** (ICF) and one-third is **extracellular** (ECF); the ECF is then split roughly three-to-one into **interstitial** fluid and **plasma**. That last slice matters enormously out of proportion to its size, because plasma is the only compartment that perfuses organs. Water crosses these compartments passively, by osmosis, chasing solute — it does not move because we want it to. The compartments therefore stay in osmotic equilibrium at all times, and the only way to shift water is to change the osmolality of one side.

MEMORY HOOK **60–40–20**: 60% of body weight is water, 40% is intracellular, 20% is extracellular. Split the 20 again into 15% interstitial and 5% plasma. Infants run nearer 70–75% water with proportionally more ECF (which turns over fast), so they dehydrate in hours; the elderly run nearer 50% with less reserve, so they have less margin before a deficit becomes symptomatic.

Tonicity: what a fluid does to a cell

Tonicity is not a property of the bag — it is a prediction about the cell. Compare the infused fluid to the ECF it is entering, then ask which way water must move to equalise the two.

Fluid / state	Effect on ECF osmolality	Water movement	Cell result and classic use
Isotonic (0.9% NaCl, lactated Ringer's)	Unchanged	Stays in the ECF	Cells unchanged; expands vascular volume — haemorrhage, vomiting, isotonic dehydration
Hypotonic (0.45% NaCl; D5W once the dextrose is metabolised)	Falls	ECF → ICF	Cells swell; rehydrates cells in hypernatraemia, but risks cerebral oedema if given fast
Hypertonic (3% NaCl, D10W, D5 in normal saline)	Rises	ICF → ECF	Cells shrink; pulls water off a swollen brain and corrects severe hyponatraemia, but risks vascular overload

Free water excess → serum osmolality falls → water enters neurons → cerebral oedema → headache, lethargy, confusion, seizure

The four causes of oedema

Oedema is excess fluid in the **interstitium**, and there are only four ways to put it there. Three of them are Starling-force problems at the capillary (the push-out force is too high, the pull-in force is too low, or the wall has become leaky) and the fourth is a drainage problem (the lymphatics that normally mop up the small daily surplus are blocked).

Mechanism	Force disturbed	Classic example
Increased capillary hydrostatic pressure	Push-out force too high	Heart failure, venous obstruction or DVT, sodium and water retention
Decreased plasma oncotic pressure	Pull-in force too low (low albumin)	Cirrhosis (albumin not made), nephrotic syndrome (albumin lost in urine), severe malnutrition
Increased capillary permeability	Wall leaks protein into the interstitium	Inflammation, burns, sepsis, anaphylaxis
Lymphatic obstruction	Return route blocked	Post-mastectomy node dissection, tumour, filariasis — classically **non-pitting** lymphoedema

> **Pearl —** Oedema and hypovolaemia are not opposites — they routinely coexist. In **third spacing** (burns, sepsis, ascites, bowel obstruction) the fluid is still inside the body but has left the vasculature, so the patient can be visibly swollen while the blood pressure falls and the urine output stops. Judge volume status by perfusion — heart rate, blood pressure, urine output, capillary refill — not by how puffy the legs look.

Electrolyte imbalances and their signature findings

Sodium disorders are almost always **water** disorders in disguise, and they show up in the brain because neurons swell or shrink with osmolality. Potassium and calcium disorders show up on the **ECG** because both ions set the resting membrane potential and the speed of repolarisation.

Imbalance	Core mechanism	Signature ECG or neuro finding
Hyponatraemia (< 135)	Water excess relative to sodium — SIADH, heart failure, excess hypotonic fluid, psychogenic polydipsia	Neurons swell: headache, lethargy, confusion, seizure, coma
Hypernatraemia (> 145)	Water deficit — diabetes insipidus, no access to water, tube feeds, osmotic diuresis	Neurons shrink: intense thirst, agitation, muscle twitching, seizure, coma
Hypokalaemia (< 3.5)	GI and diuretic loss, or shift into cells during alkalosis and insulin therapy	Flattened or inverted T wave, prominent **U wave**, ST depression; flaccid weakness, ileus
Hyperkalaemia (> 5.0)	Retention (renal failure, ACE inhibitors, potassium-sparing diuretics) or shift out of cells (acidosis, crush injury, cell lysis)	**Tall peaked T wave** → widened QRS → sine wave → asystole
Hypocalcaemia	Loss of PTH (thyroidectomy), vitamin D deficiency, alkalosis increasing the protein-bound fraction	Prolonged QT; Chvostek and Trousseau signs, tetany, laryngospasm
Hypercalcaemia	Bone breakdown — hyperparathyroidism, bony malignancy, prolonged immobility	Shortened QT; "stones, bones, groans and moans" — renal calculi, bone pain, constipation, lethargy

RED FLAG

Potassium above 6.5 mmol/L, or **any** QRS that is widening on the monitor, is a same-minute emergency: the myocardium is losing its ability to conduct and can go to a sine wave and then asystole without warning. IV calcium first to stabilise the membrane (it does not lower potassium), then insulin with dextrose and a nebulised beta-agonist to shift potassium into cells, then something that actually removes it — dialysis or a binder.

Pearl — Correction speed is its own disease. Chronic hyponatraemia lets the brain extrude osmoles to shrink back toward normal size; raise the sodium faster than about 8 mmol/L in 24 hours and you pull water out of those adapted neurons and cause **osmotic demyelination** days later. The mirror error is correcting hypernatraemia too fast, which drives water into adapted neurons and causes **cerebral oedema**. Slow problems need slow fixes, in both directions.

Acidosis → H+ moves into cells → K+ moves out to balance the charge → serum potassium rises → peaked T waves (treat the acidosis and the potassium follows it back in)

PRACTICE

Q1. A patient with cirrhosis has 4+ pitting oedema of both legs, tense ascites, and a serum albumin of 1.8 g/dL. Which Starling force best explains the peripheral oedema?

A. Increased capillary hydrostatic pressure

B. Decreased plasma oncotic pressure

C. Increased capillary permeability

D. Lymphatic obstruction

Show answer

B is correct. The diseased liver cannot synthesise albumin, and albumin is what generates the pull-in (oncotic) force that returns fluid to the capillary. Without it, fluid filtered at the arterial end is never reclaimed and pools in the interstitium. Increased hydrostatic pressure is a genuinely tempting answer because portal hypertension is present and does drive the ascites — but portal pressure is confined to the splanchnic bed and cannot explain generalised pitting oedema of both legs; the albumin of 1.8 can.

PRACTICE

Q2. A patient in acute kidney injury has tall, narrow, peaked T waves and a QRS that has widened from 90 to 140 ms over an hour. What is the correct interpretation?

A. Hypokalaemia — replace potassium orally

B. Hypocalcaemia — the QT interval is prolonged

C. Hyperkalaemia — this is a life-threatening membrane emergency

D. Hypercalcaemia — the QT interval is shortened

Show answer

C is correct. Failing kidneys cannot excrete potassium. Rising extracellular potassium raises the resting membrane potential toward threshold, which inactivates fast sodium channels and slows conduction — that is the widening QRS. Peaked T waves alone are the early warning; a widening QRS means the next steps are a sine wave and asystole. Hypokalaemia is the wrong direction entirely and produces flat T waves with a U wave, and the calcium disorders alter the QT interval, not the QRS width.

PRACTICE

Q3. A nursing home resident is found with a serum sodium of 116 mmol/L that has developed gradually over about a week. She is confused but not seizing. What is the greatest hazard of treatment?

A. Cerebral oedema caused by correcting the sodium too slowly

B. Osmotic demyelination caused by correcting the sodium too quickly

C. Hyperkalaemia caused by the sodium-containing fluid load

D. Rebound hypernatraemia within the first hour of therapy

Show answer

B is correct. Because the fall was gradual, the brain has already adapted by extruding osmoles and shrinking back to near-normal volume. Raising the serum sodium quickly now makes the ECF suddenly hypertonic relative to those adapted neurons, water leaves them, and central pontine myelin is injured — often presenting two to six days later, after the sodium looks 'fixed'. Option one inverts the physiology: cerebral oedema is the hazard of correcting **hypernatraemia** too fast, not of going slowly here.

14 Module 1 — Acid–Base Balance and ABG Interpretation

> **Key idea —** There are only two levers on pH. **CO2 is the respiratory lever** — an acid, controlled by the lungs, adjustable within minutes. **HCO3 is the metabolic lever** — a base, controlled by the kidneys, adjustable over hours to days. The primary disorder is whichever value is abnormal in the direction that would itself produce the observed pH; the other value moving the opposite way is compensation.

Four moves, every single time

- **Move 1 — pH (7.35–7.45).** Below 7.35 is acidosis, above 7.45 is alkalosis. If it is inside the range but not exactly 7.40, note which side of 7.40 it sits on; that side names the primary disorder even when compensation has hidden it.
- **Move 2 — pCO2 (35–45 mmHg).** High CO2 means retained acid from hypoventilation; low CO2 means acid blown off by hyperventilation.
- **Move 3 — HCO3 (22–26 mEq/L).** Low bicarbonate means acid was gained or base was lost; high bicarbonate means acid was lost or base was gained.
- **Move 4 — match and compensate.** Whichever value explains the pH is primary. If the other value has moved in the opposite direction, compensation has begun: pH still abnormal = **partially** compensated, pH back in range = **fully** compensated, other value untouched = **uncompensated**.

MEMORY HOOK **ROME — Respiratory Opposite, Metabolic Equal.** In respiratory disorders the pH and the pCO2 move in opposite directions (pH down, CO2 up = respiratory acidosis). In metabolic disorders the pH and the HCO3 move in the same direction (pH down, HCO3 down = metabolic acidosis).

Disorder	pH	Primary change	Compensation	Typical causes
Respiratory acidosis	Low	pCO2 > 45 from hypoventilation	Kidneys retain HCO3 (hours to days, so it is slow)	COPD exacerbation, opioid or sedative overdose, neuromuscular weakness, chest trauma, atelectasis
Respiratory alkalosis	High	pCO2 < 35 from hyperventilation	Kidneys excrete HCO3	Anxiety, pain, fever, early sepsis, pulmonary embolism, high altitude, early salicylate toxicity
Metabolic acidosis	Low	HCO3 < 22 — acid gained or base lost	Lungs blow off CO2 as deep rapid Kussmaul breathing (minutes, so it is fast)	DKA, lactic acidosis and shock, renal failure, severe diarrhoea (loses bicarbonate)
Metabolic alkalosis	High	HCO3 > 26 — acid lost or base gained	Lungs hypoventilate, but only a little — hypoxia overrides it	Vomiting or NG suction, loop and thiazide diuretics, antacid overuse, hyperaldosteronism

> **Pearl —** Compensation restores the pH but never overshoots the midline — the body will not turn an acidosis into an alkalosis to fix itself. That is the trick for reading a **fully compensated** gas: if the pH is 7.37 with a CO2 of 60 and a bicarbonate of 34, the pH is on the acidotic side of 7.40, so the acidosis was the original problem and the alkalotic-looking bicarbonate is the kidneys' repair work.

The anion gap sorts the acidoses

Once you have a metabolic acidosis, one calculation tells you the mechanism: **anion gap = Na − (Cl + HCO3)**, normally about 8–12. A **high gap** means an unmeasured acid was added to the blood — ketones, lactate, uraemic acids, toxic alcohols — and consumed bicarbonate on the way in. A **normal gap** means bicarbonate was simply lost, most often through the stool in diarrhoea or through the tubules in renal tubular acidosis; chloride is reabsorbed to fill the vacated anion slot, which is why this is also called a hyperchloraemic acidosis.

MEMORY HOOK High-gap acids to recall: **ketoacids** (DKA, starvation, alcohol), **lactate** (shock, sepsis, any tissue starved of oxygen), **uraemic acids** (renal failure) and **ingested toxins** (salicylates, methanol, ethylene glycol). Normal-gap causes are the two great bicarbonate leaks: **diarrhoea** below and **renal tubular acidosis** in the kidney.

DKA → ketoacids accumulate → HCO3 consumed buffering them → anion gap widens → chemoreceptors drive Kussmaul respirations → pCO2 falls as compensation

RED FLAG

A severe asthmatic whose pCO2 has 'normalised' from 26 to 42 is not improving — the respiratory muscles are fatiguing and can no longer sustain the hyperventilation that was keeping the CO2 low. A normal gas in a patient who is visibly working hard is the gas that precedes respiratory arrest. Prepare for intubation rather than reassuring the family.

Pearl — Acid–base and potassium are one system, not two. In acidosis, hydrogen ions move into cells and potassium moves out to preserve electroneutrality, so the serum potassium reads high even when total body potassium is depleted — treat DKA and the potassium will fall sharply as insulin drives it back in. In alkalosis the trade runs the other way and potassium falls. Always read a gas and a potassium together.

PRACTICE

Q1. A patient with long-standing COPD has pH 7.30, pCO2 58 mmHg, HCO3 30 mEq/L. Which interpretation is correct?

A. Uncompensated metabolic acidosis

B. Partially compensated respiratory acidosis

C. Fully compensated respiratory alkalosis

D. Partially compensated metabolic alkalosis

Show answer

B is correct. The pH is acidotic. The pCO2 of 58 would by itself lower the pH, so the respiratory arm is primary — retained CO2 from chronic hypoventilation. The bicarbonate of 30 has moved in the opposite (alkalotic) direction, which is renal compensation, but since the pH has not returned into range the compensation is only partial. Option four is the tempting one because that bicarbonate genuinely is elevated — but a **primary** metabolic alkalosis would push the pH up, and this patient's pH is down.

PRACTICE

Q2. A patient in DKA has Na 138, Cl 98, HCO3 8, pH 7.15, and deep laboured breathing at 36 per minute. Which statement is correct?

A. Anion gap 32 — high-gap metabolic acidosis with respiratory compensation

B. Anion gap 32 — metabolic acidosis with a superimposed respiratory acidosis

C. Anion gap 12 — normal-gap acidosis from bicarbonate loss

D. Anion gap 12 — respiratory acidosis with metabolic compensation

Show answer

A is correct. 138 − (98 + 8) = 32, a markedly elevated gap, because ketoacids are unmeasured anions that consumed the bicarbonate. The deep rapid breathing is Kussmaul respiration, the fast respiratory compensation — it is **lowering** the pCO2, so it cannot be a second acidosis, which is why option two fails. A normal gap of 12 would point to bicarbonate simply leaking out, as in diarrhoea, where chloride rises to take its place.

PRACTICE

Q3. After three days of vomiting and nasogastric suction a patient has pH 7.50, pCO2 47 mmHg, HCO3 34 mEq/L, and K 3.0 mEq/L. Which explanation fits?

A. Loss of gastric hydrochloric acid leaves bicarbonate unopposed; hypoventilation is the compensation

B. Loss of gastric bicarbonate lowers the buffer pool; hyperventilation is the compensation

C. Retention of CO2 from abdominal splinting, with the kidneys retaining bicarbonate in response

D. Renal loss of bicarbonate, with a compensatory rise in pCO2

Show answer

A is correct. The stomach secretes hydrochloric acid, so vomiting and suction strip out hydrogen and chloride ions and leave bicarbonate relatively in excess — a metabolic alkalosis. Volume depletion then raises aldosterone, which reabsorbs sodium in exchange for potassium and hydrogen, deepening both the alkalosis and the hypokalaemia. The pCO2 of 47 is appropriate hypoventilation, not a primary respiratory acidosis; note that this compensation is always limited, because hypoventilating far enough to fix the pH would drop the oxygen level to a point the body will not tolerate.

15 Module 2 — Immunity, Hypersensitivity and Immune Failure

Key idea — Innate immunity is **fast, fixed and forgetful**; adaptive immunity is **slow the first time, exquisitely specific, and never forgets**. Every immune disorder on the final is one of three failures: overreaction to something harmless (hypersensitivity), reaction against self (autoimmunity), or no reaction at all (immunodeficiency).

Feature	Innate immunity	Adaptive immunity
Speed	Minutes to hours — already in place before exposure	Days on first exposure, hours on re-exposure
Specificity	Pattern recognition; the same response to any invader	Antigen-specific receptors, one clone per antigen
Memory	None — the thousandth exposure looks exactly like the first	Yes — memory B and T cells, which is why vaccination works
Key players	Skin and mucosa, neutrophils, macrophages, NK cells, complement, fever, inflammation	B cells and antibody (humoral), CD4 helper and CD8 cytotoxic T cells (cell-mediated)
What you observe clinically	Redness, heat, swelling, pain, fever, leucocytosis with a left shift	Rising antibody titres, positive serology, delayed skin test reactions

Within the adaptive arm, split **humoral** from **cell-mediated**. Humoral immunity is antibody made by plasma cells and works against things floating in extracellular fluid — bacteria, toxins, free virus. Cell-mediated immunity is CD8 cytotoxic T cells killing host cells that have been taken over, and it is what handles intracellular viruses, fungi, tuberculosis, tumour cells and transplanted tissue. That division explains the infections you see when each arm fails: antibody deficiency gives recurrent encapsulated bacterial infections, whereas T-cell deficiency gives opportunistic and fungal disease.

MEMORY HOOK The five antibody classes are **GAMED**. **IgM** is iMMediate — the first antibody made in a new infection, so a positive IgM means acute. **IgG** is the Greatest in quantity, carries long-term memory, and is the only one that crosses the placenta. **IgA** guards mucosal surfaces — sAliva, teArs, breAst milk. **IgE** is the allergy and parasite antibody, the one sitting on mast cells. **IgD** stays on the B-cell surface as a receptor.

The four hypersensitivity reactions

Hypersensitivity is not broken immunity — it is normal immune machinery pointed at the wrong target or dialled up too far. Sort the four types by asking two questions: which effector is doing the damage (IgE, antibody against a cell, deposited immune complexes, or T cells), and how fast does it appear.

Type	Mechanism	Time course	Examples
I — Immediate / anaphylactic	Antigen cross-links IgE bound to mast cells → degranulation, histamine release, vasodilation and bronchoconstriction	Seconds to minutes	Anaphylaxis, allergic asthma, hay fever, food and bee-sting allergy, latex anaphylaxis
II — Cytotoxic	IgG or IgM binds an antigen on a cell surface → complement and phagocytes destroy that cell, or the antibody blocks or stimulates its receptor	Minutes to hours	ABO transfusion reaction, haemolytic disease of the newborn, Graves disease, myasthenia gravis, immune thrombocytopenia
III — Immune complex	Antigen–antibody complexes form in the circulation and deposit in vessel walls, joints and glomeruli → complement and neutrophils damage the bystander tissue	Hours to days	Systemic lupus erythematosus, rheumatoid arthritis, post-streptococcal glomerulonephritis, serum sickness
IV — Delayed / cell-mediated	Sensitised T cells recruit macrophages; **no antibody is involved at all**	48 to 72 hours	Tuberculin skin test, poison ivy and contact dermatitis, transplant rejection, coeliac disease

MEMORY HOOK **ACID** — Anaphylactic, Cytotoxic, Immune complex, Delayed. Types I through III are antibody-mediated, so serum from a sensitised person could transfer them; type IV needs the living T cells themselves, and that requirement for cell recruitment is exactly why it takes two to three days to appear.

Pearl — Latex shows up in two different rows on purpose. The same allergen produces a type I IgE anaphylaxis within minutes (urticaria, wheeze, hypotension) in one patient and a type IV contact dermatitis at 48 hours (itchy vesicular rash confined to where the glove touched) in another. Identical exposure, different arm of the immune system, different time course, and completely different management — which is why timing is the single most useful piece of data in a hypersensitivity question.

RED FLAG

Stridor, swelling of the lips or tongue, wheeze and falling blood pressure after an exposure is **anaphylaxis** — a type I reaction closing the airway and dumping vascular tone at the same time. Give intramuscular epinephrine into the lateral thigh immediately. Antihistamines and steroids are adjuncts that work far too slowly to save the airway, and waiting to see whether the rash progresses is how patients are lost.

Autoimmunity versus immunodeficiency

Feature	Autoimmunity	Immunodeficiency
Core defect	Loss of self-tolerance — the immune system attacks host tissue as though it were foreign	A missing, destroyed or suppressed immune component
Typical origin	Genetic susceptibility (HLA types), molecular mimicry after an infection, hormonal influence — strong female predominance	Primary/congenital (SCID, selective IgA deficiency) or secondary/acquired (HIV, chemotherapy, corticosteroids, malnutrition, advanced age)
What you see	Chronic relapsing inflammation, autoantibodies (ANA, anti-dsDNA, rheumatoid factor), often several organs at once	Recurrent, severe, prolonged or opportunistic infections that do not clear and come back with the same organisms
Treatment logic	Turn the immune response **down** — corticosteroids, DMARDs, biologics	Turn it **up** or replace what is missing — immunoglobulin replacement, antiretrovirals, prophylactic antimicrobials

Pearl — The two conditions meet in the middle, and that is where exam questions live. Treating autoimmunity with immunosuppression manufactures a **secondary immunodeficiency**, so a lupus patient on high-dose steroids is now at risk for opportunistic infection. Worse, suppressed patients may not mount fever, pus or the usual inflammatory signs, because those signs are the immune response — so an infection that would look dramatic in a healthy person can look like nothing more than mild fatigue.

Group A streptococcal pharyngitis → anti-M-protein antibodies cross-react with cardiac myosin and valve tissue (molecular mimicry, **type II**) → rheumatic carditis weeks later. Separately: streptococcal pharyngitis or skin infection → circulating antigen–antibody complexes form → they deposit in the glomerular basement membrane (**type III**) → post-streptococcal glomerulonephritis 1–3 weeks later

PRACTICE

Q1. Fifteen minutes into a transfusion of packed red cells, a patient develops fever, flank pain, and dark red-brown urine. Which hypersensitivity reaction is occurring?

A. Type I — IgE-mediated

B. Type II — cytotoxic

C. Type III — immune complex

D. Type IV — delayed cell-mediated

Show answer

B is correct. An ABO mismatch means the recipient already has preformed IgM antibodies against antigens sitting on the donor red cell membranes. Those antibodies bind the cell surface and activate complement, lysing the cells inside the vessels — hence the haemoglobin spilling into the urine and the flank pain as it passes through the kidney. Antibody directed at an antigen on a cell surface is the definition of type II. Type I is tempting because it is equally fast, but it is IgE and mast cells and produces urticaria, wheeze and angio-oedema, not haemoglobinuria.

PRACTICE

Q2. A nurse develops an itchy, weeping, vesicular rash across both hands roughly 48 hours after a shift spent wearing latex gloves. Which statement is correct?

A. Type I IgE reaction — this presentation means she is in immediate danger of anaphylaxis

B. Type IV T-cell reaction — the delay and the absence of antibody are the diagnostic clues

C. Type II cytotoxic reaction against antigens on the keratinocyte surface

D. Type III immune complex deposition within dermal vessels

Show answer

B is correct. A localised rash appearing two days later, confined to the contact area, is delayed cell-mediated hypersensitivity: sensitised T cells recruit macrophages, and that recruitment is what takes 48 to 72 hours. No antibody is involved. Option one is a genuinely plausible trap, because latex truly does cause type I anaphylaxis in other patients — but that reaction begins within minutes and presents as urticaria, bronchospasm and hypotension, not as a delayed vesicular dermatitis.

PRACTICE

Q3. A patient taking high-dose prednisone long-term for systemic lupus erythematosus develops Pneumocystis pneumonia. Which statement best explains what has happened?

A. The lupus has evolved into a primary immunodeficiency disorder

B. The corticosteroids have produced a secondary immunodeficiency, and opportunistic organisms are its signature

C. This is a type III immune complex reaction involving the lung parenchyma

D. Innate immunity is unaffected by steroids, so an alternative diagnosis should be sought

Show answer

B is correct. Lupus is a type III immune complex disease, so treatment means suppressing the immune response — and the price of that suppression is acquired immunodeficiency. Organisms like Pneumocystis, CMV and Candida cause disease only when cell-mediated immunity is impaired, which is why they are the fingerprint of a suppressed host. Option one confuses the categories: primary means congenital, whereas this deficiency is clearly acquired. Option four is also wrong in an important way — corticosteroids blunt innate immunity too, suppressing neutrophil migration and the febrile response, so the patient may look deceptively well while the infection advances.

16 Cellular Adaptation, Reversible Injury and the Two Ways a Cell Dies

Adaptation is the cell buying time

A cell under stress has three options: adapt, die, or be injured somewhere in between. Adaptation is a **reversible, controlled change in size, number or type** that lets the cell keep functioning under a new workload. The critical teaching point is that adaptation is not disease — it is successful compensation. It only becomes dangerous when the stimulus persists so long that the adapted cell is itself abnormal (metaplasia drifting into dysplasia) or when the compensating tissue creates a new problem (the hypertrophied ventricle that outgrows its own coronary supply).

> **Key idea —** Remove the stimulus early and the cell walks all the way back to normal. Every adaptation on this list is reversible except the last one, and dysplasia is only reversible while the basement membrane is still intact — once cells breach it, you are no longer describing adaptation, you are describing cancer.

Adaptation	What actually changes	Driving stimulus	Classic example	Reversible?
Atrophy	Cell **shrinks** — autophagy and ubiquitin-proteasome degrade organelles	Disuse, denervation, ischaemia, loss of hormone or nutrition	Quadriceps after 6 weeks in a cast; endometrium after menopause	Yes, with reloading or hormone restoration
Hypertrophy	Cell **enlarges** — more protein and organelles, no new cells	Increased workload or hormonal drive in cells that cannot divide	Left ventricle in chronic hypertension; skeletal muscle in a lifter	Yes, if afterload or load is reduced
Hyperplasia	**More cells** — requires a mitotically capable cell	Hormonal stimulation or compensatory demand	Breast in pregnancy; benign prostatic hyperplasia; liver regeneration	Yes, but pathologic hyperplasia is fertile ground for cancer
Metaplasia	One **mature cell type replaced by another** mature type that tolerates the stress better	Chronic irritation — smoke, acid, stone	Ciliated respiratory epithelium to squamous in smokers; Barrett oesophagus (squamous to columnar)	Yes, but you lose the original function (mucus clearance)
Dysplasia	**Disordered** growth — variable size, shape, nuclear changes, loss of orderly maturation	Persistent stress on already-metaplastic tissue	Cervical dysplasia on Pap; high-grade Barrett dysplasia	Mild yes; severe dysplasia often behaves as carcinoma in situ

Pearl — Metaplasia is the cell choosing the lesser evil. Squamous epithelium survives cigarette smoke better than ciliated epithelium does — but it has no cilia, so the smoker loses the mucociliary escalator and gains chronic bronchitis and retained carcinogens. The adaptation itself sets up the next disease.

Reversible versus irreversible injury: where the line sits

Almost every cell injury on an exam starts with **hypoxia**, because oxygen is what oxidative phosphorylation needs to make ATP. When ATP falls, the Na+/K+-ATPase stops running, sodium and water pour into the cell, and the cell swells — the earliest and still-reversible change. The cell also flips to anaerobic glycolysis, lactate accumulates, intracellular pH drops, and ribosomes detach from rough ER so protein synthesis stalls. All of that reverses if perfusion returns. The point of no return is **membrane integrity**: once lysosomal and mitochondrial membranes rupture and calcium floods the cytosol, activated proteases, phospholipases and endonucleases digest the cell from the inside and nothing brings it back.

Hypoxia → ATP falls → Na+/K+ pump fails → cell swells (REVERSIBLE) → lactic acidosis → massive Ca2+ influx → lysosomal and mitochondrial membrane rupture → enzymatic digestion (IRREVERSIBLE = necrosis)

Pearl — Every cardiac enzyme you have ever ordered is a report on failed membranes. Troponin, CK-MB, ALT, AST and lipase are intracellular proteins — they can only appear in serum if the plasma membrane has been breached. A rising troponin is literally laboratory proof that the injury crossed from reversible to irreversible.

Necrosis versus apoptosis

Feature	Necrosis	Apoptosis
Trigger	Pathologic — ischaemia, toxins, trauma, infection	Physiologic or pathologic — programmed, signal-driven
Energy	Passive; occurs because ATP failed	**ATP-dependent** — the cell spends energy to die
Cell size	Swells, then bursts	**Shrinks**, condenses
Membrane	Ruptures — contents spill	Stays intact; cell fragments into membrane-bound apoptotic bodies
Inflammation	**Yes** — spilled contents are damage signals	**No** — bodies are quietly phagocytosed
Extent	Groups of contiguous cells or whole tissue	Single scattered cells
Examples	Myocardial infarction, gangrene, pancreatitis	Menstrual endometrial shedding, webbing loss in fetal fingers, cytotoxic T-cell killing, apoptosis loss in cancer

- **Coagulative** — protein denaturation preserves tissue outline; the ischaemic infarct pattern of heart, kidney, spleen
- **Liquefactive** — enzymes dissolve tissue to fluid; brain infarcts and any bacterial abscess
- **Caseous** — cheesy crumbly centre inside a granuloma; tuberculosis, some fungi
- **Fat** — lipase releases fatty acids that bind calcium (saponification); acute pancreatitis, breast trauma
- **Gangrenous** — a clinical term: dry (coagulative, ischaemic limb) versus wet (superimposed bacteria, liquefactive) versus gas (Clostridium perfringens)

MEMORY HOOK "Apoptosis is a planned demolition; necrosis is a building collapse." A demolition is scheduled, funded (ATP), contained and leaves no mess for the neighbours (no inflammation). A collapse is unplanned, spills debris everywhere, and the whole block reacts.

RED FLAG

Escalating pain out of proportion to findings in a tense, cool extremity, with rising CK, rising lactate and dark urine, is ischaemic muscle necrosis (compartment syndrome or rhabdomyolysis). This is irreversible injury unfolding in real time — it needs urgent pressure measurement, fasciotomy consideration and aggressive fluids before myoglobin destroys the kidneys.

PRACTICE

Q1. A 62-year-old with 20 years of untreated hypertension has an echocardiogram showing a markedly thickened left ventricular wall. The myocardial cell count is unchanged. Which cellular adaptation does this represent?

A. Hyperplasia, because chronic pressure load stimulates myocyte mitosis

B. Hypertrophy, because cardiac myocytes cannot divide and can only enlarge

C. Metaplasia, because myocytes convert to a more durable fibroblast phenotype

D. Dysplasia, because the disordered wall thickening is premalignant

Show answer

B is correct. Cardiac myocytes are permanent (post-mitotic) cells. Faced with chronic pressure overload they cannot make more of themselves, so they synthesise more contractile protein and organelles and enlarge — hypertrophy. Hyperplasia is the tempting distractor, but it is only available to cells capable of mitosis (glandular, epithelial, hepatic, marrow). Note the clinical consequence: the enlarged muscle outgrows its coronary supply, which is why hypertrophy predisposes to ischaemia and diastolic failure.

PRACTICE

Q2. Which finding indicates that a hypoxic cell has crossed from reversible injury into irreversible injury?

A. Cellular and mitochondrial swelling from Na+/K+-ATPase failure

B. Detachment of ribosomes from the rough endoplasmic reticulum

C. Rupture of lysosomal and mitochondrial membranes with massive cytosolic calcium influx

D. Accumulation of intracellular lactate with a falling intracellular pH

Show answer

C is correct. Swelling, ribosomal detachment and lactic acidosis are all early, fully reversible consequences of ATP depletion — restore perfusion and the cell recovers. The threshold event is loss of **membrane integrity**. Once lysosomal enzymes escape and calcium floods in, calcium-activated proteases, phospholipases and endonucleases autodigest the cell, and enzymes leak outward into serum. That is necrosis, and no intervention reverses it.

PRACTICE

Q3. Which statement correctly contrasts apoptosis with necrosis?

A. Apoptosis affects contiguous groups of cells and triggers a brisk neutrophil response

B. Necrosis is an ATP-dependent, genetically programmed process affecting single cells

C. Apoptosis requires ATP, affects scattered single cells, and does not incite inflammation

D. Necrosis causes cell shrinkage with preserved plasma membrane integrity

Show answer

C is correct. Apoptosis is deliberate and expensive: the cell spends ATP to shrink, condense its chromatin and package itself into membrane-bound apoptotic bodies that macrophages clear silently — no inflammation, and only scattered individual cells are affected. Necrosis is the opposite on every axis: it happens **because** ATP failed, involves swathes of contiguous tissue, and ruptures the membrane so intracellular contents act as danger signals that recruit neutrophils. Options 1, 2 and 4 each swap one feature between the two processes.

17 Neoplasia: Benign versus Malignant, the Genes Behind It, and How Cancer Spreads

What actually makes a tumour malignant

Students lose points here by grading tumours on size or growth rate. A uterine fibroid can be the size of a grapefruit and a basal cell carcinoma can be 4 mm; neither fact settles the question. The defining property of malignancy is **invasion** — the ability to breach the basement membrane, chew through surrounding stroma, and seed distant sites. Everything else on the comparison table (poor differentiation, high nuclear-to-cytoplasmic ratio, loss of a capsule, abnormal mitoses) is a morphologic clue that predicts invasive behaviour, not the definition of it.

> **Key idea —** Cancer is a genetic disease of somatic cells. It is not one mutation but an accumulation of them, in genes that control the cell cycle, apoptosis and DNA repair — which is why cancer incidence rises steeply with age and with anything that raises mutation rate (tobacco, UV, radiation, chronic inflammation, oncogenic viruses).

Feature	Benign	Malignant
Differentiation	Well differentiated — resembles tissue of origin	Poorly differentiated to anaplastic; may not resemble origin at all
Growth rate	Slow, often plateaus	Rapid, erratic, with abnormal mitotic figures
Border	Encapsulated, cohesive, pushes tissue aside	Non-encapsulated, infiltrative, crab-like extensions
Basement membrane	**Intact**	**Breached** — the defining event
Metastasis	**Never**	Yes — the single most reliable malignant feature
Systemic effect	Local mass effect only	Cachexia, anaemia, paraneoplastic syndromes, organ failure
Naming	Tissue + **-oma** (lipoma, adenoma)	Epithelial = **carcinoma**; connective/mesenchymal = **sarcoma**; blood/lymph = leukaemia and lymphoma

Pearl — Benign does not mean harmless — it means non-invasive. A benign meningioma inside a fixed cranial vault, or a benign pituitary adenoma compressing the optic chiasm, can kill by location alone. Teach location and mass effect alongside the histology.

Oncogenes versus tumour suppressor genes

	Proto-oncogene to oncogene	Tumour suppressor gene
Normal job	Accelerator — drives normal, regulated proliferation	Brake — halts the cycle, forces repair, or triggers apoptosis
Mutation type	**Gain** of function; the accelerator jams on	**Loss** of function; the brake line is cut
Alleles needed	**One** mutated copy is enough (dominant at cell level)	**Both** copies must be lost — Knudson two-hit hypothesis
Inheritance pattern	Rarely inherited	Commonly the basis of inherited cancer syndromes
Examples	RAS, MYC, HER2/neu, BCR-ABL	TP53 ("guardian of the genome"), RB1, BRCA1/BRCA2, APC
Clinical relevance	Targetable — trastuzumab for HER2, imatinib for BCR-ABL	Explains familial risk, earlier onset, bilateral or multifocal disease

Pearl — The two-hit model is why hereditary cancers appear younger and in both organs. A child who inherits one defective RB1 allele already carries hit one in every retinal cell — a single somatic hit anywhere finishes the job, so retinoblastoma appears early and often bilaterally. A sporadic case needs both hits in the same cell by chance, which takes decades and gives one tumour.

Staging versus grading — the pair students reverse most

	Grading	Staging
Question answered	How abnormal do the cells look?	How far has the tumour travelled?
Basis	Microscopic — differentiation, mitotic rate, nuclear atypia	Clinical and pathologic — size, nodes, distant spread
Scale	Grade 1 (well differentiated) to Grade 4 (anaplastic)	**TNM**: T = tumour size/extent, N = regional nodes, M = distant metastasis; rolled into Stage 0-IV
Weight in decisions	Contributes to prognosis	**The dominant driver** of prognosis and treatment plan

MEMORY HOOK **G**rade = **G**lass slide (what the pathologist sees under the microscope). **S**tage = **S**pread (how far it has gone). And when the two disagree, stage wins the argument.

How metastasis happens and where it goes

Local invasion → enzymatic breach of basement membrane → intravasation into vessel or lymphatic → survival in circulation as a tumour embolus → adhesion and extravasation at a distant site → angiogenesis → clinically detectable metastasis

Route	Mechanism	Typically used by	Signature presentation
Lymphatic	Cells enter lymphatics and lodge in the first draining node	Most **carcinomas**	Sentinel node positivity; palpable regional nodes
Haematogenous	Cells enter veins and travel to the first capillary bed downstream	Most **sarcomas**, plus renal, thyroid, hepatocellular	Liver and lung metastases (portal and venous drainage); bone, brain
Seeding (transcoelomic)	Direct shedding into a body cavity	Ovarian, gastric, colonic, some CNS tumours	Peritoneal studding with malignant ascites; drop metastases in CSF

RED FLAG

In any patient with known malignancy: new severe back pain with leg weakness, saddle anaesthesia or bladder retention is **spinal cord compression** from vertebral metastasis — imaging and steroids within hours or the deficit becomes permanent. Likewise new confusion, constipation and polyuria in a cancer patient is hypercalcaemia of malignancy until proven otherwise.

PRACTICE

Q1. A pathology report describes a 9 cm uterine mass that is well circumscribed, encapsulated, composed of well-differentiated smooth muscle, with no nodal involvement. Which single feature most reliably establishes that this lesion is benign?

A. Its well-differentiated appearance under the microscope

B. Its large size relative to normal myometrium

C. Its encapsulation and absence of invasion beyond the basement membrane

D. Its location within a hormonally responsive organ

Show answer

C is correct. Malignancy is defined by **invasion and metastatic capacity**, not by size, differentiation alone, or hormone responsiveness. A 9 cm leiomyoma is benign; a 6 mm melanoma can be lethal — size is a trap. Differentiation is a useful clue but some malignancies are well differentiated. The intact capsule with no breach of the basement membrane is the structural statement that this mass cannot seed elsewhere.

PRACTICE

Q2. A child is diagnosed with bilateral retinoblastoma at age 14 months. Genetic testing shows a germline RB1 mutation. Which principle best explains the early, bilateral presentation?

A. RB1 is an oncogene, and a single gain-of-function mutation is sufficient to drive tumour formation

B. RB1 is a tumour suppressor, and the inherited first hit means only one additional somatic hit is needed in any retinal cell

C. RB1 mutations block apoptosis only in tissues with high mitotic rates, so bilateral disease is coincidental

D. Bilateral disease indicates metastasis from one eye to the other via the optic chiasm

Show answer

B is correct. RB1 is the prototype **tumour suppressor** and follows Knudson's two-hit rule — both alleles must be inactivated. A germline mutation preloads hit one into every cell in the body, so a single random somatic hit in any retinal cell completes the process; with millions of retinal cells that happens early and often in both eyes. option B describes oncogene behaviour (gain of function, one hit), which is precisely the concept being contrasted. Option 4 is anatomically implausible and confuses bilateral primary tumours with metastasis.

PRACTICE

Q3. A pathology report reads "Grade 3, Stage I." Which interpretation is correct?

A. The cells appear poorly differentiated, but the tumour is still localised and prognosis is driven mainly by that limited spread

B. The tumour has spread to three regional lymph node groups but the cells are well differentiated

C. Grade describes extent of spread and stage describes cellular appearance, so this tumour is widely metastatic

D. Grade 3 automatically upgrades the tumour to Stage III for treatment purposes

Show answer

A is correct. Grade is the **microscopic** statement — Grade 3 means poorly differentiated, aggressive-appearing cells. Stage is the **anatomic** statement — Stage I means small and localised, no nodal or distant disease. The two are independent scales, and staging carries more prognostic and therapeutic weight, which is why a Grade 3 Stage I tumour is still generally favourable. Options 2 and 3 reverse the definitions (the single most common error on this topic) and option 4 invents a conversion that does not exist.

18 Inflammation: The Cardinal Signs, Acute versus Chronic, and Wound Healing

Every cardinal sign traces back to two vascular events

Inflammation is the vascularised tissue's stereotyped response to injury, and it is fundamentally a **vascular** event before it is a cellular one. Injured cells and resident mast cells release histamine, bradykinin, prostaglandins, leukotrienes, nitric oxide and complement fragments. Two things follow: arterioles **dilate**, and capillaries and venules become **leaky**. If a student can hold those two changes in mind, all five cardinal signs can be derived rather than memorised.

Cardinal sign	Latin	Why it happens
Redness	Rubor	Arteriolar vasodilation (histamine, NO, prostaglandins) floods the area with oxygenated blood
Heat	Calor	The same increased flow of warm core blood into a peripheral site; systemically, pyrogens reset the hypothalamic set point
Swelling	Tumor	Endothelial cells contract, forming gaps; protein-rich **exudate** escapes into the interstitium, raising interstitial oncotic pressure and pulling more fluid out
Pain	Dolor	**Bradykinin** and prostaglandins sensitise nociceptors, and the expanding oedema stretches and compresses nerve endings
Loss of function	Functio laesa	Pain plus mechanical splinting from oedema; the part is guarded and immobilised, which is protective in the short term

> **Key idea —** Inflammation is not infection. It is the response to **any** injury — burn, trauma, ischaemia, autoimmunity, foreign body — and infection is only one trigger. That distinction explains why a patient can be floridly inflamed with sterile cultures, and why anti-inflammatory therapy and antimicrobial therapy answer different questions.

MEMORY HOOK **R**ed, **H**ot, **S**wollen, **P**ainful, **L**ost function — or the Latin set **Rubor, Calor, Tumor, Dolor, Functio laesa**. Anchor them to the two mechanisms: the first two are **vasodilation**, the middle two are **permeability**, and the fifth is the consequence of the other four.

Pearl — The exudate that causes the swelling is doing useful work. It dilutes toxins, delivers complement and antibody, brings fibrinogen that walls off the area with a fibrin mesh, and drains antigen to regional lymph nodes. The oedema is not a side effect — it is the delivery mechanism.

Acute versus chronic inflammation

Feature	Acute	Chronic
Onset and duration	Minutes to hours; days to a few weeks	Insidious; weeks to years
Dominant cell	**Neutrophil** (first 24-48 h)	**Macrophage**, lymphocyte, plasma cell
Vascular changes	Prominent — dilation, exudate, oedema	Minimal; angiogenesis instead
Tissue response	Exudation, possible suppuration	**Fibrosis**, granulation tissue, tissue destruction running alongside repair
Cause	Infection, trauma, burn, ischaemia, foreign body	Persistent infection (TB), unremoved foreign body, autoimmunity, chronic irritation
Systemic markers	Neutrophilia with left shift, fever, rising CRP	Elevated ESR, normocytic anaemia of chronic disease, weight loss, low-grade fever
Outcome	Complete resolution, abscess, or transition to chronic	Scarring, organ dysfunction, granuloma formation

Pearl — A granuloma is a macrophage admitting defeat. When macrophages cannot digest an agent — mycobacteria, fungi, a suture, silica — they fuse into epithelioid and multinucleated giant cells and wall it off behind a lymphocyte collar. Think TB (with caseous centre), sarcoidosis and Crohn disease (non-caseating). The lesion is containment, not clearance, which is why latent TB can reactivate decades later when immunity falls.

Wound healing and what derails it

Haemostasis (minutes) → Inflammatory phase, neutrophils then macrophages (day 0-3) → Proliferative phase, fibroblasts lay collagen III, angiogenesis, granulation tissue,

epithelialisation (day 3-21) → Remodelling, collagen III replaced by stronger collagen I (3 weeks to 2 years, plateauing near 80% of original tensile strength)

Healing by **primary intention** is a clean, approximated surgical incision: minimal tissue loss, minimal granulation, a fine scar. **Secondary intention** is an open wound with tissue loss — a pressure injury or a dehisced abdomen — which must fill from the base with granulation tissue, contracts by myofibroblast action, and leaves a large scar with a longer, more infection-prone course. Tertiary (delayed primary) intention is a contaminated wound deliberately left open and closed later once infection risk has fallen.

- **Poor perfusion or oxygenation** — PAD, venous insufficiency, pressure, smoking (nicotine vasoconstricts and CO displaces oxygen); collagen synthesis is oxygen-dependent
- **Diabetes and hyperglycaemia** — impaired neutrophil chemotaxis and phagocytosis, glycosylated microvasculature, neuropathy that hides the injury
- **Infection or foreign body** — holds the wound in the inflammatory phase and consumes the oxygen and nutrients repair needs
- **Malnutrition** — protein for collagen substrate, **vitamin C** for hydroxylation and collagen cross-linking, **zinc** for epithelial and fibroblast proliferation, vitamin A opposing steroid effects
- **Corticosteroids and immunosuppressants** — blunt the inflammatory phase, inhibit fibroblasts and collagen deposition, and mask the fever and erythema that would announce infection
- **Chemotherapy and radiation** — suppress the rapidly dividing fibroblasts and epithelium repair depends on
- **Advanced age and obesity** — slower epithelialisation; adipose is poorly vascularised and tensions the wound edges
- **Mechanical stress** — coughing, straining, obesity or early heavy lifting on a wound still at a fraction of its final strength

> **RED FLAG**
> Sudden increase in **serosanguineous drainage** from an abdominal incision around post-op day 5-7, or a patient reporting "something gave way," signals impending **dehiscence**. Keep the patient still and supine with knees flexed, cover any exposed viscera with sterile saline-soaked gauze, do not attempt to reinsert them, keep the patient NPO and call the surgeon — evisceration is a surgical emergency.

PRACTICE

Q1. A patient's ankle is swollen 6 hours after a sprain. Which mechanism directly explains the swelling?

A. Vasoconstriction raising capillary hydrostatic pressure upstream of the injury

B. Increased capillary permeability allowing a protein-rich exudate into the interstitium

C. Histamine-mediated obstruction of lymphatic valves preventing all drainage

D. A rise in plasma albumin that osmotically draws fluid into the tissue

Show answer

B is correct. Histamine and bradykinin cause endothelial cells to contract, opening interendothelial gaps. Protein escapes into the interstitium, which raises interstitial oncotic pressure and drags water with it — that protein-rich fluid is **exudate**, and it is what makes the joint swell. option B inverts the vascular event (inflammation causes vaso**dilation**). Option 4 is backwards: plasma protein falls locally as it leaks out, and a high plasma albumin would hold fluid in the vessel.

PRACTICE

Q2. Six weeks after an injury, biopsy of the site shows macrophages, lymphocytes and plasma cells, with fibroblast proliferation and new capillary formation. Which process does this describe?

A. Acute inflammation still in its exudative phase

B. Chronic inflammation with concurrent tissue destruction and repair

C. Complete resolution with restoration of normal architecture

D. A serous exudate typical of an early blister

Show answer

B is correct. Neutrophils are short-lived — they dominate the first 24 to 48 hours and are gone. A mononuclear infiltrate (macrophage, lymphocyte, plasma cell) accompanied by fibroblasts and angiogenesis at six weeks is the defining picture of **chronic** inflammation, in which destruction and repair proceed simultaneously and produce fibrosis. option B is tempting only if you forget the neutrophil timeline; resolution would show restored normal tissue with no infiltrate at all.

PRACTICE

Q3. On post-operative day 6, a patient who is obese, diabetic and taking chronic prednisone reports that something "gave way" while coughing, and the abdominal dressing is suddenly saturated with serosanguineous fluid. What is the priority interpretation and action?

A. Expected remodelling-phase drainage; reinforce the dressing and document

B. Impending wound dehiscence; position supine with knees flexed, cover with sterile saline-soaked gauze, keep NPO and notify the surgeon

C. Early granulation tissue formation; irrigate the wound and apply a dry dressing

D. Wound infection; obtain a culture and await antibiotic orders before further action

Show answer

B is correct. A sudden surge of serosanguineous drainage between days 5 and 7, especially after a Valsalva event, is the classic herald of **dehiscence** — the wound is only at a small fraction of its final tensile strength while collagen III is still being laid down. This patient stacks three delaying factors: corticosteroids suppress fibroblast activity and collagen synthesis, hyperglycaemia impairs neutrophil function and microvascular flow, and obesity adds poorly perfused tissue plus mechanical strain. Infection is a reasonable competing thought, but purulence and spreading erythema, not a sudden gush of serosanguineous fluid after coughing, would point there — and waiting on a culture delays care for a wound that may eviscerate.

RAPID REVIEW

Modules 5–6 — Neurology, Mental Health, MSK & Skin

19 Module 5 — The Pressurized Brain: ICP, Stroke, TBI and Seizures

> **Key idea —** The skull is a rigid box with a fixed volume, so every neurologic emergency in this module is a variation on one theme: either a volume the cranium cannot accommodate, or a perfusion pressure it can no longer maintain. Neurons hold no fuel reserve — they fail within minutes of losing flow, and the tissue that dies does not come back. Everything you do for a neuro patient is aimed at protecting the tissue that is still salvageable.

Monro-Kellie doctrine: why compensation ends all at once

The cranial vault holds roughly 1,400 to 1,700 mL of three things: brain tissue (about 80%), blood (10%) and cerebrospinal fluid (10%). The Monro-Kellie doctrine states that if any one of the three grows, another must shrink or the pressure climbs. The brain buys time by shunting CSF into the spinal subarachnoid space, absorbing CSF faster, and squeezing blood out of the cerebral veins. Those compensations are finite and non-linear: while they hold, an expanding hematoma barely moves the ICP at all; once they are spent, a few more milliliters send the pressure up a cliff. That is exactly why a head-injured patient can look deceptively stable and then deteriorate in minutes.

Expanding mass → CSF displaced to spinal canal and venous blood squeezed out → compensation exhausted → small added volume produces a steep ICP rise → CPP falls → ischemia → cytotoxic edema → still more volume

Normal ICP is 5 to 15 mmHg, but what actually perfuses the brain is cerebral perfusion pressure: CPP = MAP − ICP, and it must stay above roughly 60 mmHg. Below 50 autoregulation fails and ischemia begins; below 30, infarction is essentially assured. Every ICP intervention is really a CPP intervention — head of bed at 30 degrees with the head midline so the jugular veins drain, no neck or hip flexion, suctioning clustered and brief, fever and pain and seizures treated because each raises metabolic demand, ventilation kept normocapnic (carbon dioxide is the most potent cerebral vasodilator there is), and water pulled out of the brain with mannitol or hypertonic saline. Never hang hypotonic fluid — free water follows the osmotic gradient straight into injured tissue.

RED FLAG

The earliest sign of rising ICP is a change in level of consciousness — restlessness, confusion, harder to rouse — because the reticular activating system is exquisitely pressure-sensitive. Do not wait for the famous signs. By the time you see Cushing's triad (rising systolic pressure with a WIDENING pulse pressure, bradycardia, and irregular or Cheyne-Stokes respirations), a fixed dilated pupil, or decerebrate posturing, the brainstem is already being compressed and herniation is minutes away.

Pearl — Cushing's triad looks backwards on purpose. In shock, blood pressure falls and heart rate rises; here blood pressure climbs and heart rate falls. The brainstem, starved because ICP is consuming CPP, fires a massive sympathetic surge to raise MAP and force blood back into the skull; the aortic and carotid baroreceptors read that hypertension and reflexly slow the heart, while the compressed medulla disorganizes the respiratory rhythm. A unilateral blown pupil is anatomy talking in the same way: an expanding temporal mass pushes the uncus over the tentorium onto cranial nerve III, whose parasympathetic fibers ride on the outside and die first — so the pupil dilates and stops reacting on the SAME side as the lesion.

MEMORY HOOK Posturing: de**cor**ticate means arms flexed in toward the **core** — injury above the midbrain, bad. De**ce**rebrate means arms **e**xtended — injury at the midbrain or pons, worse. A patient who moves from decorticate to decerebrate is telling you the damage is descending through the brainstem.

Stroke: blocked pipe versus burst pipe

Feature	Ischemic stroke (about 87%)	Hemorrhagic stroke (about 13%)
Mechanism	A thrombus forms on an atherosclerotic plaque, or an embolus (classically from atrial fibrillation) lodges in a cerebral vessel; the tissue downstream starves	A vessel ruptures — chronic hypertension bursting a Charcot-Bouchard microaneurysm (intracerebral, often basal ganglia) or a berry aneurysm (subarachnoid). The blood itself is both toxic and space-occupying
Onset	Thrombotic: stuttering, stepwise, often noticed on waking. Embolic: deficit maximal within seconds	Sudden, with headache patients describe as the worst of their life, and deficits that keep worsening over minutes to hours as the hematoma grows
Headache, vomiting, seizure at onset	Usually absent early	Common — blood irritates the meninges and drives ICP up fast
Level of consciousness	Usually preserved unless the infarct is large	Declines early and rapidly
Blood pressure	May be elevated, rarely extreme	Often markedly elevated; it is both the cause and the consequence
Meningeal signs	Absent	Nuchal rigidity and photophobia with subarachnoid bleeding
First test and first move	Non-contrast CT to exclude blood, then thrombolytic within the 4.5 hour window if eligible	Non-contrast CT shows blood immediately; control blood pressure, reverse anticoagulation, involve neurosurgery. Thrombolytics are absolutely contraindicated

Around any occlusion, perfusion falls off as a gradient rather than an on-off switch. In the center — the infarct core — flow is too low to keep membranes running: ATP fails, the sodium-potassium pump stops, sodium and water rush in (cytotoxic edema), the depolarized cell dumps glutamate, glutamate opens calcium channels on its neighbors, and calcium activates the proteases, lipases and free radicals that finish the cell off. Encircling that core is the penumbra: tissue supplied by collaterals with just

enough flow to stay alive but not enough to fire. It is electrically silent, which is why the visible deficit is larger than the dead tissue, and it is the entire target of reperfusion therapy.

Vessel occlusion → ATP failure in the core → pump failure, cytotoxic edema, glutamate release → calcium influx and free radical injury → penumbra recruited into the core minute by minute → deficit becomes permanent

> **Pearl —** "Time is brain" is arithmetic, not a slogan: roughly 1.9 million neurons are lost per minute of untreated large-vessel occlusion. The one thing that must happen before a thrombolytic is given is a non-contrast CT — not to see the ischemic stroke, since early ischemia is often invisible on CT, but to prove there is no blood. Dissolving clot into a hemorrhage converts a survivable bleed into a fatal one.

Traumatic brain injury: the primary hit and the preventable second one

TBI splits into primary injury — the tearing, bruising and bleeding that happened at impact and cannot be undone — and secondary injury, the cascade of hypoxia, hypotension, edema, seizure, fever and rising ICP that unfolds over the following hours to days. Secondary injury is the entire reason for serial neuro checks: it is preventable, and it is what actually determines the outcome.

Injury	Vessel or mechanism	Imaging signature	Bedside giveaway
Epidural hematoma	Middle meningeal ARTERY torn beneath a temporal skull fracture	Biconvex lens shape; stops at suture lines	The lucid interval - brief knockout, wakes and talks normally, then deteriorates over 1 to 4 hours. A fixed dilated pupil on the SAME side as the clot
Subdural hematoma	Bridging VEINS torn as the brain shifts inside the skull	Crescent shape that crosses suture lines	Older adults, alcohol use, anticoagulation - atrophy stretches those veins so trivial trauma bleeds. The chronic form shows up weeks later as confusion or personality change, not coma
Subarachnoid hemorrhage	Ruptured berry aneurysm or trauma; blood mixes with CSF	Blood tracking into the sulci and basal cisterns	Thunderclap headache with nuchal rigidity and photophobia. Watch for vasospasm on days 3 to 14, which delivers a second, delayed stroke
Concussion and diffuse axonal injury	Acceleration-deceleration shearing of axons; no focal clot	CT is frequently normal; the injury is microscopic and diffuse	Coma or deficit wildly out of proportion to a clean scan
Coup-contrecoup contusion	Brain bruised at the impact site and again as it rebounds against the opposite skull	Contusions at both poles	Deficits that worsen over 24 to 72 hours as contusion edema peaks

Seizures: excitation outrunning inhibition

A seizure is a paroxysmal, hypersynchronous discharge of cortical neurons — glutamate-driven excitation overwhelming GABA-mediated inhibition in a population of cells that normally fire independently. Focal seizures begin in one hemisphere and may preserve or impair awareness; generalized seizures engage both hemispheres from the outset and always impair it. The tonic-clonic pattern is stiffening (tonic, often

with a cry as air is forced through the larynx) then rhythmic jerking (clonic), followed by a postictal period of confusion, headache and heavy sleepiness while the exhausted cortex recovers. That postictal state is itself diagnostic — non-epileptic events usually lack it. Childhood absence seizures are the opposite picture: a blank three-to-ten-second stare with no aura and no postictal phase, routinely mistaken for inattention.

- A new seizure is metabolic until proven otherwise: hyponatremia, hypoglycemia, hypocalcemia, hypoxia, fever in small children, alcohol or benzodiazepine WITHDRAWAL, stimulant or local anesthetic toxicity, eclampsia. Correct the cause and the seizure stops; anticonvulsants alone will not hold it.
- Epilepsy means two or more UNPROVOKED seizures — a chronic tendency, not a single event.
- During the event: nothing in the mouth, no restraint, side-lying to protect the airway, pad and lower to the floor, and note the time plus the pattern of onset — where it starts tells you where the focus is.
- Check glucose and sodium at the bedside first; EEG identifies the discharge and imaging looks for a structural lesion afterward.

> **RED FLAG**
> Status epilepticus is any seizure lasting 5 minutes or longer, or repeated seizures without full recovery of consciousness in between. Sustained excitotoxicity kills neurons outright, and the systemic cost stacks up fast: hyperthermia, lactic acidosis, rhabdomyolysis, hypoxia and aspiration. Airway, oxygen, glucose and sodium, IV benzodiazepine first line, then a loading anticonvulsant. Do not wait to see whether it self-terminates — past about 5 minutes, it usually will not.

PRACTICE

Q1. A 24-year-old is admitted after a motorcycle crash. Over 30 minutes his blood pressure rises from 124/78 to 190/62, his heart rate falls from 92 to 46, and his respirations become irregular with long pauses. What is the mechanism behind this pattern?

A. Brainstem ischemia triggers a sympathetic surge to raise MAP and defend cerebral perfusion pressure; the resulting baroreceptor reflex slows the heart while the compressed medulla disorganizes breathing

B. Spinal cord injury above T6 has produced neurogenic shock with loss of sympathetic tone

C. Hypovolemia from internal bleeding is being compensated through vagal withdrawal

D. An expanding hematoma is compressing the carotid sinus and causing reflex hypertension

Show answer

A is correct. This is Cushing's triad: a widening pulse pressure (systolic climbing while diastolic falls), bradycardia, and irregular respirations. As ICP rises toward MAP, cerebral perfusion pressure collapses; the ischemic brainstem fires a sympathetic discharge to force blood back into the skull, and the baroreceptors answer that hypertension with reflex bradycardia. Neurogenic shock is the tempting distractor because it also produces bradycardia — but it produces HYPOtension, never a systolic of 190. Hypovolemia raises heart rate, it does not drop it. Remember the timing: Cushing's triad is a LATE sign of impending herniation; the earliest sign of rising ICP is a change in level of consciousness.

PRACTICE

Q2. In acute ischemic stroke, reperfusion therapy is aimed at the ischemic penumbra. Which statement best describes that tissue?

A. The irreversibly infarcted core, which will liquefy regardless of treatment

B. Hypoperfused tissue surrounding the core, kept marginally alive by collateral flow — electrically silent but still metabolically viable, and salvageable if blood flow is restored quickly

C. Vasogenic edema around the infarct that resolves once the blood-brain barrier reseals

D. The zone in which hemorrhagic transformation is most likely to occur after thrombolysis

Show answer

B is correct. Blood flow falls off as a gradient around an occlusion. In the core, flow drops below the threshold for membrane function: ATP fails, the sodium-potassium pump stops, the cell swells, glutamate floods out, calcium pours in, and the cell digests itself. In the penumbra, collateral vessels supply just enough flow to keep membranes intact but not enough for neurons to fire — so the tissue is dysfunctional, which is why the clinical deficit looks larger than the dead tissue, yet still alive. Restore flow and it recovers; leave it and it is recruited into the core minute by minute. Option A is what has already been lost, and option D is a complication of treatment, not its target.

PRACTICE

Q3. A 19-year-old is struck in the left temple by a baseball. He is briefly unconscious, wakes up, walks to the car and jokes with friends, then 90 minutes later becomes drowsy, vomits, and develops a fixed dilated LEFT pupil. Which injury is this, and why is it moving so fast?

A. Chronic subdural hematoma from torn bridging veins

B. Diffuse axonal injury from rotational shearing

C. Epidural hematoma from a torn middle meningeal artery beneath a temporal fracture

D. Subarachnoid hemorrhage from a ruptured berry aneurysm

Show answer

C is correct. The lucid interval is the signature: the concussion causes the first brief loss of consciousness, then ARTERIAL bleeding between skull and dura accumulates silently until compensation is exhausted and the patient crashes. It moves fast precisely because it is arterial and under systemic pressure. The blown pupil is ipsilateral because the expanding temporal clot pushes the uncus over the tentorium onto cranial nerve III, whose superficial parasympathetic fibers fail first. Subdural bleeding is venous and low pressure, so it evolves over days to weeks, classically in an older or anticoagulated patient whose atrophied brain has stretched the bridging veins. Diffuse axonal injury gives neither a lucid interval nor a focal pupil — it gives coma out of proportion to a normal CT.

20 Module 6 — Bone: Osteoporosis versus Osteomalacia, and the Three Fracture Emergencies

> **Key idea —** Bone disease sorts into a quantity problem and a quality problem. Osteoporosis is too little bone that is otherwise normally mineralized; osteomalacia is a normal amount of bone matrix with too little mineral driven into it. That distinction predicts everything downstream — who has pain before the fracture, and what the labs will show. After the fracture, a different clock starts: three complications each strike in their own time window.

The remodeling balance and why estrogen matters

Bone is never static — osteoclasts excavate and osteoblasts refill, with peak bone mass banked by roughly age 30. Osteoblasts actually control their own demolition crew through two signals: RANKL, which activates osteoclasts, and osteoprotegerin, a decoy receptor that soaks RANKL up. Estrogen keeps osteoprotegerin high and osteoclast lifespan short. After menopause that brake is released, resorption outruns formation, and the thin trabeculae of cancellous bone — vertebral bodies, femoral neck, distal radius — are eaten first. Glucocorticoids reach the same endpoint by a different road: they suppress osteoblasts, blunt intestinal calcium absorption and increase urinary calcium loss, which is why steroid therapy expected to run beyond three months demands bone protection from the start.

Menopause → estrogen loss → osteoprotegerin falls and RANKL goes unopposed → osteoclast number and lifespan ↑ → resorption outpaces formation → trabeculae thin and perforate → fragility fracture

Feature	Osteoporosis	Osteomalacia (rickets in children)
Core defect	Too little bone; what remains is normally mineralized — a QUANTITY problem	Adequate osteoid matrix, insufficient mineral laid into it — a QUALITY problem, so the bone is soft rather than scarce
Driver	Resorption outpaces formation: estrogen loss, aging, glucocorticoids, immobility, smoking, low calcium intake, hyperthyroidism, prolonged PPI use	Vitamin D deficiency, malabsorption (bariatric surgery, celiac), phosphate wasting or chronic kidney disease — calcium and phosphate are simply not available to mineralize new osteoid
Serum calcium and phosphate	Both normal	Calcium low or low-normal, phosphate low
Alkaline phosphatase	Normal	Elevated — osteoblasts working overtime on matrix they cannot harden
Parathyroid hormone	Normal	Elevated (secondary hyperparathyroidism), which then strips still more bone to defend serum calcium
Symptoms before a fracture	None. Completely silent until something breaks	Diffuse aching bone pain, tenderness on pressing the bone, proximal muscle weakness, waddling gait
Signature findings	Vertebral compression fractures with height loss and kyphosis, hip and Colles fractures; DEXA T-score of -2.5 or lower	Looser zones (pseudofractures) and bowing of soft bone; in children, bowed legs, rachitic rosary and widened growth plates because the physes are still open

MEMORY HOOK Let alkaline phosphatase sort the bone diseases. **Normal ALP** = not enough otherwise-normal bone (osteoporosis). **High ALP with low calcium and low phosphate** = bone being built and never hardened (osteomalacia or rickets). **Very high ALP with normal calcium and phosphate** = bone built far too fast and disorganized (Paget disease).

Pearl — Osteoporosis is silent until it breaks; osteomalacia hurts before it breaks. That one sentence resolves most exam questions. The other giveaway is force: a fragility fracture is one produced by a fall from standing height or less — a vertebra collapsing while lifting groceries, a wrist snapping on a curb — and it indicates fragile bone whatever the DEXA number says. A T-score of -1 to -2.5 is osteopenia; -2.5 or lower is osteoporosis. A hip fracture in an older adult carries roughly 20 to 30% one-year mortality, and the killer is almost never the bone — it is the immobility that follows: pneumonia, DVT and pulmonary embolism, pressure injury, delirium and deconditioning.

After the fracture: three emergencies, three clocks

A fracture heals in overlapping stages — hematoma, inflammation, soft fibrocartilaginous callus, hard woven-bone callus, then remodeling to lamellar bone — and that takes weeks. The complications that kill or maim do not wait for it. Each of the three owns a distinct time window, and knowing the clock is usually enough to name the diagnosis before any test returns.

Emergency	When it strikes	Mechanism in one line	What tells you
Compartment syndrome	Hours - typically within the first 6 to 8, with irreversible muscle and nerve loss beginning at 4 to 6 hours of ischemia	Bleeding and edema inside a closed fascial compartment raise tissue pressure above capillary perfusion pressure; the resulting ischemia produces more edema, which raises pressure further	Escalating pain out of proportion to the injury and unrelieved by opioids, severe pain on PASSIVE STRETCH of the distal muscles, a wood-hard compartment, early paresthesia. Pallor, paralysis and pulselessness are late and mean muscle is already dying
Fat embolism syndrome	24 to 72 hours after a long-bone or pelvic fracture	Marrow fat globules enter torn venous sinusoids and lodge in pulmonary capillaries; lipase liberates free fatty acids that inflame and leak the endothelium	The triad of hypoxemia and dyspnea, new confusion or restlessness, and a petechial rash across chest, axillae and conjunctivae
Rhabdomyolysis	Immediately, and most dangerously at the moment a crushed limb is decompressed	Sarcolemma ruptures and dumps myoglobin, potassium, phosphate and creatine kinase into the circulation; myoglobin casts obstruct and directly poison renal tubules	Tea- or cola-colored urine, dipstick positive for blood with NO red cells on microscopy, CK in the thousands, and a climbing potassium

RED FLAG

Compartment syndrome is the one you can prevent at the bedside, and never by checking a pulse. Escalating pain out of proportion to the injury plus pain on passive stretch, with a tense compartment, is enough to act on. Bivalve or remove the cast, loosen the dressing, keep the limb at HEART LEVEL — elevating it lowers arterial inflow and worsens the ischemia — never apply ice, and escalate immediately for pressure measurement. A delta pressure (diastolic minus compartment pressure) under 30 mmHg means fasciotomy. The endpoint of a missed diagnosis is Volkmann ischemic contracture, a permanently clawed, useless limb.

Pearl — Here is why the pulse fools people. Compartment pressure only has to exceed CAPILLARY perfusion pressure, around 30 mmHg, to shut down the microcirculation — it does not have to approach systolic pressure. So a large artery keeps pulsing straight through a compartment full of dying muscle, and the foot stays warm and pink. Pallor, paralysis, pulselessness and poikilothermia are not the diagnosis; they are the autopsy of the limb.

Crush or prolonged compression → sarcolemma ruptures → myoglobin, potassium, phosphate and CK flood the circulation → hyperkalemic arrhythmia (the immediate killer) plus myoglobin casts obstructing the tubules → acute kidney injury → aggressive IV fluid is both the treatment and the prevention

PRACTICE

Q1. A 66-year-old with a history of gastric bypass reports months of aching bones, tenderness when the tibia is pressed, and difficulty rising from a chair. Labs show calcium low-normal, phosphate low, alkaline phosphatase markedly elevated, PTH elevated, and 25-OH vitamin D very low. Which process best explains her disease?

A. Osteoclastic resorption outpacing formation, producing a reduced mass of normally mineralized bone

B. Vitamin D deficiency leaving newly laid osteoid unmineralized, producing soft bone

C. Disordered, accelerated remodeling producing thickened but structurally chaotic bone

D. Metastatic bone destruction releasing calcium into the circulation

Show answer

B is correct. Bariatric surgery is a classic route to vitamin D and calcium malabsorption. Without vitamin D the gut cannot absorb enough calcium and phosphate, so osteoblasts keep laying down osteoid that never mineralizes — bone that is soft, not scarce. That softness is why it HURTS on pressure, and the proximal muscle weakness (also a vitamin D effect) is why she cannot rise from a chair. Option A is osteoporosis, and the labs are the discriminator: in osteoporosis calcium, phosphate, ALP and PTH are all NORMAL and there is no pain until something breaks. The high ALP here is osteoblasts working furiously on matrix they cannot harden; the high PTH is the parathyroid defending serum calcium by stripping still more bone. Option C describes Paget disease, where ALP is very high but calcium and phosphate are normal, and option D would raise calcium, not lower it.

PRACTICE

Q2. Six hours after casting a tibial fracture, a patient reports escalating pain not controlled by opioids. The foot is warm, capillary refill is 2 seconds, and the dorsalis pedis pulse is easily palpable. Which finding should be assessed next, and why?

A. Recheck the pedal pulse in 30 minutes, since compartment syndrome is excluded while a pulse is present

B. Pain on passive extension of the toes, because stretching ischemic muscle reproduces pain long before any pulse is lost

C. Ability to wiggle the toes, since paralysis is the earliest reliable sign

D. Pallor and coolness of the skin, the first indicators of compartment compromise

Show answer

B is correct. Compartment syndrome is a capillary problem, not an arterial one. Pressure inside the closed fascial compartment only needs to exceed capillary perfusion pressure — around 30 mmHg — to starve muscle and nerve, which is nowhere near systolic pressure. A bounding pulse, warm skin and normal capillary refill are therefore entirely compatible with muscle that is infarcting. Waiting for pulselessness, pallor or paralysis means waiting for necrosis; those are the late Ps, and by then you are managing a Volkmann contracture rather than preventing one. Pain out of proportion plus pain on passive stretch, with a tense compartment and early paresthesia, is the window to act: bivalve or remove the cast, keep the limb at heart level (elevation lowers arterial inflow and worsens ischemia), no ice, and escalate for pressure measurement and possible fasciotomy.

PRACTICE

Q3. A 21-year-old with a femoral shaft fracture is stable for the first day. Thirty-six hours after injury he becomes restless and confused, his SpO2 falls to 86% on room air, and scattered petechiae appear over his chest, axillae and conjunctivae. What is happening?

A. Pulmonary embolism arising from a deep vein thrombosis

B. Fat embolism syndrome

C. Rhabdomyolysis with acute kidney injury

D. Early sepsis from an occult wound infection

Show answer

B is correct. Marrow fat is forced into torn venous sinusoids at the fracture site and lodges in the pulmonary microcirculation; lipase then liberates free fatty acids that inflame and leak the endothelium, producing hypoxemia, cerebral microemboli and thrombocytopenia. The combination of hypoxemia, new neurologic change, and a petechial rash appearing 24 to 72 hours after a long-bone or pelvic fracture is the answer every time. A DVT-related pulmonary embolism is the tempting distractor because it also causes abrupt hypoxemia — but it requires days of immobility to form and it does NOT produce petechiae. Rhabdomyolysis after a fracture is real, yet it announces itself with tea-colored urine, a CK in the thousands and a rising potassium, not with dyspnea and a rash. Read restlessness in a young trauma patient as hypoxia until proven otherwise.

21 Module 6 — Joints and Skin: The Arthritis Triad, Pressure Injuries and Burns

Key idea — Joints and skin fail in ways you can read from the outside. Three arthritides run on three entirely different engines — osteoarthritis is mechanical failure of cartilage, rheumatoid arthritis is autoimmune destruction of synovium, gout is a crystal-triggered innate immune explosion — and each writes a different pattern onto the hands. Skin breakdown obeys a single rule: tissue dies wherever pressure exceeds perfusion, and both pressure injuries and burns are classified by the DEPTH of tissue lost, never by how alarming the surface appears.

Osteoarthritis versus rheumatoid arthritis versus gout

Feature	Osteoarthritis	Rheumatoid arthritis	Gout
Engine	Mechanical wear plus chondrocyte failure; matrix metalloproteinases degrade cartilage faster than it can be rebuilt, exposing bone to bone	Autoimmune synovitis; citrullinated self-antigens in a genetically susceptible host drive a destructive pannus that creeps over cartilage and erodes bone	Monosodium urate crystals precipitate and are phagocytosed by neutrophils, firing the NLRP3 inflammasome and releasing interleukin-1 beta
Joint pattern	Asymmetric and weight-bearing - knee, hip, lumbar spine - plus DIP (Heberden) and PIP (Bouchard) nodes	Symmetric small joints: MCP, PIP and wrist, and it characteristically SPARES the DIP	Monoarticular and peripheral; the first MTP joint (podagra), then midfoot, ankle, knee
Stiffness and tempo	Under 30 minutes of stiffness; pain worsens with use and eases with rest; slow progression over years	Over 60 minutes of morning stiffness that loosens as the day goes on; flares and remissions	Not a stiffness disease but an attack - peak agony within hours, classically waking the patient overnight, resolving over days
Systemic features	None; this is a joint disease only	Fatigue, low-grade fever, weight loss, rheumatoid nodules, anemia of chronic disease, interstitial lung disease, Sjogren syndrome, accelerated cardiovascular risk	Tophi and urate nephrolithiasis in chronic disease; an acute flare can produce fever and leukocytosis and mimic septic arthritis
Labs and joint fluid	Labs normal; imaging shows asymmetric joint space narrowing, subchondral sclerosis, cysts and osteophytes	Elevated ESR and CRP, rheumatoid factor (sensitive) and anti-CCP (the specific one); symmetric marginal erosions on imaging	Aspiration is definitive: needle-shaped, NEGATIVELY birefringent crystals. Serum urate can be normal during the flare, so a normal level does not exclude gout

Deformity	Bony enlargement, crepitus, malalignment such as varus knees	Ulnar deviation, swan-neck and boutonniere deformities from tendon and capsule destruction	Chalky tophi in the helix of the ear, olecranon bursa and finger pads; joint destruction late

Pearl — Two bedside discriminators do most of the work. First, the DIP joint: osteoarthritis loves it (Heberden nodes), rheumatoid arthritis spares it. Second, the gel phenomenon: RA is at its worst on waking and loosens over an hour of movement, because inflammatory exudate pools in an immobile joint overnight, whereas OA feels best on waking and deteriorates as the day loads the cartilage. And if a single joint went from normal to unbearable in six hours overnight, at the base of the great toe, stop thinking arthritis and think crystals.

Under-excretion or over-production of urate → hyperuricemia → crystals precipitate in cooler distal joints → neutrophils phagocytose them → NLRP3 inflammasome fires → interleukin-1 beta storm → red, hot, exquisitely tender joint within hours

MEMORY HOOK Crystals: **N**eedles are **N**egatively birefringent and mean gout (monosodium urate). Rhomboids are positively birefringent and mean pseudogout (calcium pyrophosphate, usually knee or wrist). Attach the treatment rule to it: an acute flare is treated with an anti-inflammatory — NSAID, colchicine or a steroid — and you never START allopurinol during an attack, because shifting urate levels mobilize crystals and prolong the flare. If the patient is already established on allopurinol, keep it running.

Pressure injuries: staging by what is exposed

Capillary closing pressure at the skin is roughly 32 mmHg, and lying or sitting on a bony prominence exceeds it easily. Compression collapses the capillaries, tissue goes ischemic, and metabolic waste accumulates. When pressure is relieved, reactive hyperemia flushes the area — redness that BLANCHES under a finger and fades within minutes is healthy compensation, not an injury. Sustained pressure beyond that point causes cell death. Three accomplices accelerate it: shear, where the skeleton slides down in bed while the skin stays put, kinking and tearing deep vessels; friction, which strips epidermis; and moisture from incontinence or sweat, which macerates the barrier. Muscle tolerates ischemia far worse than skin does, so serious injuries frequently begin at the bone-muscle interface and surface only later.

Stage	What you see	Tissue involved
Stage 1	Intact skin with localized NON-blanchable erythema. In darkly pigmented skin the color change may be subtle, so assess for warmth, coolness, firmness or a boggy texture compared with surrounding skin	Epidermis intact; the damage is in the microcirculation beneath it
Stage 2	Shallow open ulcer with a pink or red, moist base, or an intact or ruptured serum-filled blister. No slough, no granulation tissue, no visible fat	Partial thickness - epidermis lost, dermis exposed
Stage 3	Deeper crater with subcutaneous FAT visible, granulation tissue and rolled wound edges (epibole); slough or eschar may be present but does not obscure the depth; undermining and tunneling are possible	Full thickness through the dermis into adipose tissue. No muscle, tendon, fascia or bone visible
Stage 4	Exposed or directly palpable fascia, muscle, tendon, ligament, cartilage or bone	Full thickness with tissue loss into supporting structures; osteomyelitis is a genuine risk
Unstageable	The base is completely obscured by slough or eschar - you cannot see the bottom, so you cannot assign a depth	At minimum a stage 3, frequently a stage 4, once the base is debrided and visualized
Deep tissue pressure injury	Intact or non-intact skin showing persistent deep red, maroon or purple discoloration, or a blood-filled blister	Injury began at the bone-muscle interface and is working outward; may deteriorate into a stage 3 or 4 within days

RED FLAG

Two staging traps and one care rule. A deep tissue pressure injury is NOT a stage 1 — the damage is already deep and can open into a stage 3 or 4 within days, so it demands immediate offloading, not observation. An unstageable wound is not a low stage; it is a stage 3 or 4 concealed under slough or eschar. And do not debride a stable, dry, adherent eschar on an ischemic heel: it is functioning as the body's biological cover, and opening it in a limb without perfusion creates a wound that cannot heal. One more rule — wounds are never reverse-staged as they heal, because the defect fills with scar rather than with the original tissue layers.

Burns: depth, and the two phases of danger

Depth	Appearance	Pain	Course
Superficial (first degree)	Dry and red, blanches with pressure, no blisters	Painful	Heals in 3 to 7 days with no scar. NOT counted in TBSA calculations
Superficial partial thickness (second degree)	Moist and weeping with blisters, bright red, blanches briskly	Exquisitely painful - nerve endings are exposed and intact	Heals in 2 to 3 weeks with minimal scarring
Deep partial thickness (second degree)	Mottled red and waxy white, drier, sluggish or absent blanching	Less painful than superficial partial thickness because some nerve endings are destroyed	Slow healing with scarring and contracture; frequently grafted
Full thickness (third degree)	Leathery, dry, waxy white, tan or charred; no blanching; thrombosed vessels may be visible beneath	INSENSATE at the center, ringed by a painful partial-thickness margin	Cannot re-epithelialize; requires excision and grafting
Fourth degree	Charred, exposing fat, muscle, tendon or bone	Insensate	Requires reconstruction and sometimes amputation

A burn is dangerous twice, in opposite directions, and the electrolytes flip between the two. The emergent or resuscitative phase runs from injury through roughly the first 24 to 48 hours: histamine, bradykinin, prostaglandins and oxygen radicals make capillaries leak, plasma pours into the interstitium, and the patient develops burn shock that is hypovolemic and distributive at once — hemoconcentration with a HIGH hematocrit despite losing volume, hypotension, oliguria, HYPERkalemia from lysed cells, hyponatremia as sodium follows water into the tissues, and metabolic acidosis. The acute or diuretic phase begins around 48 to 72 hours when capillary integrity is restored: interstitial fluid is remobilized into the vasculature and the danger inverts to circulatory overload and pulmonary edema, with HYPOkalemia as potassium re-enters cells and is lost in urine. Behind both sits the hypermetabolic phase, in which resting energy expenditure can double, protein catabolism is relentless, immunity is suppressed, and infection becomes the leading cause of death after resuscitation.

Burn → inflammatory mediator release → capillary permeability ↑ → plasma shifts into the interstitium → massive edema with intravascular volume ↓ → burn shock, hemoconcentration, hyperkalemia, oliguria → (48 to 72 h) capillaries seal → fluid returns to the vessels → diuresis, overload risk, hypokalemia

Pearl — Fluid resuscitation is timed from the moment of the BURN, not from arrival. The Parkland formula gives 4 mL x body weight in kg x %TBSA of lactated Ringer's over 24 hours, with half delivered in the first 8 hours from the time of injury — so a patient who arrives two hours late still has to receive that half in the remaining six. The formula is only a starting estimate; the real endpoint is urine output of about 0.5 mL/kg/h, or 30 to 50 mL/h in an adult. Depth also affects pain in a counterintuitive way: the burn that does not hurt is the worst one, because full-thickness injury has destroyed the nerve endings, and whatever the patient feels is coming from the partial-thickness ring around it. Rule of nines in an adult: head 9, each arm 9, each leg 18, anterior trunk 18, posterior trunk 18, perineum 1 — and superficial burns are never counted.

RED FLAG

Two burn findings outrank everything else. Inhalation injury — facial burns, singed nasal hairs, carbonaceous sputum, hoarseness or stridor, or any fire in an enclosed space — means the airway will swell shut as resuscitation fluid goes in; intubate EARLY while it is still easy, and suspect carbon monoxide poisoning, remembering that a reassuring pulse oximeter reading is meaningless because the oximeter cannot distinguish carboxyhemoglobin from oxyhemoglobin. A circumferential full-thickness burn is a tourniquet: rigid eschar around a limb strangles perfusion as edema builds beneath it, and around the chest it prevents the ribs from expanding. The treatment is escharotomy, and it cannot wait.

PRACTICE

Q1. A 48-year-old woman reports 90 minutes of morning stiffness that eases as she works, with symmetric swelling of the MCP and PIP joints of both hands and sparing of the DIP joints, plus fatigue and low-grade fevers. Which statement best captures her disease?

A. Cartilage has worn away mechanically, so her pain will worsen through the day with use

B. Autoimmune synovitis has formed a pannus that erodes cartilage and bone, which is why the disease is symmetric and systemic

C. Urate crystals have deposited in the small joints of the hand, triggering inflammasome activation

D. Subchondral sclerosis and osteophyte formation are narrowing the joint spaces asymmetrically

Show answer

B is correct. Every element points at rheumatoid arthritis: symmetry, because an immune process circulating in the blood strikes both sides equally; MCP and PIP involvement with DIP SPARING; stiffness lasting over an hour that improves with movement; and systemic features such as fatigue and fever. Options A and D describe osteoarthritis, whose fingerprint is the mirror image — asymmetric weight-bearing joints, DIP nodes, under 30 minutes of stiffness, and pain that worsens with use and improves with rest. The gel phenomenon is the fastest bedside discriminator: RA is worst on waking and loosens up, OA is best on waking and worsens as the day loads the joint. Gout is monoarticular, explosive within hours, and starts at the first MTP rather than across both hands.

PRACTICE

Q2. A sacral wound has a crater base in which yellow subcutaneous fat is clearly visible, along with granulation tissue and rolled edges. A small amount of slough sits at one margin, but the base and full depth are visible. No bone, tendon or muscle is exposed. How should this be staged?

A. Stage 2, because there is an open ulcer with visible slough

B. Stage 3, because the tissue loss is full thickness with adipose visible but no deeper structures exposed

C. Stage 4, because the presence of slough implies deep structural involvement

D. Unstageable, because slough is present in the wound

Show answer

B is correct. Staging reflects which tissue layer is exposed, not how bad the wound looks. Visible fat with granulation tissue and epibole is the definition of stage 3: through epidermis and dermis into subcutaneous tissue but stopping short of fascia, muscle, tendon or bone, which would make it stage 4. Stage 2 is partial thickness only — a shallow pink moist ulcer or a serum-filled blister, and by definition with NO fat, granulation or slough. Unstageable is the trap answer: slough alone does not make a wound unstageable; a wound is unstageable only when slough or eschar OBSCURES the base so depth cannot be determined. Remember also that wounds are never reverse-staged — a healing stage 3 remains a stage 3, because the defect fills with scar rather than with the original tissue.

PRACTICE

Q3. A 70 kg patient with 40% TBSA burns was fluid-resuscitated appropriately. On day 3 he develops bibasilar crackles, jugular venous distention, a brisk urine output, and a serum potassium of 3.1 mEq/L. What is the mechanism?

A. Ongoing burn shock; capillary leak continues and he needs a fluid bolus

B. Capillary integrity has been restored, so interstitial fluid is remobilizing into the vasculature — producing circulatory overload while potassium shifts back into cells and is diuresed

C. Acute kidney injury from myoglobin, causing volume retention

D. Early burn wound sepsis with distributive shock

Show answer

B is correct. Burns are dangerous in two distinct phases, and the electrolytes invert between them. During the emergent phase, roughly the first 24 to 48 hours, inflammatory mediators make capillaries leak plasma into the interstitium: the patient is hypovolemic and hemoconcentrated, and potassium runs HIGH because damaged cells spill it. From about 48 to 72 hours the capillaries seal and that interstitial fluid pours back into circulation — the diuretic phase — so the danger flips to fluid overload and pulmonary edema, and potassium falls as it re-enters cells and is excreted. Giving more fluid, as option A proposes, would deepen the overload. The brisk urine output argues directly against acute kidney injury. Sepsis remains the leading cause of death after resuscitation and stays on the differential, but it does not explain a low potassium accompanied by signs of volume excess.

22 Module 7 — Inheritance Patterns, Chromosomal Error, and Recurrence Risk

> **Key idea —** Every pedigree question is really two questions: is the gene on an autosome or on the X, and does one bad copy cause disease (dominant) or do you need two (recessive)? Skipped generations, male predominance, and father-to-son transmission are all downstream consequences of those two switches, so decide the switches first and the pattern names itself.

Reading a pedigree in four moves

Work the chart in a fixed order rather than hunting for a pattern you already have in mind. The order matters because each move eliminates whole categories, and the fourth move is the single most powerful discriminator on the exam.

- **Does it appear in every generation?** A vertical, unbroken line of affected people means one mutant allele is enough — dominant. If the trait vanishes for a generation and then reappears among the children of two unaffected parents, those parents were silent carriers — recessive.
- **What is the sex ratio?** Roughly equal male and female involvement points to an autosome. A pedigree that is almost entirely male, with the disease travelling through unaffected women, is X-linked recessive: a male has only one X, so a single mutant allele is unopposed.
- **Is there father-to-son transmission?** A father gives his son a Y, never his X. Any single male-to-male transmission **rules out both X-linked patterns permanently**.
- **Check the affected father's daughters.** In X-linked dominant disease an affected man transmits to **all** of his daughters and **none** of his sons, because every daughter receives his one X. That fan-shaped pattern is unmistakable once you look for it.

Pattern	Pedigree signature	Recurrence risk (typical cross)	Classic examples
Autosomal dominant	Present in every generation, males and females equally, an affected child usually has an affected parent	Affected heterozygote × unaffected = **50% each pregnancy**	Huntington disease, Marfan syndrome, familial hypercholesterolaemia, neurofibromatosis 1
Autosomal recessive	Skips generations, clusters within one sibship, parents unaffected carriers, consanguinity raises the odds	Carrier × carrier = **25% affected, 50% carrier, 25% clear**	Cystic fibrosis, sickle cell disease, phenylketonuria, Tay-Sachs
X-linked recessive	Almost all affected are male; transmitted through unaffected carrier mothers; never father to son	Carrier mother × unaffected father = 50% of **sons** affected, 50% of daughters carriers	Haemophilia A, Duchenne muscular dystrophy, red-green colour blindness, G6PD deficiency
X-linked dominant	Affected father → all daughters, no sons; affected mother → 50% of children of either sex; often more severe in males	Depends entirely on which parent is affected	Vitamin D-resistant (hypophosphataemic) rickets, fragile X (with repeat-expansion caveats)
Multifactorial	Familial clustering with no clean Mendelian ratio; risk rises with the number and closeness of affected relatives	Empiric, commonly 2-5% for a first-degree relative	Cleft lip/palate, neural tube defects, pyloric stenosis, type 2 diabetes

MEMORY HOOK Dominant = **every** generation; recessive = **skips** a generation. And the veto rule: if you ever see father → son, cross the X off the list.

When whole chromosomes go wrong: nondisjunction

Homologues (meiosis I) or sister chromatids (meiosis II) fail to separate → gamete carrying 24 or 22 chromosomes → fertilisation → trisomic or monosomic zygote → every cell carries a

whole-chromosome dosage error

Nondisjunction is a mechanical failure of separation, not an inherited allele — which is why these disorders usually appear in families with no history at all. The maternal-age effect exists because a woman's oocytes arrest in prophase I during her own fetal life and sit there for decades; the protein complexes holding homologues together degrade over time, so separation fails more often in an older egg. Because a trisomy misdoses hundreds of genes at once, the phenotype is always multi-system: face, heart, brain, and growth together.

Disorder	Karyotype	Mechanism note	Signature findings
Down syndrome	Trisomy 21 (47,XX or XY, +21)	~95% maternal meiotic nondisjunction; a small share from Robertsonian translocation	Flat facial profile, upslanting palpebral fissures, single transverse palmar crease, hypotonia, intellectual disability, endocardial cushion (AV canal) defects, early Alzheimer-type changes
Edwards syndrome	Trisomy 18	Nondisjunction	Clenched fists with overlapping fingers, rocker-bottom feet, micrognathia, severe cardiac and renal defects; most die within the first year
Patau syndrome	Trisomy 13	Nondisjunction	Midline failures — cleft lip/palate, holoprosencephaly, microphthalmia — plus polydactyly and cardiac defects; survival usually days to months
Turner syndrome	45,X (monosomy X)	Often loss of the **paternal** sex chromosome; **no** maternal-age effect	Short stature, webbed neck, shield chest, streak ovaries with primary amenorrhoea and infertility, coarctation or bicuspid aortic valve, lymphoedema of hands and feet at birth; intelligence typically normal
Klinefelter syndrome	47,XXY	Nondisjunction in either parent	Tall with long limbs, small firm testes, gynaecomastia, sparse body hair, infertility; low testosterone with high LH and FSH (a primary gonadal failure pattern)

> **Pearl —** Turner is the only monosomy compatible with life, and X-inactivation explains why: in every female cell one X is already silenced as a Barr body, so losing it costs far less than losing an autosome. The features that do appear — short stature and ovarian failure — come from the minority of genes that normally **escape** inactivation and therefore truly need two copies.

Recurrence risk: what carries over to the next pregnancy

Recurrence risk is calculated per pregnancy and has no memory. Three affected children do not use up the 25% for a carrier couple; each conception is an independent roll. For aneuploidy caused by sporadic nondisjunction, recurrence is low — roughly 1% or the mother's age-related risk, whichever is greater — precisely because the error was a one-off mechanical event rather than a transmitted allele. The exception that changes counselling entirely is a parent carrying a **balanced Robertsonian translocation**: that parent is phenotypically normal but produces unbalanced gametes repeatedly, so recurrence is substantial and parental karyotyping is warranted.

- **Penetrance** — the proportion of people with the genotype who show any phenotype. Incomplete penetrance makes a dominant pedigree appear to skip a generation, a classic trap.
- **Expressivity** — how severely the disease shows in those who do express it. The same neurofibromatosis mutation may cause a few café-au-lait spots in one relative and disfiguring tumours in another.
- **Anticipation** — an unstable trinucleotide repeat expands as it passes through the germline, so each generation is affected earlier and more severely (Huntington disease, fragile X).
- **Mosaicism** — a mutation arising after fertilisation, so only some cell lines carry it; findings are patchy and typically milder.
- **Heterozygote advantage** — sickle trait resists falciparum malaria, which is why a lethal recessive allele persists at high frequency in endemic regions.

> **RED FLAG**
> A newborn with hypotonia, a flat facial profile, and a single palmar crease needs an **echocardiogram before discharge**. Up to half of infants with trisomy 21 have a congenital heart defect, most often an AV canal, and the murmur can be silent until pulmonary vascular resistance falls over the first weeks — by which time the infant presents in heart failure.

PRACTICE

Q1. A pedigree shows the trait in every generation, affects males and females about equally, and includes several affected fathers with affected sons. Which inheritance pattern fits best?

A. Autosomal dominant

B. X-linked dominant

C. X-linked recessive

D. Autosomal recessive

Show answer

A is correct. Appearance in every generation with an equal sex ratio means one mutant allele suffices on an autosome. X-linked dominant is the tempting answer because it also shows up in every generation, but it is excluded by the male-to-male transmission: a father passes a Y to his son, so an affected man transmits an X-linked trait to all of his daughters and none of his sons. X-linked recessive is excluded for the same reason plus the equal sex ratio, and autosomal recessive would skip generations.

PRACTICE

Q2. Two unaffected parents have a child with cystic fibrosis. What is the probability that their next child is affected?

A. 0% — the affected child has already occurred

B. 25%

C. 50%

D. 67%

Show answer

B is correct. Both parents must be obligate heterozygous carriers, so each conception is an independent 1-in-4 chance of receiving both mutant alleles. Prior outcomes do not change the odds. The 67% distractor is a real number attached to a different question — it is the chance that an **already unaffected** sibling is a carrier (2 of the 3 remaining genotypes) — and 50% is the carrier fraction of all offspring, not the affected fraction.

PRACTICE

Q3. Which statement best distinguishes Turner syndrome (45,X) from trisomy 21?

A. Turner syndrome arises from nondisjunction, whereas trisomy 21 does not

B. Turner syndrome incidence rises sharply with advancing maternal age

C. Turner syndrome is the only monosomy compatible with life, and its incidence is not related to maternal age

D. Turner syndrome characteristically produces severe intellectual disability

Show answer

C is correct. Both result from nondisjunction, so the first option is false. The maternal-age effect belongs to the trisomies, because oocytes arrested in prophase I since fetal life progressively lose the cohesion needed for clean separation; 45,X most often reflects loss of the paternal sex chromosome and shows no such age relationship. Intelligence in Turner syndrome is usually normal with a selective visuospatial weakness — the defining features are short stature, streak ovaries, and aortic anomalies. Monosomy X survives only because one X is normally inactivated anyway.

23 Module 8 (Part 1) — Endocrine Feedback Logic, ADH Disorders, and the Thyroid

> **Key idea —** Every endocrine diagnosis reduces to one loop and one rule. The loop: hypothalamus → pituitary tropic hormone → target gland hormone → negative feedback onto both. The rule: never interpret a single hormone level — read the **pair**. If the target hormone and its tropic hormone move in **opposite** directions, the target gland itself is diseased (primary). If they move in the **same** direction, the pituitary or hypothalamus is driving the abnormality (secondary or tertiary).

Hypothalamic releasing hormone → anterior pituitary tropic hormone (TSH, ACTH, LH/FSH) → target gland hormone (T4, cortisol, sex steroids) → negative feedback suppressing both levels above

Work through it once and it becomes automatic. High free T4 with a **suppressed** TSH means the thyroid is producing hormone on its own and the intact pituitary is trying to shut it off — primary hyperthyroidism. High free T4 with a normal or high TSH is nonsensical for a working feedback loop, so the pituitary must be the culprit (a TSH-secreting adenoma). Low cortisol with a **high** ACTH means the adrenal cortex has failed while the pituitary shouts at it — primary adrenal insufficiency. Low cortisol with a **low** ACTH means the shouting has stopped — the lesion is at the top. The same arithmetic works for every axis you will be asked about.

MEMORY HOOK **Opposite = organ, same = summit.** Levels moving in opposite directions incriminate the target organ; levels moving together incriminate the pituitary at the summit of the axis.

ADH and the two water disorders

Antidiuretic hormone acts on V2 receptors of the collecting duct and inserts aquaporin-2 channels into the luminal membrane. It moves **water, not sodium** — which is the entire key to both diseases. Too much ADH means free water is reabsorbed and retained, diluting the plasma: sodium falls while the urine comes out inappropriately concentrated. Too little ADH (or a duct that cannot answer it) means free water pours out: litres of dilute urine, rising serum sodium and osmolality, and unrelenting thirst that is the only thing keeping the patient compensated. An awake patient with an intact thirst mechanism and access to water can hide diabetes

insipidus for a long time; sedate that patient, or intubate them, and the sodium climbs fast.

Feature	SIADH	Central diabetes insipidus	Nephrogenic diabetes insipidus
Core defect	ADH released without an osmotic or volume stimulus	Pituitary/hypothalamus cannot make or release ADH	Collecting duct cannot respond to circulating ADH
Serum sodium and osmolality	**Low** Na (often under 130), low serum osmolality	**High** or high-normal Na, high serum osmolality	Same as central
Urine	Low volume, inappropriately **concentrated** (urine osm high, urine Na high)	Massive volumes of very dilute urine, 3-20 L/day, low urine osmolality	Same as central
Volume status	Euvolaemic — water is retained but pressure natriuresis prevents oedema	Hypovolaemic if thirst cannot keep pace	Hypovolaemic if thirst cannot keep pace
Classic causes	Small cell lung carcinoma (ectopic ADH), CNS injury or infection, pneumonia, pain, surgery, SSRIs and carbamazepine	Head trauma, pituitary surgery or tumour, infiltrative disease	Lithium, hypercalcaemia, hypokalaemia, chronic kidney disease
Management	Fluid restriction first; hypertonic saline only for seizures or coma; vaptans; treat the cause	Desmopressin (DDAVP) replaces the missing hormone — and confirms the diagnosis by concentrating the urine	Remove the offending drug; thiazide plus low-solute diet; DDAVP does **not** work

Pearl — The two diseases are mirror images of the same molecule, and the old names still teach the difference. Insipidus means tasteless — the urine is so dilute it has no flavour, which is literally how clinicians separated it from diabetes mellitus (honeyed) before laboratories existed. If the urine is dilute while the blood is concentrated, ADH is missing; if the urine is concentrated while the blood is dilute, ADH is in excess.

RED FLAG

Correct chronic hyponatraemia **slowly** — no more than roughly 8 mEq/L in 24 hours. Brain cells adapt to a low sodium by extruding osmolytes; raising serum sodium faster than they can be replaced dehydrates and demyelinates the pons. The patient looks improved, then one to three days later develops dysarthria, dysphagia, and quadriparesis from osmotic demyelination, often permanently. A sodium under about 120 with seizures or coma still earns hypertonic saline — but in small measured boluses with sodium rechecked every one to two hours.

Graves versus Hashimoto — same organ, opposite antibodies

Both are autoimmune, both can enlarge the gland, and both can begin with a thyrotoxic phase — so the discriminator is what the antibody **does**. In Graves disease, thyroid-stimulating immunoglobulin binds the TSH receptor and activates it, so the gland behaves as though TSH were permanently high: hormone output soars, pituitary TSH is suppressed, and the gland grows diffusely. Because the same TSH receptor sits on retro-orbital and pretibial fibroblasts, stimulated cells lay down glycosaminoglycans behind the eye and in the shin — producing exophthalmos and pretibial myxoedema, findings no other thyroid disease creates. In Hashimoto thyroiditis, anti-thyroid peroxidase and antithyroglobulin antibodies accompany lymphocytic destruction of follicles; stored hormone leaking from dying cells can cause a brief hashitoxicosis, after which the gland burns out into permanent hypothyroidism.

Feature	Graves disease	Hashimoto thyroiditis
Immune mechanism	TSI/TRAb **stimulate** the TSH receptor (type II hypersensitivity, stimulatory)	Anti-TPO and antithyroglobulin antibodies with cytotoxic lymphocytic destruction of follicles
Hormone pattern	Free T4 and T3 high, TSH **suppressed**	TSH high, free T4 low (may transiently reverse early)
Gland on examination	Diffuse, smooth, non-tender, sometimes with a bruit from hypervascularity	Firm, rubbery, painless enlargement that eventually atrophies
Findings unique to it	Exophthalmos, lid lag and stare, pretibial myxoedema	None of those; strongly associated with other autoimmune disease and a small risk of thyroid lymphoma
Radioactive iodine uptake	Diffusely increased	Low or patchy
Clinical picture	Heat intolerance, weight loss despite good appetite, tachycardia and atrial fibrillation, fine tremor, anxiety, frequent stools, insomnia	Cold intolerance, weight gain, bradycardia, constipation, dry coarse skin, hair loss, menorrhagia, delayed relaxation of deep tendon reflexes
Treatment	Methimazole (PTU in first trimester), beta-blocker for symptom control, radioiodine ablation or thyroidectomy	Levothyroxine for life, taken fasting and separated from calcium and iron

RED FLAG

Both ends of the thyroid have a lethal decompensation. **Thyroid storm** — temperature above 40°C, tachyarrhythmia or high-output failure, agitation progressing to delirium, vomiting and diarrhoea — is precipitated in an untreated hyperthyroid patient by surgery, infection, trauma, or iodinated contrast. **Myxoedema coma** — hypothermia without shivering, hypoventilation with CO_2 retention, hypoglycaemia, hyponatraemia, bradycardia, and obtundation — typically strikes an elderly patient in winter who stopped taking levothyroxine. Both are treated on clinical suspicion; waiting for confirmatory labs costs lives.

MEMORY HOOK In thyroid storm, give the **thionamide before the iodine** — at least an hour apart. Iodine exploits the Wolff-Chaikoff effect to block hormone release, but given first it simply supplies fresh substrate for synthesis. Order of operations: propylthiouracil or methimazole → beta-blocker → iodine → glucocorticoid (which also blocks peripheral T4-to-T3 conversion) → cooling and treating the trigger.

PRACTICE

Q1. A patient has a low free T4. Which additional result indicates that the lesion lies in the pituitary rather than in the thyroid gland?

A. TSH markedly elevated at 42 mIU/L

B. TSH low or inappropriately normal

C. Positive anti-thyroid peroxidase antibodies

D. Diffusely increased radioactive iodine uptake

Show answer

B is correct. Apply the pair rule. A failing thyroid with an intact pituitary gives opposite movement — low T4 with a high TSH — which is primary hypothyroidism, so the markedly elevated TSH points at the gland, not the pituitary. Both hormones low means the tropic drive has been lost at the summit. Anti-TPO antibodies indicate Hashimoto disease, which is a primary gland problem, and increased radioiodine uptake belongs to Graves-type hyperthyroidism, the opposite clinical state.

PRACTICE

Q2. Three days after transsphenoidal pituitary surgery, a patient produces 500 mL of urine per hour. Serum sodium is 152 mEq/L, serum osmolality 310 mOsm/kg, and urine osmolality 90 mOsm/kg. What is the most likely diagnosis?

A. SIADH

B. Central diabetes insipidus

C. Nephrogenic diabetes insipidus

D. Primary (psychogenic) polydipsia

Show answer

B is correct. Dilute urine in the face of concentrated blood proves the kidney is failing to reabsorb free water, and the setting — pituitary surgery with abrupt onset — localises the failure to ADH production. Nephrogenic DI produces identical fluid numbers but follows lithium, hypercalcaemia, or chronic kidney disease, and it can be separated at the bedside by giving desmopressin: the central form concentrates the urine, the nephrogenic form does not. SIADH is the mirror image, with low sodium and concentrated urine, and psychogenic polydipsia drives serum sodium **down**, not up.

PRACTICE

Q3. Which finding occurs in Graves disease but never in Hashimoto thyroiditis?

A. A diffusely enlarged goitre

B. Exophthalmos and pretibial myxoedema

C. Circulating antithyroid antibodies

D. A transient period of thyrotoxicosis

Show answer

B is correct. Only Graves antibodies bind and activate the TSH receptor, and that receptor is also expressed on retro-orbital and pretibial fibroblasts, which respond by depositing glycosaminoglycans — hence the proptosis and shin plaques. The distractors are all shared: Hashimoto enlarges the gland, is defined by antibodies (anti-TPO), and produces an early hashitoxicosis when stored hormone spills from dying follicles, which is exactly why an early Hashimoto patient can be mistaken for a Graves patient on symptoms alone.

24 Module 8 (Part 2) — Cortisol Extremes and the Glucose Emergencies

> **Key idea —** Cortisol is the catabolic stress hormone: it raises blood glucose, breaks down muscle protein and peripheral fat and redeposits it centrally, retains sodium and water, and suppresses immunity and inflammation. Read Cushing syndrome as too much of every one of those actions and Addison disease as none of them **plus** the loss of aldosterone. Do that and both sign lists become derivable rather than memorised.

CRH → ACTH (cleaved from POMC, the same precursor as melanocyte-stimulating hormone) → adrenal cortisol → negative feedback on hypothalamus and pituitary

That shared POMC precursor is worth holding on to, because it explains a physical sign. Whenever ACTH is chronically **high** — primary adrenal failure or an ectopic ACTH-secreting tumour — the excess precursor also yields MSH, and the patient darkens in the palmar creases, buccal mucosa, old scars, and pressure points. Hyperpigmentation therefore is not a random finding; it is a direct readout that the pituitary is intact and shouting at a cortex that cannot answer.

Feature	Cushing syndrome (cortisol excess)	Addison disease (primary adrenal insufficiency)
Usual cause	Exogenous glucocorticoids by far the most common; then a pituitary ACTH adenoma (Cushing **disease**), an adrenal tumour, or ectopic ACTH from small cell lung carcinoma	Autoimmune adrenalitis in developed countries, tuberculosis worldwide; also adrenal haemorrhage, metastasis, or infiltration
Habitus and skin	Central obesity with thin extremities, moon face, dorsocervical fat pad, thin skin, wide purple striae, easy bruising, poor wound healing	Weight loss and anorexia, hyperpigmentation of creases, scars, and buccal mucosa (present only when ACTH is high, i.e. primary disease)
Glucose, pressure, electrolytes	Hyperglycaemia, hypertension, sodium and water retention, **hypo**kalaemia and metabolic alkalosis	Hypoglycaemia, hypotension with orthostasis, hyponatraemia, **hyper**kalaemia, salt craving
Musculoskeletal	Proximal muscle wasting and weakness, osteoporosis with vertebral compression fractures	Diffuse fatigue, myalgia and arthralgia, profound weakness
Immune and mood	Infection risk with a blunted fever response, mood lability, insomnia, frank psychosis at high doses	Depression and apathy; frequently accompanied by vitiligo and other autoimmune disease
Confirming workup	Late-night salivary cortisol, 24-hour urine free cortisol, or low-dose dexamethasone suppression; then measure ACTH to localise the source	Low morning cortisol with a **high** ACTH; cosyntropin stimulation test shows no cortisol rise

Pearl — Aldosterone is the switch that separates primary from secondary adrenal insufficiency. In Addison disease the entire cortex is destroyed, so aldosterone fails with cortisol: hyperkalaemia, hyponatraemia, salt craving, and orthostatic hypotension. In secondary insufficiency — pituitary failure or, far more commonly, a prednisone taper done too quickly — the zona glomerulosa is untouched and still answers to angiotensin II and potassium, so aldosterone is preserved. That patient has a **normal potassium and no hyperpigmentation**, because ACTH is low, not high.

RED FLAG

Addisonian crisis. Hypotension or shock that does not respond to fluids alone, with vomiting, abdominal pain that can mimic an acute abdomen, fever, confusion, hypoglycaemia, and hyperkalaemia — typically in a patient with known adrenal insufficiency who developed influenza, missed doses, or went to surgery. Give IV hydrocortisone plus normal saline with dextrose **immediately**; do not wait for the cortisol result to return. Every patient on chronic glucocorticoids needs stress dosing for illness, injury, or surgery, and no chronic steroid is ever stopped abruptly.

Diabetes: why type 1 makes ketones and type 2 usually does not

Type 1 diabetes is autoimmune destruction of beta cells (anti-GAD and islet cell antibodies), producing an **absolute** insulin deficiency. With no insulin at all to restrain hormone-sensitive lipase, lipolysis runs unopposed, free fatty acids flood the liver, and ketoacids accumulate. Type 2 diabetes is insulin resistance with a **relative** deficiency: circulating insulin is inadequate for glucose uptake but is still enough to suppress lipolysis. That single residual function is the whole reason type 2 patients typically do not generate ketoacidosis and instead drift over days into a hyperosmolar state as glucose climbs and osmotic diuresis dehydrates them.

Feature	Type 1 diabetes	Type 2 diabetes
Pathogenesis	Autoimmune beta-cell destruction; absolute insulin deficiency	Insulin resistance in muscle, liver and fat with progressive beta-cell exhaustion
Onset and habitus	Usually childhood or adolescence, abrupt, lean, with polyuria, polydipsia and weight loss over weeks	Usually adult, insidious over years, often obese with acanthosis nigricans; frequently found on screening
Laboratory markers	C-peptide low or absent; islet autoantibodies present	C-peptide normal or high early; no autoantibodies
Characteristic acute crisis	Diabetic ketoacidosis	Hyperosmolar hyperglycaemic state
Treatment	Insulin from diagnosis, always — it is replacement, not intensification	Lifestyle plus metformin and other agents; insulin later as beta cells fail

Insulin deficiency → unopposed glucagon and catecholamines → lipolysis → free fatty acids to liver → ketoacid production → anion-gap metabolic acidosis → Kussmaul respirations and fruity breath as the lungs blow off CO_2 to compensate

Feature	DKA	HHS
Typical patient	Type 1, often younger, precipitated by infection, missed insulin, or new onset	Type 2, often elderly with limited access to water, precipitated by infection or a new medication
Onset	Hours to a day or two	Days to weeks
Glucose	Usually 350-800 mg/dL	Often above 600, frequently over 1000 mg/dL
Acid-base and ketones	pH below 7.30, bicarbonate low, **elevated anion gap**, ketones strongly positive	pH above 7.30, bicarbonate near normal, minimal or absent ketones
Osmolality and mental state	Osmolality moderately raised; alertness usually preserved until late	Osmolality above 320 mOsm/kg; obtundation, focal deficits, and seizures are typical
Signature examination findings	Kussmaul respirations, acetone (fruity) breath, nausea, vomiting, abdominal pain	Profound dehydration with the largest fluid deficits seen in medicine (often 8-12 L), no Kussmaul breathing
Treatment sequence	Aggressive isotonic fluids first, potassium assessed before insulin, then an insulin infusion continued until the **anion gap closes** — add dextrose when glucose reaches ~200	Fluids are the mainstay, with lower-dose insulin; correct osmolality slowly to avoid cerebral oedema

MEMORY HOOK **Potassium lies.** Total body potassium is severely depleted in both crises through osmotic diuresis and vomiting, but the measured serum value looks normal or high because acidosis and insulin lack drive potassium out of cells. Give insulin and that potassium marches straight back in, taking the serum level with it.

RED FLAG

Check potassium **before** the first dose of insulin. If serum potassium is under 3.3 mEq/L, hold the insulin and replace potassium first — insulin given at that level can precipitate fatal arrhythmia and respiratory muscle paralysis. Two other trip-wires: do not stop the insulin infusion when glucose normalises, since you are treating the anion gap and not the glucose number (add dextrose and keep insulin running); and in children, drop glucose and osmolality slowly, because rapid correction is the setting for cerebral oedema — headache, falling heart rate with rising blood pressure, and deteriorating consciousness during treatment.

PRACTICE

Q1. A 34-year-old has months of fatigue, weight loss, and dizziness on standing. Sodium is 128 mEq/L, potassium 5.8 mEq/L, glucose 62 mg/dL, and the palmar creases and buccal mucosa are darkly pigmented. What is the most likely diagnosis?

A. Secondary adrenal insufficiency from abrupt prednisone withdrawal

B. Primary adrenal insufficiency (Addison disease)

C. Cushing syndrome

D. SIADH

Show answer

B is correct. Hyperkalaemia and hyperpigmentation together localise the lesion to the adrenal cortex itself. The high potassium says aldosterone has been lost, which only happens when the whole cortex fails; the pigmentation says ACTH is high, because ACTH and MSH are cleaved from the same POMC precursor. Steroid withdrawal is the tempting answer given the similar fatigue and hypotension, but there the pituitary is suppressed — ACTH is **low**, so there is no pigmentation, and the renin-driven zona glomerulosa still makes aldosterone, so potassium is normal. SIADH also lowers sodium but leaves potassium normal and the patient euvolaemic without pigment.

PRACTICE

Q2. A 78-year-old nursing home resident with type 2 diabetes is obtunded. Glucose is 1080 mg/dL, pH 7.34, bicarbonate 22 mEq/L, serum ketones trace, and calculated osmolality 355 mOsm/kg. Which crisis is this?

A. Diabetic ketoacidosis

B. Hyperosmolar hyperglycaemic state

C. Lactic acidosis from metformin

D. Hypoglycaemic coma

Show answer

B is correct. Extreme hyperglycaemia and osmolality with a nearly normal pH and no meaningful ketones is the defining pattern of HHS. The residual insulin present in type 2 diabetes is too little to move glucose but sufficient to suppress lipolysis, so ketoacids never accumulate; instead glucose climbs over days while osmotic diuresis strips volume, and the rising osmolality itself produces the obtundation. DKA is the reflex answer to a very high glucose, but it requires an anion-gap acidosis with pH under 7.30, a low bicarbonate, and strongly positive ketones — and its glucose is usually far lower, in the 350-800 range.

PRACTICE

Q3. A patient in new-onset DKA has a serum potassium of 3.1 mEq/L. What should be done first?

A. Begin the insulin infusion immediately to close the anion gap

B. Replace potassium and withhold insulin until the level exceeds 3.3 mEq/L

C. Give sodium bicarbonate to correct the acidosis

D. Give an ampoule of 50% dextrose

Show answer

B is correct. Insulin shifts potassium into cells along with glucose. Total body potassium is already badly depleted by osmotic diuresis and vomiting, so a measured 3.1 represents an extreme deficit; starting insulin at that level can drop it into the range of ventricular arrhythmia and respiratory muscle weakness. Closing the gap is the goal but not at that cost, and it can wait the short time replacement takes. Bicarbonate is reserved for a pH below roughly 6.9 because it worsens hypokalaemia and can paradoxically acidify the CNS, and dextrose is added later, once glucose falls to about 200 while the insulin infusion continues.

25 Anaemia, Haemolysis, and Sickle Cell Disease

> **Key idea —** Anaemia is never a diagnosis on its own — it is a signal that red cells are being **underproduced, lost, or destroyed**. The MCV tells you what the marrow was able to build, the reticulocyte count tells you whether the marrow is even trying, and the iron studies tell you whether the raw material or the delivery system is the problem.

Sorting anaemia by cell size (MCV)

Cell size is not cosmetic — it is a fingerprint of the step that failed. A developing erythroblast divides until its cytoplasm is packed with haemoglobin. If haemoglobin cannot be built (no iron, no heme, too few globin chains), the cell keeps dividing while it waits, and the final product is **small**. If instead DNA synthesis is blocked (B12 or folate), the nucleus cannot keep pace with a cytoplasm that keeps maturing, so the cell grows **large** before it can divide. If the cells that emerge are normal size, then manufacturing is intact and the problem is loss, destruction, or a marrow that simply is not being told to work.

Class	Why the cell ends up that size	Classic causes	Test that settles it
Microcytic (under 80 fL)	Too little haemoglobin to fill the cytoplasm, so the precursor divides an extra time and finishes small	Iron deficiency (most common worldwide), thalassaemia, late anaemia of chronic disease, sideroblastic anaemia and lead poisoning	Ferritin and TIBC first; haemoglobin electrophoresis if the iron studies come back normal
Normocytic (80 to 100 fL)	The cells that are made are built correctly — the failure is downstream of manufacturing	Acute blood loss, haemolysis, chronic kidney disease (no erythropoietin), early anaemia of chronic disease, aplastic anaemia	Reticulocyte count: high means losing or destroying, low means not producing
Macrocytic (over 100 fL)	DNA synthesis is blocked, so nuclear division lags behind a cytoplasm that keeps maturing	B12 and folate deficiency (megaloblastic); alcohol, liver disease, hypothyroidism (non-megaloblastic)	B12 and folate levels; hypersegmented neutrophils on the smear confirm megaloblastic change

> **Pearl —** B12 and folate deficiency produce identical blood pictures, but only **B12 deficiency damages the nervous system** — B12 is required to maintain myelin, so the patient gets paraesthesias, loss of vibration and position sense, and ataxia. Giving folate to a B12-deficient patient corrects the anaemia while the neuropathy silently progresses, which is why you check both before treating either.

Reading the iron studies as a set

The four numbers only make sense together. **Serum iron** is what is circulating right now. **Ferritin** is the storage form and mirrors total body iron. **TIBC** (transferrin) is the delivery fleet — when the liver senses empty stores it builds **more** transferrin, so TIBC moves in the opposite direction to ferritin. **Saturation** is simply how much of that fleet is loaded. Anaemia of chronic disease is the great mimic: inflammation drives hepcidin, which locks iron inside macrophages, so serum iron falls even though the stores are full — ferritin is normal or high (it is also an acute-phase reactant) and TIBC is low because the body is not iron-starved, it is iron-blocked.

Pattern	Serum iron	Ferritin (stores)	TIBC / transferrin	Saturation
Iron-deficiency anaemia	Low	Low — this is the discriminator	High	Low
Anaemia of chronic disease	Low	Normal or high	Low	Low or normal
Thalassaemia trait	Normal	Normal	Normal	Normal
Sideroblastic anaemia / lead poisoning	High	High	Normal or low	High

> **Pearl —** A markedly low MCV with **normal iron studies, a high red cell count, and only mild anaemia** is thalassaemia trait, not iron deficiency. The body is not short of iron — it is short of globin chains. Treating that patient with iron does nothing except risk overload.

The haemolysis signature

When red cells die early, two things happen everywhere: the marrow ramps up (**reticulocytes rise**), and the contents of the broken cells spill out. Intracellular **LDH rises**, heme is catabolised to **unconjugated (indirect) bilirubin**, and free plasma haemoglobin is scavenged by **haptoglobin**, which is therefore consumed and falls. That is the whole signature: three up, one down. Where the cells die then determines the rest of the picture.

Feature	Intravascular haemolysis	Extravascular haemolysis
Where the cell dies	Inside the vessel — the membrane ruptures and free haemoglobin spills into plasma	Inside the spleen and liver, where macrophages phagocytose antibody-coated or misshapen cells
Haptoglobin	Very low or undetectable — rapidly consumed mopping up free haemoglobin	Normal to mildly reduced
Urine	Haemoglobinuria then haemosiderinuria; tea- or cola-coloured urine	Clear urine; urobilinogen may rise
Bilirubin and spleen	Indirect bilirubin rises; spleen usually not enlarged	Indirect bilirubin rises with jaundice; splenomegaly is common
Typical causes	ABO transfusion reaction, mechanical shear (DIC, prosthetic valve, TTP), G6PD crisis	Hereditary spherocytosis, autoimmune haemolytic anaemia, sickle cell disease, malaria

MEMORY HOOK Haemolysis labs: **three up, one down** — reticulocytes up, LDH up, indirect bilirubin up, haptoglobin down. Haptoglobin is the haemoglobin sponge, so it is the one thing that gets used up rather than released.

Sickle cell disease

A single point mutation swaps valine for glutamic acid at position 6 of the beta-globin chain. The resulting HbS behaves normally while oxygenated, but when it gives up its oxygen the abnormal surface lets HbS molecules **polymerise into rigid rods** that deform the cell into a crescent. Those cells are stiff, sticky to endothelium, and short-lived (10 to 20 days instead of 120), so the patient gets two diseases at once: **vaso-occlusion** (ischaemic pain crises, stroke, priapism, renal papillary necrosis) and **chronic haemolytic anaemia**. Anything that increases the deoxygenated fraction or concentrates the haemoglobin inside the cell triggers a crisis: hypoxia, dehydration, acidosis, fever and infection, cold-induced vasoconstriction, and physiologic stress.

Deoxygenation → HbS polymerises → red cell sickles and stiffens → microvascular occlusion + early destruction → ischaemic pain crisis and haemolytic anaemia

Repeated splenic infarction leaves the patient **functionally asplenic** by early childhood, which removes the body's main defence against encapsulated organisms. This is why vaccination and prompt antibiotics matter so much, and why fever is treated as an emergency rather than a symptom. Hydroxyurea works by raising fetal haemoglobin, which physically dilutes HbS and interrupts polymer formation.

RED FLAG

Fever in a sickle cell patient is sepsis until proven otherwise — they are functionally asplenic. And chest pain with fever, hypoxia, and a new pulmonary infiltrate is **acute chest syndrome**, the leading cause of death in sickle cell disease: it needs oxygen, hydration, analgesia, antibiotics, and often transfusion immediately, not a wait-and-see approach. Also act now on sudden focal neurological deficit (stroke), rapidly enlarging spleen with falling haemoglobin (splenic sequestration), and priapism.

PRACTICE

Q1. A 34-year-old woman with years of heavy menstrual bleeding has Hb 9.1 g/dL, MCV 74 fL, ferritin 6 ng/mL, and an elevated TIBC. Which mechanism explains her anaemia?

A. Hepcidin-driven sequestration of iron inside macrophages during chronic inflammation

B. Depleted iron stores limiting haem synthesis, so precursors divide again while waiting to fill with haemoglobin

C. Impaired DNA synthesis causing nuclear-cytoplasmic asynchrony in the erythroblast

D. Reduced beta-globin chain production with normal iron availability

Show answer

B is correct. Low ferritin means the storage tank is genuinely empty, and the high TIBC is the liver building extra transferrin because it senses iron starvation — that combination is diagnostic of true iron deficiency. Anaemia of chronic disease — the hepcidin-sequestration option — is the tempting distractor because serum iron is low in both, but there the iron is trapped rather than absent, so ferritin is normal or high and TIBC is LOW. Impaired DNA synthesis would make cells macrocytic, and thalassaemia trait would leave iron studies normal.

PRACTICE

Q2. Which laboratory combination best confirms that a patient's normocytic anaemia is due to haemolysis?

A. High reticulocytes, high LDH, high indirect bilirubin, low haptoglobin

B. Low reticulocytes, high LDH, high direct bilirubin, high haptoglobin

C. High reticulocytes, low LDH, high direct bilirubin, low haptoglobin

D. Low reticulocytes, low LDH, normal bilirubin, low haptoglobin

Show answer

A is correct. Early red cell destruction spills intracellular LDH and heme, and heme is catabolised to UNCONJUGATED (indirect) bilirubin because it has not yet been conjugated by the liver — so direct bilirubin (options 1 and 2) points to a hepatic or obstructive problem instead. Haptoglobin is consumed binding free plasma haemoglobin, so it falls; a rising haptoglobin is what you would see in inflammation, since it is also an acute-phase reactant. A LOW reticulocyte count would tell you the marrow is failing to produce rather than the periphery destroying.

PRACTICE

Q3. Which set of conditions most directly promotes HbS polymerisation and a vaso-occlusive crisis?

A. Hyperoxia, alkalosis, and volume expansion

B. Hypoxia, dehydration, and acidosis

C. Hypercalcaemia, hyperglycaemia, and hypertension

D. Iron overload, folate excess, and leukocytosis

Show answer

B is correct. Only deoxygenated HbS polymerises, so hypoxia supplies the trigger; acidosis right-shifts the oxyhaemoglobin curve and increases the deoxygenated fraction; and dehydration concentrates HbS inside the cell so polymer forms faster. option A is the exact opposite and describes the therapeutic goal — oxygenate, hydrate, correct acidosis. Iron overload is a consequence of chronic transfusion, not a sickling trigger, and folate is actually supplemented in these patients to support high red cell turnover.

26 Coagulation Patterns, DIC, and Thrombosis

> **Key idea —** Haemostasis happens in two stages, and every bleeding disorder is a failure of one of them. **Primary haemostasis** is platelets plugging the hole — it fails with mucosal, superficial, immediate bleeding (petechiae, gums, epistaxis, menorrhagia). **Secondary haemostasis** is the coagulation cascade weaving fibrin through that plug — it fails with deep, delayed, rebleeding-type haemorrhage (joints, muscle, retroperitoneum). Four numbers tell you which arm broke: **platelet count, PT/INR, aPTT, and fibrinogen**.

What each number is actually measuring

PT/INR tests the tissue-factor (extrinsic) pathway and the common pathway, and it is dominated by factor VII. **aPTT** tests the contact-activation (intrinsic) pathway — factors XII, XI, IX, and VIII — plus the common pathway. So an isolated aPTT prolongation points at VIII or IX (haemophilia) or heparin; an isolated PT prolongation points at VII (warfarin, vitamin K deficiency, early liver disease); and **both** prolonged means either the shared final pathway (X, V, prothrombin, fibrinogen) or a global catastrophe such as DIC, liver failure, or massive transfusion. **Fibrinogen** is the substrate itself and stays normal in almost everything except DIC and end-stage liver failure — which is exactly what makes it so useful. **D-dimer** is not a clotting test at all: it is a fibrin breakdown fragment, so it tells you clot has been formed and then dissolved.

Clinical picture	Platelets	PT / INR	aPTT	Fibrinogen	Mechanism to say out loud
Warfarin or vitamin K deficiency	Normal	High	Normal or slightly high	Normal	Factors II, VII, IX, X cannot be carboxylated; factor VII has the shortest half-life so the PT moves first
Unfractionated heparin	Normal (falls in HIT)	Normal	High	Normal	Antithrombin is accelerated against thrombin and factor Xa — an intrinsic-pathway effect
Haemophilia A or B	Normal	Normal	High	Normal	Isolated factor VIII or IX deficiency; deep bleeding into joints and muscle, not mucosal ooze
Immune thrombocytopenia (ITP)	Very low	Normal	Normal	Normal	Antibody-coated platelets are cleared by the spleen; the cascade is untouched, so only the count is abnormal
Von Willebrand disease	Normal count, poor function	Normal	Normal or high	Normal	Platelets cannot adhere to damaged endothelium; aPTT rises only when the carried factor VIII also falls
Liver failure	Low (splenic sequestration)	High	High	Low in late disease	The liver synthesises nearly every factor, so

					the PT/INR is the most sensitive real-time marker of synthetic function
DIC	Low and falling	High	High	Low	Simultaneous consumption of platelets and factors plus runaway fibrinolysis; D-dimer is markedly high

MEMORY HOOK The vitamin K-dependent factors are **1972**: factors **10, 9, 7, and 2** (plus proteins C and S). Warfarin blocks 1972; because factor VII has the shortest half-life, the **PT/INR** is what you monitor.

> **Pearl —** In liver disease the PT/INR rises **before** albumin falls, because factor VII turns over in about six hours while albumin takes about three weeks. The INR in a cirrhotic patient is therefore a live readout of hepatic synthetic function, not a warfarin number — and it is one reason the INR is built into the MELD score.

Disseminated intravascular coagulation

DIC is not a disease; it is a complication of something else — sepsis (the most common), obstetric catastrophe (abruption, amniotic fluid embolism, retained fetus), major trauma or crush injury, burns, malignancy, and severe transfusion reactions. Each floods the circulation with **tissue factor**, generating thrombin everywhere at once. Microthrombi lodge in the small vessels of the kidneys, lungs, brain, and digits, causing organ ischaemia, and they shear passing red cells into **schistocytes**. Meanwhile the platelets and clotting factors used to build all that fibrin are consumed faster than the liver can replace them, and plasmin activates to dissolve the clot burden. The result is the paradox students must be able to explain: **the patient is clotting and haemorrhaging simultaneously**.

Systemic tissue factor release → uncontrolled thrombin generation → widespread microthrombi → consumption of platelets and factors + secondary fibrinolysis → organ ischaemia AND uncontrollable bleeding

RED FLAG

A septic or post-partum patient who starts **oozing from every IV site, the gums, and old puncture wounds**, with a falling platelet count, prolonged PT and aPTT, falling fibrinogen and a soaring D-dimer, is in DIC. Escalate immediately — and remember the definitive treatment is **eliminating the trigger** (source control of sepsis, delivery of the fetus); products are supportive only.

Virchow's triad — why thrombi form where they do

Pathological thrombosis needs at least one of three conditions. **Stasis**: immobility, long flights, atrial fibrillation, varicosities, or a dilated failing ventricle — slow blood lets clotting factors accumulate locally instead of being washed away. **Endothelial injury**: surgery, indwelling catheters, trauma, smoking, hypertension, hyperglycaemia, or a ruptured atherosclerotic plaque — an intact endothelium is actively anticoagulant, so exposing subendothelial collagen and tissue factor removes that protection. **Hypercoagulability**: pregnancy and the post-partum period, oestrogen-containing contraceptives, malignancy, factor V Leiden, antiphospholipid syndrome, nephrotic syndrome (antithrombin III is lost in the urine), and the post-operative acute-phase rise in fibrinogen.

MEMORY HOOK Virchow's triad: **SIT still** — **S**tasis, **I**njury to endothelium, **T**hrombophilia. The classic post-operative hip replacement patient collects all three simultaneously, which is precisely why that is the textbook DVT scenario.

Pearl — The triad also explains drug choice. **Venous** thrombi form in slow, low-shear blood and are fibrin- and red-cell-rich, so you fight them with **anticoagulants** (heparin, warfarin, DOACs). **Arterial** thrombi form on ruptured plaque in fast, turbulent flow and are platelet-rich, so you fight them with **antiplatelets** (aspirin, P2Y12 inhibitors). Same word, two different clots, two different drug classes.

RED FLAG

Heparin-induced thrombocytopenia is the trap: platelets fall 5 to 10 days into heparin therapy, yet the patient **clots** rather than bleeds, because the antibody-PF4-heparin complex activates the remaining platelets. Stop all heparin, do **not** transfuse platelets (you are adding fuel), and switch to a non-heparin anticoagulant such as a direct thrombin inhibitor.

PRACTICE

Q1. A patient in septic shock develops oozing from every venipuncture site. Platelets have fallen from 250,000 to 38,000, PT and aPTT are both prolonged, fibrinogen is low, and D-dimer is markedly elevated. Schistocytes appear on the smear. What is happening?

A. Immune thrombocytopenic purpura triggered by infection

B. Disseminated intravascular coagulation

C. Haemophilia A unmasked by critical illness

D. Von Willebrand disease with stress-related decompensation

Show answer

B is correct. Only DIC prolongs BOTH pathway times while simultaneously consuming platelets and fibrinogen and raising D-dimer — that combination is the fingerprint of thrombin generated everywhere at once, followed by fibrinolysis. ITP is the plausible distractor because thrombocytopenia dominates the picture, but in ITP the cascade is untouched, so PT, aPTT, and fibrinogen are all normal. Haemophilia gives an isolated prolonged aPTT with normal platelets, and von Willebrand disease causes mucocutaneous bleeding with a largely normal panel.

PRACTICE

Q2. A 6-year-old boy has recurrent bleeding into his knees. Platelet count is normal, PT is normal, aPTT is prolonged, and fibrinogen is normal. Which mechanism fits?

A. Deficiency of a vitamin K-dependent factor from poor nutrition

B. Deficiency of factor VIII or IX in the intrinsic pathway

C. Antibody-mediated platelet destruction in the spleen

D. Consumption of fibrinogen by ongoing intravascular clotting

Show answer

B is correct. The aPTT interrogates the intrinsic pathway (XII, XI, IX, VIII), so an isolated prolongation with a normal PT and a normal platelet count localises the defect to factor VIII (haemophilia A) or IX (haemophilia B) — and deep joint bleeding is the classic secondary-haemostasis pattern. Vitamin K deficiency is tempting, but it depletes factors II, VII, IX, and X, and because factor VII has the shortest half-life the PT/INR would rise first. Platelet destruction would drop the count and cause mucosal rather than joint bleeding, and fibrinogen consumption would show a low fibrinogen with both times prolonged.

PRACTICE

Q3. Three days after a total hip replacement, a patient develops a unilateral swollen, painful calf. Which explanation best accounts for the thrombus?

A. Platelet aggregation on a ruptured atherosclerotic plaque, best treated with aspirin alone

B. Simultaneous venous stasis from immobility, surgical endothelial trauma, and the post-operative acute-phase rise in clotting factors

C. Isolated endothelial injury from the incision, with no contribution from flow or coagulability

D. Impaired fibrinolysis indicated by the low D-dimer typical of acute DVT

Show answer

B is correct. The post-operative orthopaedic patient satisfies all three arms of Virchow's triad at once, which is why this population has the highest DVT risk and receives routine prophylaxis. option A describes an ARTERIAL thrombus — plaque rupture with a platelet-rich clot — which is why antiplatelet therapy is the wrong tool here; venous thrombi are fibrin-rich and require anticoagulation. option D also inverts the physiology: D-dimer is a fibrin degradation product and is typically ELEVATED once a clot is forming and lysing.

27 Cardiac Output, Ischaemia, Heart Failure, and Shock

> **Key idea —** Every cardiovascular disease on this exam reduces to one equation — **CO = HR x SV**, with stroke volume set by **preload, afterload, and contractility**, and perfusion pressure set by **MAP = CO x SVR**. Disease either steals one of those variables, or the body compensates so aggressively for the loss that the **compensation itself becomes the disease**.

The determinants and their levers

Determinant	What it is	Raised by	Lowered by
Preload	End-diastolic stretch on the myocardial fibres — essentially venous return	Fluid loading, venous constriction, regurgitant valves, renal salt and water retention	Diuretics, nitrates (venodilation), haemorrhage, tachycardia that shortens filling time
Afterload	The wall tension the ventricle must generate to open the semilunar valve and eject	Hypertension, aortic stenosis, vasoconstriction, high SVR	ACE inhibitors and ARBs, arterial vasodilators, the pathologically low SVR of sepsis
Contractility	Force of contraction independent of stretch — the calcium available to the sarcomere	Sympathetic stimulation, catecholamines, digoxin, dobutamine	Myocardial infarction, cardiomyopathy, acidosis, hypoxia, beta-blockade
Heart rate	Beats per minute; the multiplier on stroke volume	Sympathetic tone, fever, hypovolaemia, anaemia, hyperthyroidism	Vagal tone, beta-blockers, conduction block — and note extreme tachycardia lowers CO by abolishing diastolic filling and coronary perfusion

> **Pearl —** In a healthy heart, more preload means more output (Frank-Starling). In a failing heart the curve is **flattened**, so extra preload buys no stroke volume at all — it simply raises filling pressure, which is transmitted backwards into the pulmonary capillaries. That is why the same litre of saline that rescues a hypovolaemic patient drowns a heart-failure patient.

From atherosclerosis to infarction

It begins with **endothelial injury** from hypertension, smoking, hyperglycaemia, or high LDL. Damaged endothelium becomes permeable and sticky: LDL enters the intima and is oxidised, monocytes migrate in and become macrophages that gorge on oxidised LDL to form **foam cells**, and the resulting fatty streak recruits smooth muscle that lays down a **fibrous cap** over a soft lipid core. As long as that cap holds, the plaque only narrows the lumen — the patient has symptoms only when demand outruns a fixed supply, which is **stable angina** and generally requires around 70 percent stenosis. The catastrophe is **rupture**: a thin cap tears, exposing the thrombogenic lipid core and tissue factor to flowing blood, platelets adhere and aggregate within seconds, and a fibrin-platelet thrombus can occlude the vessel completely.

Endothelial injury → LDL entry and oxidation → foam cells and fatty streak → fibrous plaque → cap rupture → platelet-fibrin thrombus → coronary occlusion → myocardial necrosis

> **Pearl —** The plaque that kills is often **not** the tightest one. Severely stenotic plaques tend to be old, calcified, and stably capped, and they have had time to grow collaterals. The lesion that causes sudden death is frequently a soft, lipid-rich, thin-capped plaque causing only modest stenosis — which is why a patient can pass a stress test and infarct weeks later, and why statins earn their survival benefit by **stabilising** plaque, not just shrinking it.

Angina versus infarction, and what troponin actually tells you

Feature	Stable angina	Unstable angina	NSTEMI	STEMI
Coronary lesion	Fixed plaque, roughly 70 percent or greater stenosis	Plaque disruption with partial, intermittent thrombus	Partial or transient occlusion	Complete, persistent occlusion of an epicardial artery
When pain occurs	Predictably with exertion or emotion	At rest, newly onset, or accelerating from a known pattern	At rest, prolonged	At rest, prolonged and crushing, with diaphoresis and nausea
Response to rest or nitrates	Relieved within minutes	Incomplete or fleeting relief	Not relieved	Not relieved
ECG	Normal at rest; ST depression only during the episode	ST depression or T-wave inversion, transient	ST depression or T-wave inversion; no ST elevation	ST elevation, then pathological Q waves
Troponin	Negative	Negative — this is the defining feature	Elevated	Elevated
Tissue outcome	Reversible ischaemia, no cell death	Reversible ischaemia, no cell death	Subendocardial (partial-thickness) necrosis	Transmural necrosis; emergent reperfusion is the goal

Cardiac troponin I and T are structural proteins of the myocyte contractile apparatus. They can only reach the bloodstream if the cell membrane has failed — that is, if the cell has **died**. Troponin therefore draws the line between unstable angina (ischaemia, still reversible) and NSTEMI (necrosis, irreversible), even though the two look identical at the bedside. It rises within about 3 to 4 hours, peaks around 24 hours, and stays elevated for 7 to 14 days, which is excellent for late diagnosis but poor for detecting re-infarction. Critically, troponin means **myocyte injury**, not **coronary occlusion**: it also rises in sepsis, pulmonary embolism, myocarditis, decompensated heart failure, and renal failure.

RED FLAG

Crushing substernal chest pain with **ST elevation** is a STEMI — an artery is completely shut and myocardium is dying by the minute, so reperfusion is measured in minutes, not hours. Equally dangerous is the presentation you almost miss: women, older adults, and people with diabetes frequently present with **dyspnoea, unexplained fatigue, epigastric discomfort, or nausea**, and diabetic autonomic neuropathy can make an infarct entirely painless.

Left versus right heart failure

Heart failure is best understood as a plumbing problem: whichever ventricle fails, blood dams up **behind** it while output falls in front of it. The left ventricle receives from the lungs, so left failure floods the pulmonary circuit. The right ventricle receives from the body, so right failure floods the systemic veins.

Feature	Left-sided failure	Right-sided failure
Where blood backs up	Into the pulmonary veins and capillaries	Into the systemic venous system
Cardinal findings	Dyspnoea, orthopnoea, paroxysmal nocturnal dyspnoea, bibasilar crackles, S3 gallop, frothy pink-tinged sputum, nocturia	Jugular venous distension, dependent pitting oedema, hepatomegaly with right upper quadrant pain, ascites, rapid weight gain
Usual cause	Hypertension, ischaemic heart disease, aortic or mitral valve disease	Left-sided failure first and foremost, then cor pulmonale from COPD, pulmonary hypertension, or recurrent PE
What to teach the patient to watch	Increasing breathlessness on exertion and needing extra pillows at night	Daily weights — over 2 to 3 lb in a day or 5 lb in a week — plus tighter shoes and rings

MEMORY HOOK **L**eft failure floods the **L**ungs; **R**ight failure floods the **R**est of the body. And because the right ventricle's job is to push into the lungs, the most common cause of right heart failure is a failing left heart raising pulmonary pressures.

Systolic versus diastolic failure

Feature	Systolic failure (HFrEF)	Diastolic failure (HFpEF)
Core defect	The ventricle cannot contract and empty — a pump-strength problem	The ventricle cannot relax and fill — a compliance and stiffness problem
Ejection fraction	Reduced, 40 percent or less	Preserved, 50 percent or more — the EF looks reassuring while the patient is frankly congested
Ventricular geometry	Dilated, thin-walled chamber (eccentric hypertrophy)	Thick-walled chamber with a small cavity (concentric hypertrophy)
Typical causes	Prior myocardial infarction, dilated cardiomyopathy, chronic volume overload, myocarditis	Long-standing hypertension, aortic stenosis, hypertrophic cardiomyopathy, ageing, restrictive disease
Why congestion develops	Blood left behind after each systole raises end-systolic then filling pressures	Very high pressure is required to fill a stiff chamber, and that pressure is transmitted backwards
Rate tolerance	Tolerates modest tachycardia reasonably	Tachycardia is poorly tolerated — shortening diastole removes exactly the filling time this ventricle depends on

When compensation becomes the pathology

A falling cardiac output triggers three ancient reflexes that were designed for haemorrhage, not for a chronically weak pump. **Sympathetic activation** raises heart rate and vasoconstricts — which raises afterload and myocardial oxygen demand while shortening the diastole in which coronary perfusion actually occurs. **RAAS activation** retains salt and water, adding preload to a ventricle already stretched past its useful limit, and angiotensin II and aldosterone drive interstitial **fibrosis**. **Remodelling** follows: concentric hypertrophy thickens a wall that then outgrows its own blood supply and stiffens, while eccentric dilation stretches sarcomeres past optimal actin-myosin overlap so each fibre generates less force. Every compensation raises wall stress, and wall stress is the thing killing the ventricle.

Falling cardiac output → sympathetic + RAAS activation → more preload, more afterload, remodelling and fibrosis → higher wall stress and oxygen demand → further fall in output (self-

reinforcing spiral)

> **Pearl —** This is why the drugs that **blunt** compensation prolong life — beta-blockers, ACE inhibitors, ARBs and ARNIs, aldosterone antagonists, SGLT2 inhibitors — while drugs that simply flog the myocardium harder relieve symptoms without improving survival. It also explains **BNP**: stretched ventricular myocytes release it as the body's own counter-hormone (natriuresis and vasodilation), so we measure it as a stretch gauge. A low BNP in acute dyspnoea essentially rules heart failure out as the cause.

The shock categories

All shock is one thing — **oxygen delivery below cellular demand** — which forces anaerobic metabolism and generates **lactate**. What separates the categories is which part of the delivery system failed, and the haemodynamic profile tells you which.

Type	Core defect	Preload (CVP/JVP)	Cardiac output	SVR	Bedside clue
Hypovolaemic	Not enough circulating volume — haemorrhage, burns, vomiting, diuresis, DKA	Low	Low	High (compensatory)	Cold, pale, clammy; tachycardia and a narrowing pulse pressure appear well before the blood pressure falls
Cardiogenic	Pump failure — the volume is present but cannot be moved (large MI, arrhythmia, acute valve rupture)	High	Low	High	Cold and mottled WITH crackles and JVD — the only shock that is congested and underperfused at once
Obstructive	Mechanical block to filling or ejection (tamponade, tension pneumothorax, massive PE)	High	Low	High	JVD with clear lungs; hunt for muffled heart sounds, tracheal deviation, or sudden pleuritic dyspnoea
Distributive	Vessels dilate and leak, so volume is misplaced rather than lost (sepsis, anaphylaxis, neurogenic)	Low	High early, low late	Low	Warm, flushed, bounding pulses early; neurogenic is the exception — bradycardia with warm, dry skin

MEMORY HOOK The skin sorts shock at the bedside: **cold and clammy** means a volume or pump problem (hypovolaemic, cardiogenic, obstructive, all with high SVR); **warm and flushed** means distributive — until it decompensates late and goes cold too.

RED FLAG

A rising lactate with tachycardia and a **narrowing pulse pressure** in a patient whose blood pressure still reads normal is **compensated shock**. Vasoconstriction props up the systolic reading until roughly 30 to 40 percent of volume is gone, and young patients compensate longest and then crash hardest. Act on the heart rate, the pulse pressure, the mental status, and the lactate — not on the blood pressure, which fails last.

PRACTICE

Q1. A patient with an ejection fraction of 25 percent is given a rapid 1-litre saline bolus for a soft blood pressure and within 20 minutes becomes hypoxic with diffuse crackles. What explains the deterioration?

A. The bolus diluted haemoglobin enough to cause anaemic hypoxia

B. The failing ventricle sits on a flattened Frank-Starling curve, so the added preload raised filling pressure without raising stroke volume, and that pressure was transmitted back to the pulmonary capillaries

C. The bolus raised systemic vascular resistance and therefore afterload, blocking ejection

D. The increased contractility from the volume load raised myocardial oxygen demand and caused ischaemia

Show answer

B is correct. In a healthy ventricle, added stretch produces added stroke volume; in a severely failing ventricle the curve is flat, so the volume has nowhere to go forward and instead raises left atrial and then pulmonary capillary hydrostatic pressure until fluid is forced into the alveoli. option C is the classic trap — a saline bolus does not meaningfully raise SVR, and it is preload, not afterload, that was manipulated. Dilutional anaemia from a single litre is far too small and far too slow to explain acute pulmonary oedema.

PRACTICE

Q2. A patient has 30 minutes of crushing chest pain at rest that is not relieved by nitroglycerin. The ECG shows ST depression and T-wave inversion with no ST elevation, and the troponin is elevated. Which is correct?

A. Stable angina, because the ECG lacks ST elevation

B. Unstable angina, because the coronary artery is only partially occluded

C. NSTEMI, because the elevated troponin proves myocyte necrosis without transmural injury

D. STEMI, because the pain lasted longer than 20 minutes and did not respond to nitrates

Show answer

C is correct. Troponin is a structural protein that can only escape the myocyte if the membrane has failed, so its elevation is the exact boundary between reversible ischaemia and infarction. Unstable angina is the tempting answer because the presentation is clinically indistinguishable — rest pain, unrelieved, with ST depression — but by definition unstable angina has a NEGATIVE troponin. STEMI requires ST elevation reflecting full-thickness injury, and duration of pain alone never defines a STEMI. Stable angina is excluded by pain at rest without relief.

PRACTICE

Q3. A patient with a complete cervical spinal cord injury has a blood pressure of 78/45, a heart rate of 52, and warm, dry, pink skin. Which mechanism is responsible?

A. Occult haemorrhage causing hypovolaemic shock

B. Loss of sympathetic outflow causing distributive (neurogenic) shock with unopposed vagal tone

C. Pump failure from myocardial contusion causing cardiogenic shock

D. Mechanical obstruction to ventricular filling from a tension pneumothorax

Show answer

B is correct. A high cord injury severs the sympathetic chain, so arterioles and veins lose all vasomotor tone — SVR collapses and blood pools peripherally — while the vagus is left unopposed. That gives the one combination unique in all of shock: hypotension WITH bradycardia and warm, dry skin. Occult haemorrhage is the crucial distractor in a trauma patient, but hypovolaemic shock drives a compensatory tachycardia with cold, clammy, vasoconstricted skin. Cardiogenic and obstructive shock both raise venous pressures and produce JVD with cold, mottled skin.

RAPID REVIEW

Modules 11–12 — Renal & Respiratory

28 Renal Function, GFR, and Acute Kidney Injury

> **Key idea —** Glomerular filtration is a **hydraulic** event, not a chemical one. Anything that lowers the pressure inside the glomerular capillary lowers GFR, and because surviving nephrons hyperfilter to cover for dead ones, serum creatinine stays deceptively normal until roughly half the filtering capacity is already gone.

Where GFR actually comes from

Each kidney holds about a million nephrons, and the glomerulus is the one capillary bed in the body sandwiched between two arterioles rather than between an arteriole and a venule. That arrangement is the whole trick: the kidney can tune inflow and outflow independently. Net filtration pressure is glomerular hydrostatic pressure (about 55 mmHg) minus capsular hydrostatic pressure (about 15) minus glomerular oncotic pressure (about 30), leaving roughly 10 mmHg of usable driving force and a normal GFR near 125 mL/min. To defend that number when perfusion drops, the kidney dilates the **afferent** arteriole using locally made prostaglandins and constricts the **efferent** arteriole using angiotensin II, effectively damming the blood inside the tuft.

This is why two very common drugs become dangerous in a dehydrated patient. NSAIDs block prostaglandin synthesis, so the afferent arteriole cannot dilate; ACE inhibitors and ARBs block angiotensin II, so the efferent arteriole cannot constrict. Either alone is usually tolerated. Together, in someone who is already volume down, both autoregulatory arms are amputated and glomerular pressure collapses. Downstream, the proximal tubule reclaims about 65 percent of the filtrate plus all the glucose, the loop of Henle builds the medullary concentration gradient (the target of the potent loop diuretics), the distal tubule fine-tunes sodium and calcium, and the collecting duct is the hormone-controlled final say, where ADH inserts aquaporins and aldosterone drives the sodium-potassium exchange.

Hormone (source)	Trigger for release	Action	Consequence when it is lost
Renin (juxtaglomerular cells)	Fall in renal perfusion pressure, low distal NaCl, sympathetic stimulation	Starts RAAS: angiotensin II causes vasoconstriction, efferent constriction, aldosterone and ADH release	Deficiency gives hypotension and salt wasting; inappropriate excess (renal artery stenosis) gives secondary hypertension
Erythropoietin (peritubular interstitial cells)	Tissue hypoxia sensed in the renal cortex	Drives red cell production in bone marrow	Normocytic normochromic anaemia of CKD: fatigue, pallor, exercise intolerance, with normal iron studies
Calcitriol via 1-alpha-hydroxylase (proximal tubule)	PTH, low serum phosphate	Converts vitamin D to its active form so the gut can absorb calcium	Hypocalcaemia, then secondary hyperparathyroidism and renal osteodystrophy
Prostaglandins PGE2 and PGI2 (medulla, glomerulus)	Low perfusion, angiotensin II itself	Dilate the afferent arteriole to protect GFR under stress	Blocked by NSAIDs: unopposed afferent constriction and prerenal AKI
Kidney as target, not source: ADH, aldosterone, PTH	Osmolality, angiotensin II and potassium, low calcium	Water reabsorption, sodium retention with potassium excretion, calcium reclamation and phosphate excretion	In CKD the hormones are present but the tubule cannot answer them, producing resistance rather than deficiency

MEMORY HOOK The kidney does **A WET BED**: **A**cid-base balance, **W**ater balance, **E**lectrolytes, **T**oxin (waste) removal, **B**lood pressure control, **E**rythropoietin, vitamin **D** activation. Every complication of chronic kidney disease is one of those seven jobs failing.

Fall in renal perfusion → renin → angiotensin II → efferent constriction plus aldosterone and ADH → GFR defended, but the urine becomes concentrated and sodium-poor

The three acute kidney injuries

AKI is defined as a creatinine rise of at least 0.3 mg/dL within 48 hours, or 1.5 times baseline within 7 days, or urine output under 0.5 mL/kg/hr for 6 hours. Classifying it is simply asking where the problem sits relative to the nephron. **Prerenal** is a perfusion problem with structurally intact, starving tubules, so those tubules hoard sodium and water furiously. **Intrarenal**, most often acute tubular necrosis, means the tubular cells are dead or dying, so they can neither reclaim sodium nor build a concentration gradient. **Postrenal** is a plumbing blockage that transmits back-pressure up to the glomerulus and cancels the filtration gradient.

Feature	Prerenal azotaemia	Intrarenal (acute tubular necrosis)	Postrenal (obstructive)
Core problem	Not enough blood reaching an intact nephron	Tubular epithelium is necrotic and non-functional	Outflow blocked; back-pressure abolishes the filtration gradient
BUN to creatinine ratio	Greater than 20:1	Roughly 10:1 to 15:1	High early, falls toward 15:1 as tubules are damaged
Urine sodium	Under 20 mEq/L	Over 40 mEq/L	Variable; over 40 once tubules are injured
FENa	Under 1 percent	Over 2 percent	Under 1 percent early, over 2 percent if prolonged
Urine osmolality	Over 500 mOsm/kg, concentrated	Around 300 mOsm/kg, isosthenuric (same as plasma)	Variable, trends toward isosthenuric
Urine sediment	Bland, occasional hyaline casts	Muddy brown granular casts and renal tubular epithelial cells	Bland; crystals or red cells if stone
Typical causes	Haemorrhage, vomiting or diarrhoea, heart failure, sepsis, NSAIDs plus ACE inhibitor	Prolonged ischaemia, aminoglycosides, IV contrast, myoglobin from rhabdomyolysis	Prostatic hypertrophy, stones, pelvic tumour, blocked or kinked catheter
Reversibility	Rapid and complete with volume restoration	Days to weeks, sometimes incomplete	Complete if relieved quickly; permanent scarring if left for weeks

Pearl — The lab indices all measure one thing: is the tubule still working? Urea is passively dragged back into the blood alongside sodium and water, while creatinine is filtered and essentially abandoned. So when a starving prerenal tubule reabsorbs avidly, urea is dragged back with it and BUN climbs out of proportion to creatinine, giving that ratio above 20:1. When the tubule is dead, nothing is dragged back and the ratio normalises. FENa quantifies the same idea directly: FENa equals urine sodium times plasma creatinine, divided by plasma sodium times urine creatinine, times 100. Under 1 percent means the tubule is still clinging to sodium and is therefore alive.

Acute tubular necrosis runs a predictable three-act course, and each act has its own danger. The **oliguric phase** lasts one to three weeks and is where patients die: potassium, acid and fluid accumulate with nowhere to go. The **diuretic phase** is deceptive good news, because filtration recovers before reabsorption does, so the patient pours out litres of dilute urine and can crash into hypovolaemia and hypokalaemia while everyone is celebrating. The **recovery phase** takes months, and tubular concentrating ability is the last function to return. Post-obstructive diuresis after relieving a blocked bladder behaves exactly the same way.

RED FLAG

Anuria plus a palpable or bladder-scanned distended bladder is the one AKI you can cure in sixty seconds: place a catheter. And in any AKI, a potassium above 6.5 mEq/L, peaked T waves, a widening QRS or a sine-wave pattern is an immediate arrest risk. Give calcium gluconate to stabilise the myocardium first, then insulin with glucose to shift potassium intracellularly, then a binder or dialysis to actually remove it.

PRACTICE

Q1. A 78-year-old with three days of vomiting and diarrhoea takes lisinopril daily and has been using ibuprofen for knee pain. Creatinine has risen from 0.9 to 2.1 mg/dL. Urine sodium is 12 mEq/L, FENa 0.4 percent, BUN to creatinine ratio 28:1, urine osmolality 620 mOsm/kg, sediment bland. What best explains this injury?

A. Both arms of glomerular autoregulation are blocked: the NSAID prevents afferent dilation while the ACE inhibitor prevents efferent constriction, so glomerular pressure collapses in a volume-depleted patient

B. Direct necrosis of proximal tubular epithelium caused by ibuprofen metabolites concentrated in the medulla

C. Obstruction of the collecting system by uric acid crystals precipitated during dehydration

D. Immune-complex deposition along the glomerular basement membrane triggered by the drug combination

Show answer

A is correct. Every index says the tubule is alive and working overtime: FENa under 1 percent, urine sodium under 20, urine osmolality over 500, and a BUN to creatinine ratio over 20:1 from urea being dragged back with sodium and water. That is prerenal physiology, and the mechanism is loss of autoregulation. Tubular necrosis is the tempting distractor because NSAIDs are nephrotoxic, but dead tubules cannot concentrate urine to 620 mOsm/kg or hold urine sodium at 12; ATN would show FENa over 2 percent, isosthenuric urine near 300, and muddy brown casts.

PRACTICE

Q2. Three days after a prolonged hypotensive episode in the operating room, a patient's urine output falls and creatinine climbs. Urine sodium is 62 mEq/L, FENa 3.4 percent, urine osmolality 295 mOsm/kg, and the sediment shows muddy brown granular casts. Which statement correctly links the findings to the mechanism?

A. Ongoing hypovolaemia is driving avid sodium reabsorption, so aggressive fluid resuscitation should reverse it within hours

B. Ischaemic tubular necrosis has destroyed both sodium reabsorption and the medullary concentrating mechanism, so sodium is wasted and urine osmolality equals plasma

C. Back-pressure from an obstructed ureter has abolished the filtration gradient, and relieving it will normalise the indices

D. Acute interstitial nephritis from a perioperative cephalosporin is producing an allergic tubular injury

Show answer

B is correct. Urine that is isosthenuric at 295 mOsm/kg means the kidney has completely lost the ability to make urine different from plasma, and a FENa above 2 percent with urine sodium above 40 means sodium is escaping unreclaimed. Muddy brown granular casts are literally sloughed tubular epithelial cells packed in Tamm-Horsfall protein, which is the histologic signature of ATN. Interstitial nephritis is plausible after surgery but would show white cell casts, sterile pyuria, eosinophils, fever or rash, not muddy brown casts; prerenal physiology is excluded by the indices pointing in exactly the opposite direction.

PRACTICE

Q3. A patient with long-standing chronic kidney disease has haemoglobin 9.2 g/dL with MCV 88 fL and entirely normal ferritin and transferrin saturation, alongside calcium 7.6 mg/dL, phosphate 6.4 mg/dL and PTH 480 pg/mL. Which pair of endocrine failures explains this picture?

A. Loss of erythropoietin production and loss of 1-alpha-hydroxylation of vitamin D

B. Loss of renin secretion and consequent aldosterone deficiency

C. Occult iron deficiency plus an autonomous parathyroid adenoma

D. Downregulation of collecting duct ADH receptors and impaired prostaglandin synthesis

Show answer

A is correct. A normocytic normochromic anaemia with genuinely normal iron studies points away from iron deficiency and toward absent erythropoietin from damaged peritubular interstitial cells. The mineral panel tells the second story: the diseased proximal tubule cannot run 1-alpha-hydroxylase, so calcitriol falls, the gut absorbs less calcium, and retained phosphate binds what calcium remains. Low calcium then whips the parathyroids, which is why PTH is 480. The adenoma distractor fails on the numbers, since primary hyperparathyroidism raises calcium and wastes phosphate, giving high calcium with low phosphate rather than the reverse.

29 Glomerular Disease and Chronic Kidney Disease

> **Key idea —** Glomerular disease splits on what the injury does to the filter. **Inflammation** that plugs and tears capillary walls lets red cells escape and drops GFR, producing nephritic syndrome. **Loss of the charge and size barrier** without inflammation lets albumin pour out through a filter that is still wide open, producing nephrotic syndrome. Every downstream finding, the oedema pattern, the blood pressure, the lipids, the clots, follows from that one distinction.

Nephritic versus nephrotic

The filter has three layers: a fenestrated endothelium, a glomerular basement membrane studded with negatively charged heparan sulphate, and podocyte foot processes joined by slit diaphragms made of nephrin. In **nephritic** disease, immune complexes lodge in the capillary wall, complement is fixed, and inflammatory cells proliferate into the tuft. The wall becomes leaky enough for red cells to squeeze through but the lumen also narrows, so filtration falls, sodium and water are retained, and blood pressure rises. In **nephrotic** disease there is no inflammatory infiltrate at all; the podocytes simply efface and the negative charge barrier is lost, so albumin, which is small and negatively charged and normally repelled, streams into the urine at more than 3.5 grams a day.

Feature	Nephritic syndrome	Nephrotic syndrome
Lesion	Inflammatory and proliferative; immune complexes, complement activation	Non-inflammatory podocyte injury; loss of charge and size selectivity
Proteinuria	Present but under 3.5 g per day	Over 3.5 g per day, the defining feature
Urine sediment	Red cells, dysmorphic red cells, and red cell casts (essentially diagnostic)	Fatty casts, oval fat bodies, maltese cross under polarised light, few cells
Blood pressure	Hypertensive from salt and water retention	Normal or low; the patient is intravascularly depleted
Oedema pattern	Periorbital and facial, worst on waking, from volume overload	Generalised and dependent, progressing to anasarca, from low oncotic pressure
GFR	Falls, with azotaemia and oliguria	Often preserved early in the course
Serum albumin	Near normal	Low, typically under 3 g/dL
Classic causes	Post-streptococcal GN, IgA nephropathy, lupus nephritis, rapidly progressive GN and Goodpasture	Minimal change disease (children), focal segmental glomerulosclerosis, membranous nephropathy, diabetic nephropathy, amyloidosis
Signature complications	Hypertensive encephalopathy, pulmonary oedema, rapid loss of function	Thrombosis, infection with encapsulated organisms, hyperlipidaemia

MEMORY HOOK Nephr**I**tic carries the **I**: **I**nflammation and haematur**I**a, so think blood and a rising blood pressure. Nephr**O**tic carries the **O**s: pr**O**teinuria, **O**edema, hyp**O**albuminaemia and l**O**ss of oncotic pressure.

> **Pearl —** The two commonest causes of glomerular haematuria are separated by a calendar, not a microscope. Post-streptococcal glomerulonephritis appears **one to three weeks after** a pharyngitis (or two to six weeks after impetigo), with tea-coloured urine, a depressed C3, a positive ASO titre, and near-universal recovery in children. IgA nephropathy produces haematuria **during or within a day or two of** the upper respiratory infection, with a normal C3, and it recurs with every subsequent infection. The latency period, long versus none, is the discriminator examiners test.

> **Pearl —** A leaky glomerulus is not selective about which protein it loses; it loses everything near albumin's size. That single fact explains the whole nephrotic complication list. Antithrombin III washes out, natural anticoagulation fails, and the patient becomes hypercoagulable, with renal vein thrombosis as the classic event (sudden flank pain, worsening haematuria, an abrupt drop in function). Immunoglobulin G washes out, so encapsulated organisms cause peritonitis and pneumonia. Meanwhile the liver, sensing collapsing oncotic pressure, ramps up synthesis of all proteins indiscriminately, including lipoproteins, which is why cholesterol soars and fat appears in the urine.

Chronic kidney disease: what dies when nephrons die

CKD means a GFR under 60 mL/min for three months or more, or persistent markers of damage such as albuminuria, and it is staged G1 through G5 with dialysis usually considered near a GFR of 15. Diabetes and hypertension together account for roughly two-thirds of cases. The engine of progression is **hyperfiltration**: as nephrons drop out, the survivors hypertrophy and filter harder to hold total GFR steady, which is exactly why creatinine looks normal for so long, and exactly why the disease is self-perpetuating. That elevated pressure inside the surviving tufts sclerose them in turn. This is the entire rationale for ACE inhibitors, ARBs and SGLT2 inhibitors in CKD: they lower intraglomerular pressure, accepting a small immediate dip in GFR in exchange for preserving nephrons over years.

Consequence	Mechanism	What you see clinically
Anaemia	Loss of erythropoietin from peritubular interstitial cells	Normocytic normochromic anaemia with normal iron studies; fatigue, pallor, dyspnoea on exertion
Mineral and bone disorder	Phosphate retention plus loss of calcitriol lowers calcium, driving secondary hyperparathyroidism	Bone pain, fractures, renal osteodystrophy, and calcium-phosphate deposition in vessels and soft tissue
Metabolic acidosis	Failure to excrete hydrogen ions and regenerate bicarbonate	Elevated anion gap, Kussmaul respirations, bone buffering that worsens osteodystrophy, muscle wasting
Hyperkalaemia	Reduced distal potassium secretion, worsened by ACE inhibitors, ARBs and potassium-sparing diuretics	Peaked T waves, widened QRS, muscle weakness, cardiac arrest
Volume overload and hypertension	Sodium and water retention with inappropriate RAAS activation	Peripheral oedema, pulmonary oedema, left ventricular hypertrophy, heart failure
Uraemia	Accumulation of nitrogenous waste and middle molecules	Nausea, anorexia, metallic taste, pruritus, uraemic frost, asterixis, encephalopathy, pericardial friction rub
Bleeding tendency	Uraemic toxins impair platelet adhesion and aggregation	Ecchymoses, epistaxis and gastrointestinal bleeding despite a normal platelet count
Accelerated cardiovascular disease	Hypertension, calcification, hyperlipidaemia, chronic inflammation	Cardiovascular events, not renal failure itself, are the leading cause of death in CKD

Nephron loss → phosphate retention plus loss of calcitriol → hypocalcaemia → secondary hyperparathyroidism → bone resorption and vascular calcification

RED FLAG

A pericardial friction rub with pleuritic chest pain in a uraemic patient is uraemic pericarditis, which can bleed into the sac and tamponade; it is an absolute indication for urgent dialysis. Remember the emergency dialysis list as **AEIOU**: severe **A**cidosis, refractory **E**lectrolyte disturbance (potassium), **I**ngestion of a dialysable toxin, fluid **O**verload with pulmonary oedema, and symptomatic **U**raemia meaning pericarditis or encephalopathy.

PRACTICE

Q1. A 7-year-old develops cola-coloured urine, periorbital puffiness and blood pressure 148/94 about ten days after a sore throat. Urinalysis shows red cell casts and 1.2 g of protein per day; C3 is low and ASO titre elevated. Which diagnosis and mechanism fit best?

A. Minimal change disease, with podocyte effacement causing selective albumin loss

B. IgA nephropathy, with mesangial IgA deposition triggered by the recent infection

C. Post-streptococcal glomerulonephritis, an immune-complex nephritic disease that consumes complement

D. Membranous nephropathy, with subepithelial immune deposits thickening the basement membrane

Show answer

C is correct. Red cell casts plus hypertension plus sub-nephrotic proteinuria is the nephritic pattern, and the ten-day latency after pharyngitis with a consumed C3 and positive ASO nails post-streptococcal disease. IgA nephropathy is the strong distractor because it also produces glomerular haematuria after an upper respiratory infection, but it appears within one to two days rather than after a one to three week latent period and characteristically leaves C3 normal. Minimal change and membranous are nephrotic diseases: they cause heavy proteinuria with a bland, blood-free sediment and no hypertension.

PRACTICE

Q2. An adult with 6 g per day of proteinuria, serum albumin 2.1 g/dL, anasarca and cholesterol 380 mg/dL suddenly develops left flank pain, gross haematuria and a rise in creatinine. What is the most likely event and why does it happen?

A. Renal vein thrombosis, because antithrombin III is lost in the urine alongside albumin and the patient becomes hypercoagulable

B. An obstructing ureteric calculus formed from the hypercalciuria of nephrotic syndrome

C. Conversion of the nephrotic process into a nephritic one with acute capillary inflammation

D. Acute pyelonephritis seeded by the immunosuppression of chronic protein loss

Show answer

A is correct. A filter that leaks albumin leaks every protein of similar molecular size, and antithrombin III is one of them. Combine that loss with haemoconcentration from intravascular depletion and a liver overproducing fibrinogen, and the nephrotic patient becomes strongly prothrombotic, with renal vein thrombosis as the signature complication presenting exactly this way. A stone would be a reasonable guess for flank pain and haematuria in isolation, but nothing in nephrotic physiology promotes stone formation, and it fails to explain the abrupt functional decline. Nephrotic and nephritic syndromes are different injury patterns, not sequential stages.

PRACTICE

Q3. A patient with stage 4 CKD has calcium 7.6 mg/dL, phosphate 6.4 mg/dL, PTH 480 pg/mL and an elevated alkaline phosphatase. Which sequence of events produced these values?

A. An autonomously secreting parathyroid adenoma is raising PTH, which raises calcium and wastes phosphate

B. Failing kidneys retain phosphate and cannot activate vitamin D, so serum calcium falls and the parathyroids are driven to oversecrete PTH, which resorbs bone

C. Loss of erythropoietin reduces marrow activity, which secondarily lowers serum calcium and stimulates PTH

D. Metabolic acidosis directly stimulates the parathyroid gland, and PTH then raises serum phosphate

Show answer

B is correct. The direction of the calcium is the entire answer. In **secondary** hyperparathyroidism the gland is responding appropriately to a low calcium, which is why calcium is 7.6 while phosphate is high from failed excretion and PTH is soaring. The adenoma option describes **primary** hyperparathyroidism, where the gland acts without permission and the labs invert: high calcium with low phosphate. Erythropoietin has no role in calcium handling, and PTH lowers rather than raises serum phosphate by blocking proximal tubular reabsorption.

30 Respiratory: Obstruction, Restriction, Hypoxaemia, and the Acute Failures

> **Key idea —** Every respiratory problem answers two questions. First, is it hard to get air **out** (obstructive, air trapping, low FEV1/FVC) or hard to get air **in** (restrictive, small stiff lungs, preserved ratio)? Second, is the oxygenation failure one of ventilation failing to reach blood (**shunt**) or blood failing to reach ventilation (**dead space**)? Fix those two axes and everything else is detail.

Obstructive versus restrictive

Spirometry answers the first question mechanically. In obstruction the airways narrow or collapse during exhalation, so forced expiratory volume in one second falls far more than forced vital capacity does and the ratio drops below 0.70; the air that cannot get out stays behind, so residual volume and total lung capacity rise and the chest becomes hyperinflated. In restriction the lung or chest wall simply will not expand, so every volume shrinks together, total lung capacity falls, and the ratio stays normal or even climbs above 0.80 because a stiff lung empties briskly. Diffusing capacity then subdivides the causes: it is low in emphysema (alveolar surface destroyed) and in fibrosis (membrane thickened), but normal or high in asthma and in purely extrapulmonary restriction such as obesity or kyphoscoliosis.

Feature	Obstructive disease	Restrictive disease
Core defect	Airflow limitation, mainly on exhalation	Reduced compliance of lung or chest wall; the lung cannot expand
FEV1/FVC ratio	Below 0.70	Normal or above 0.80
Total lung capacity and residual volume	Increased from air trapping	Decreased proportionally
Compliance	Increased in emphysema (floppy, easy to inflate, hard to empty)	Decreased (stiff, hard to inflate, empties fast)
Breathing pattern	Slow, prolonged expiration, pursed lips, accessory muscle use	Rapid and shallow: small tidal volumes at a high rate
Typical causes	Asthma, chronic bronchitis and emphysema, bronchiectasis, cystic fibrosis	Pulmonary fibrosis, ARDS, sarcoidosis, pneumoconiosis, obesity, kyphoscoliosis, neuromuscular weakness
Late gas exchange pattern	Hypercapnia as ventilation fails, with compensated respiratory acidosis	Hypoxaemia first; carbon dioxide stays low until end stage because of the fast respiratory rate

MEMORY HOOK **O**bstructive: the problem is getting air **O**ut, the lungs are **O**verinflated, and the rati**O** drops. **R**estrictive: volumes are **R**educed but the **R**atio is **R**etained.

The mechanisms of hypoxaemia

Mechanism	What is wrong	A-a gradient	Corrects with 100 percent oxygen?	Classic example
Hypoventilation	Not enough air is moved; rising alveolar CO2 physically displaces alveolar O2	Normal	Yes, readily	Opioid overdose, neuromuscular weakness, obesity hypoventilation
V/Q mismatch	Ventilation and perfusion are distributed unevenly across lung units	Widened	Yes, usually	Asthma, COPD, pneumonia, early pulmonary embolism
Shunt (V/Q equals zero)	Alveoli are perfused but not ventilated at all, so venous blood passes through unoxygenated	Widened	No, refractory	ARDS, lobar consolidation, pulmonary oedema, atelectasis, intracardiac right-to-left shunt
Diffusion impairment	The alveolar-capillary membrane is thickened or its surface area is destroyed	Widened	Yes, often dramatically	Pulmonary fibrosis, emphysema, desaturation on exertion
Low inspired oxygen tension	The atmosphere itself is oxygen-poor; the lung is normal	Normal	Yes	High altitude

Pearl — Shunt and dead space are mirror images and the pairing is worth memorising. **Shunt** is perfusion without ventilation: blood streams past a flooded or collapsed alveolus and returns to the left heart still venous. Raising the inspired oxygen cannot rescue it, because the haemoglobin coming from the alveoli that **do** work is already close to 100 percent saturated and physically cannot carry more. That refractory hypoxaemia is the fingerprint of ARDS, and it is why the treatment is PEEP to recruit collapsed units rather than simply turning up the FiO2. **Dead space** is ventilation without perfusion: air reaches alveoli with no blood flow, as in pulmonary embolism. It wastes ventilation, widens the gap between end-tidal and arterial carbon dioxide, and drives the patient to hyperventilate.

Alveolus floods or collapses → blood perfuses an unventilated unit (shunt) → venous blood mixes into the arterial circulation → hypoxaemia that will not respond to raising FiO2 → requires PEEP and recruitment

Asthma versus COPD

Feature	Asthma	COPD
Underlying process	Type I hypersensitivity with eosinophilic airway inflammation, bronchospasm, mucosal oedema and mucus plugging	Chronic irritant injury, usually smoking: bronchitic mucus hypersecretion plus emphysematous destruction of alveolar walls
Onset and course	Childhood or young adult, episodic, trigger-driven, genuinely normal between attacks	Fifth or sixth decade, insidious, relentlessly progressive, never fully normal
Reversibility	Largely reversible: FEV1 improves by at least 12 percent and 200 mL after a bronchodilator	Largely fixed, with only partial bronchodilator response
Why airways close on expiration	The lumen is narrowed from spasm, oedema and mucus	Elastic recoil and alveolar tethering are lost, so airways collapse dynamically, which is why pursed-lip breathing helps by splinting them open
Diffusing capacity	Normal	Reduced in emphysema, where alveolar surface area is destroyed
Typical gas pattern	Early attack shows low CO_2 from hyperventilation; a normalising or rising CO_2 signals exhaustion	Chronic CO_2 retention with a compensated respiratory acidosis and elevated bicarbonate
Examination and imaging	Wheeze, prolonged expiration, reduced peak flow, hyperresonance during the attack	Barrel chest, distant breath sounds, hyperresonance, flattened diaphragms and hyperinflation on film

RED FLAG

A wheezing asthmatic is still moving air. A **silent chest** is not improvement, it is the sound of airflow that has nearly stopped. Combine a quiet chest with drowsiness, inability to speak in full sentences, a falling respiratory rate, or a $PaCO_2$ that has climbed from low back into the normal range, and you are watching impending respiratory arrest; escalate and prepare for intubation. In COPD, do not withhold oxygen from a hypoxaemic patient out of fear of CO_2 retention; give it and titrate to a saturation of 88 to 92 percent.

ARDS versus cardiogenic pulmonary oedema

Feature	ARDS (non-cardiogenic)	Cardiogenic pulmonary oedema
Mechanism	Diffuse alveolar-capillary membrane injury raises permeability; a protein-rich exudate floods alveoli, surfactant is lost and hyaline membranes form	Raised pulmonary capillary hydrostatic pressure from a failing left ventricle pushes a protein-poor transudate into the interstitium and alveoli
Trigger	Sepsis (most common), aspiration, pancreatitis, major trauma, transfusion, inhalation injury, near-drowning	Myocardial infarction, hypertensive crisis, acute valve failure, arrhythmia, volume overload
Timing	Within one week of the inciting insult, typically 12 to 48 hours	Often abrupt, over minutes to hours
Cardiac findings	Normal heart size, normal or low filling pressures, no S3	Cardiomegaly, elevated jugular venous pressure, S3 gallop, peripheral oedema
BNP	Low or normal	Markedly elevated
Chest radiograph	Bilateral patchy peripheral infiltrates with air bronchograms, progressing to white-out	Perihilar bat-wing congestion, Kerley B lines, pleural effusions, upper lobe vascular redistribution
Response to oxygen	Refractory: this is true shunt, and PaO2/FiO2 of 300 or less defines the syndrome	Improves as preload falls with diuresis, nitrates and positive pressure
Management focus	Treat the underlying cause, low tidal volume ventilation at 4 to 6 mL/kg ideal body weight, PEEP, permissive hypercapnia, prone positioning	Diuretics, nitrates, afterload reduction, non-invasive positive pressure ventilation

Pearl — Both diseases flood the alveolus, but for opposite Starling reasons, and that one difference predicts every other finding. ARDS is a **permeability** leak: the barrier is broken, so protein and fluid escape even at normal pressures, which is why the heart is normal, the BNP is low and diuretics barely help. Cardiogenic oedema is a **pressure** leak: the barrier is intact but the hydrostatic pressure behind it is too high, which is why the heart is enlarged, the BNP is high and lowering preload works within minutes.

Tension pneumothorax

A tension pneumothorax is a one-way flap valve. Air enters the pleural space on inspiration and cannot escape on expiration, so intrapleural pressure climbs above atmospheric and rises with every single breath. The lung on that side collapses first, but the lethal step comes next: the mediastinum is pushed toward the opposite hemithorax, and the shift kinks the superior and inferior vena cava. Venous return falls, the right ventricle has nothing to pump, and cardiac output collapses. It is worth being precise about this, because it changes the urgency: the patient is dying of **obstructive shock**, of absent preload, not primarily of failed oxygenation.

> **RED FLAG**
> Tension pneumothorax is a clinical diagnosis and must never wait for a chest film. Look for sudden severe dyspnoea and pleuritic pain, absent breath sounds with hyperresonance on the affected side, tracheal deviation **away** from that side, distended neck veins, hypotension and tachycardia. Treat immediately with needle decompression, passing the needle over the top of the rib in the second intercostal space at the midclavicular line or the fifth space in the anterior axillary line, then place a chest tube. In a mechanically ventilated patient, a sudden spike in peak airway pressure together with hypotension is a tension pneumothorax until proven otherwise.

One-way flap → pleural pressure rises with each breath → lung collapses → mediastinum shifts → vena cava kinks → preload falls → obstructive shock and arrest

PRACTICE

Q1. Two days after severe pancreatitis a patient develops bilateral infiltrates, PaO_2 52 mmHg on FiO_2 0.8 (PaO_2/FiO_2 of 65), a normal jugular venous pressure, BNP of 40 pg/mL and a normal ejection fraction on echo. Why does increasing the FiO_2 barely change his PaO_2?

A. Flooded alveoli are perfused but not ventilated, so shunted venous blood bypasses gas exchange entirely while haemoglobin from the healthy units is already fully saturated

B. Dead space ventilation is high, so inspired oxygen is delivered to alveoli that carry no blood flow

C. Opioid analgesia is causing hypoventilation, and rising alveolar carbon dioxide is displacing oxygen

D. A thickened alveolar-capillary membrane is slowing diffusion beyond what supplemental oxygen can overcome

Show answer

A is correct. This is ARDS, and ARDS produces true shunt: alveoli filled with protein-rich exudate receive perfusion but no ventilation, so that fraction of cardiac output returns to the left heart still venous. Enriching the gas in the alveoli that still work cannot compensate, because their outgoing haemoglobin is already near 100 percent saturated and oxygen dissolved in plasma adds almost nothing. The dead space option describes the opposite lesion, ventilation without perfusion, which mainly raises carbon dioxide. Both hypoventilation and diffusion impairment are tempting, but each responds well to supplemental oxygen, which is precisely what is not happening here. The correct intervention is PEEP to recruit collapsed alveoli.

PRACTICE

Q2. A woman with progressive exertional dyspnoea, a dry cough, fine end-inspiratory crackles and digital clubbing has FVC 62 percent predicted, FEV1 66 percent predicted, FEV1/FVC 0.85, total lung capacity 61 percent predicted and DLCO 48 percent predicted. How should this be classified?

A. An obstructive defect with emphysematous destruction, given the low diffusing capacity

B. A restrictive defect with impaired diffusion, consistent with interstitial pulmonary fibrosis

C. Asthma with fixed airway remodelling

D. A normal study, since FEV1 and FVC have fallen proportionally

Show answer

B is correct. The FEV1/FVC ratio of 0.85 with every absolute volume reduced and a total lung capacity of 61 percent is the restrictive signature: the lung is small and stiff, empties briskly, and the ratio is therefore preserved or elevated. The low DLCO is the strong distractor, because emphysema also lowers DLCO, but emphysema would show a ratio below 0.70 and an **elevated** total lung capacity from air trapping. DLCO alone never distinguishes obstruction from restriction; the ratio and the total lung capacity do. Clubbing with fine end-inspiratory crackles points to interstitial fibrosis.

PRACTICE

Q3. An asthmatic has been in severe distress for two hours. His initial assessment showed loud diffuse wheeze with an ABG of pH 7.49 and PaCO2 29 mmHg. He is now drowsy, his chest is quiet, and repeat ABG shows pH 7.34 with PaCO2 44 mmHg. How should this be interpreted?

A. He is improving, since the wheeze has resolved and his carbon dioxide has returned to the normal range

B. He has developed a lactic acidosis from repeated beta-agonist nebulisers, which explains the pH change

C. The attack is worsening critically: a silent chest with a normalising carbon dioxide means airflow has fallen so far that ventilation is failing, and respiratory arrest is imminent

D. The first blood gas was drawn incorrectly and the current values represent his true baseline

Show answer

C is correct. Wheeze requires turbulent airflow, so losing the wheeze while the patient deteriorates means air is barely moving at all. In an acute severe attack the patient should be hyperventilating and blowing off carbon dioxide, so a PaCO2 that has climbed from 29 back to a supposedly normal 44 is not reassurance, it is evidence that the respiratory muscles are exhausted and minute ventilation has collapsed. Add drowsiness and this is impending arrest. High-dose beta-agonists genuinely do cause a lactic acidosis, which makes that distractor plausible, but it does not explain the silent chest, the altered mental status or the rising carbon dioxide.

PRACTICE

Q4. A ventilated trauma patient suddenly drops to 78/40 mmHg with a heart rate of 132. Peak airway pressures spike, breath sounds are absent on the left, the trachea is deviated to the right, and the neck veins are distended. What is happening and what is the immediate priority?

A. Right mainstem intubation is ventilating only one lung; withdraw the endotracheal tube a few centimetres and obtain a confirmatory film

B. Tension pneumothorax has raised intrapleural pressure enough to shift the mediastinum and kink venous return, producing obstructive shock; decompress with a needle immediately, without waiting for imaging

C. A massive haemothorax is filling the left chest; the priority is rapid transfusion and an urgent chest radiograph

D. An acute myocardial infarction is causing cardiogenic shock; obtain a twelve-lead ECG and troponin

Show answer

B is correct. Rising pleural pressure explains all four findings at once: absent breath sounds from lung collapse, high peak airway pressures because the ventilator is fighting that pressure, tracheal deviation away from the affected side, and hypotension with distended neck veins from kinked caval inflow. Right mainstem intubation is the best distractor, since it also produces absent left breath sounds and elevated airway pressures, but it does not deviate the trachea away, distend the neck veins or cause shock. A haemothorax would give dullness rather than hyperresonance and would present with flat, not distended, neck veins because the patient is hypovolaemic. Treatment is needle decompression on clinical grounds, then a chest tube; waiting for a film can be fatal.

Study aid for course review. Always follow current clinical guidelines and your instructor's stated exam scope.